ABSTRACT

EAST, JAMES DERRICK. Understanding Multiscale Air Quality Impacts of Observed and Projected Emissions Changes using Chemical Transport Models. (Under the direction of Dr. Fernando Garcia Menendez).

Many anthropogenic activities result in emissions of primary and precursor air pollutants and of greenhouse gases. Air pollutant emissions can lead to poor air quality directly through emissions of primary species and indirectly through secondary formation of particles and gases. Emissions of greenhouse gases, whose atmospheric buildup leads to climate change, indirectly contribute to air pollution through climate change's deleterious impact on air quality. Exposure to the resulting ambient air pollution is the leading environmental risk factor causing morbidity and mortality globally. The connections between emissions and poor air quality span multiple spatial and temporal scales, from urban to global, and from seasonal to multidecadal variations. To understand the causes and impacts of air pollution, assessment of emissions and resulting impacts on air pollution is needed at multiple scales. This dissertation seeks to answer the question: "How is local to regional air quality impacted by anthropogenic emissions changes occurring at a wide range of spatial and temporal scales?" Using chemical transport models, the question is addressed in four case studies of pressing air quality problems ranging from urban to global scales. First, urban air quality is analyzed in a Latin American megacity on seasonal scales and the contribution of local emissions sources to the city's $PM_{2.5}$ burden is revealed. The analysis is extended to investigate the effectiveness of future emissions control policies. Next, a chemical data assimilation and emissions inversion framework is developed to infer NO_x emissions based on satellite observations of NO_2 on hemispheric scales. The system is applied to infer lightning and anthropogenic NO_x emissions for the Northern Hemisphere and assess the impacts of changing global anthropogenic emissions on U.S. air quality. The framework advances a commonly used inversion method, and the results identify the potential for observational constraints on emissions to improve estimates of long range transported air pollution impacting the U.S. Next, the impacts of climate on air quality are assessed using a coupled global modeling framework designed to capture the impacts of climate change on air quality on multidecadal scales with an ensemble of simulations. Results show that climate change poses a risk of aggravating poor air quality during the 21st century,

independent of changes to air pollutant emissions. The simulation ensemble is leveraged to quantify climate uncertainties due to internal variability, climate sensitivity, and emissions scenario in air quality impacts for the first time. Finally, the analysis of the impact of climate on air quality is extended to assess the impact of climate change on ozone extremes. The analysis shows that climate change has the potential to increase the frequency and magnitude of future extremes, and that climate sensitivity and emissions policy scenario are key uncertainties driving the intensity of these effects. Results from each chapter reveal new insights into the impacts of emissions changes across spatial and temporal scales. Taken together, the research points to the potential value of an integrated, observationally constrained modeling system linking the global scale to the urban scale for addressing a wide range of air quality issues.

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Understanding Multiscale Air Quality Impacts of Observed and Projected Emissions Changes using Chemical Transport Models

by
James Derrick East

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APPROVED BY:

_____	_____
Dr. Fernando Garcia Menendez Committee Chair	Dr. Barron H. Henderson
_____	_____
Dr. H. Christopher Frey	Dr. Yang Zhang
_____	
Dr. Andrew Grieshop	

DEDICATION

To my parents, Brack and Amy, and my sisters, Page and Maren.

BIOGRAPHY

James D. East was born in Charlotte, North Carolina in 1993. He received a Bachelor of Science in Environmental Engineering from NC State University, graduating *summa cum laude* in 2015. While in his senior year at NC State, James became involved in research, completing an undergraduate honors thesis. After graduating, he worked at a civil engineering company in Durham, NC. In 2017, James returned to NC State to pursue a graduate degree and study air quality. In summer 2018, he was a visiting researcher at the Universidad de La Salle in Bogotá, Colombia. Starting in January 2020, James began a research fellowship at the U.S. Environmental Protection Agency's Office of Research and Development which lasted through the completion of his doctoral degree. At the EPA, he developed skills for and a research interest in chemical data assimilation and the use of satellite instruments for studying air quality.

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Chapter 1: Introduction

1.1 Sources and impacts of air pollution

Exposure to ambient air pollution is responsible for over 4 million deaths annually (Burnett et al., 2018; Cohen et al., 2017). It is the leading environmental risk factor, and the fifth largest overall risk factor for global morbidity and mortality (Murray et al., 2020). Nearly all modern anthropogenic activities that produce goods and services also result in emissions of pollutants to the air. For example, the operation of a power plant, or motor vehicle, or the manufacturing of construction materials are all activities which provide a good valued by society, while also producing emissions. Such emissions lead to air pollution directly, through emissions of primary pollutants, such as carbon monoxide, and indirectly, through emissions of chemicals and substances which interact with other substances in the air to form secondary pollutants, such as ozone and secondary organic aerosols. Beyond anthropogenic sources, emissions may also come from biogenic sources, such as lightning, plants, and soils, and geogenic sources, such as volcanoes. Natural sources would exist in the absence of anthropogenic activity, but the primary pollutants and precursor substances they emit can interact with anthropogenic emissions to further deteriorate air quality. Understanding how complex interactions among anthropogenically and naturally emitted pollutants contribute to poor air quality on local to global scales and, ultimately, to impacts on human and environmental health is a major concern for policymakers and scientists.

Regulatory actions taken by governments in recent decades have led to successes in reducing air pollutant emissions and improving ambient air quality in some parts of the world. In the U.S., average air quality has improved since the passage of the Clean Air Act in 1970 (U.S. EPA, 2021). In China, strict regulations have led to emissions decreases of the primary pollutants NO_x ($NO_x = NO + NO_2$) and $PM_{2.5}$ (particulate matter with aerodynamic diameter ≤ 2.5 μm) since the 2010's (Sun et al., 2018). Emissions of and exposure to air pollution have decreased across Europe since 2000 (Sicard et al., 2021). Despite these emissions reductions, air pollution remains a substantial burden to human health around the globe, particularly in urban areas and in low and lower-middle income countries (Brauer et al., 2012). Air quality levels in megacities across the world are substantially higher than the World Health Organization's (WHO) guidelines (Cohen et al., 2017a). Further, in the U.S., there is evidence that air pollution emissions reductions

have slowed in recent years (Jiang et al., 2018). In addition, there is a growing importance of long-range transported pollution (Jaffe et al., 2018), unregulated sources such as volatile chemical products (McDonald et al., 2018), and the impact of wildfire and prescribed burning on air quality (Jaffe et al., 2020; Afrin and Garcia-Menendez, 2020). Mitigating poor air quality on local to global scales in the face of these continuing challenges requires a deeper understanding of emissions on all scales and their contribution to air pollution.

In addition to emissions of primary and precursor pollutant emissions, climate change is projected to exacerbate air pollution (Fiore et al., 2012, 2015; Jacob and Winner, 2009). Greenhouse gases (GHG), including CO_2, CH_4, and N_2O, are co-emitted with other air pollutants. The buildup of these gases in the atmosphere leads to climate change (IPCC, 2021). Uncertainties have been quantified in climate change projections (Hawkins and Sutton, 2009; Lehner et al., 2020). While many studies have quantified the impacts of climate change on ambient pollution, termed the climate penalty (Barnes et al., 2016; Day and Pandis, 2015; Garcia-Menendez et al., 2015, 2017; Nolte et al., 2018b; Pienkosz et al., 2019; Schnell et al., 2016; Westervelt et al., 2016), and on the subsequent health and economic impacts (Fann et al., 2021; Hong et al., 2019; Silva et al., 2017) , the propagation of climate change uncertainties to the air quality penalty has not been fully quantified.

1.2 Air quality modeling

To disentangle and quantify air quality impacts caused by anthropogenic emissions, natural emissions, and GHG emissions via climate change, their effects must be analyzed in a unified framework. Chemical transport models (CTMs) are tools that can help address this challenge by simulating emissions and their impacts on atmospheric composition alongside meteorological processes. CTMs discretize the atmosphere into 3-dimensions and solve the continuity equation for atmospheric gases and particles (Brasseur and Jacob, 2017; Jacobson, 2005). To do this, CTMs simulate the transport, sources, and sinks of atmospheric constituents and calculate their time- and space-varying concentrations. Sources and sinks may include transport, surface exchange, as well as physical and chemical processes, all of which CTMs aim to capture. Realistically simulating these processes requires meteorological simulation of atmospheric state and transport variables, which may be done offline and provided as input to a CTM, or may be simulated together with a CTM in online-coupled models that

account for feedbacks between meteorological and chemical processes (Zhang, 2008). CTMs may be limited to the troposphere or may also simulate stratospheric chemistry. In the context of air pollution study in this dissertation, CTMs are referred to interchangeably as air quality models.

Air quality models are valuable tools for addressing air pollution and atmospheric chemistry questions at multiple scales. The scales of emissions changes and their impacts on air quality vary broadly in space and time, from urban-level changes with localized impacts to national-level emissions changes with global air quality impacts. Air quality models are applied across scales from urban (1km or higher resolution over an urban area), regional (10's of km resolution, covering 100s to 1000s of km), to hemispheric/global (10s to 100s km resolution, covering the globe or hemisphere) (Jacobson, 2005). Air quality models are used in the regulatory development process in the U.S. and elsewhere in the world (Carnell et al., 2019; Cohan et al., 2007; Ou et al., 2020; Palacios et al., 2002; Yang et al., 2019). In addition to simulating the impacts of emissions, air quality models are used to quantify emissions totals through inverse applications (Brasseur and Jacob, 2017; Lamsal et al., 2011; Martin et al., 2003; Miyazaki et al., 2017; Qu et al., 2019). In this approach, observations of pollutants are combined with model simulations to infer emissions amounts. Air quality models are also a primary tool used to assess the impact of climate change on air quality (Barnes et al., 2016; Fiore et al., 2022; Garcia-Menendez et al., 2015; Schnell et al., 2016; Westervelt et al., 2016). In each of the chapters of this dissertation, air quality models are the main tool used to address the chapter's motivating research question. The specific details of the implementation and advances upon current modeling methods employed are introduced here briefly and described in detail in each chapter.

1.3 Research questions

The over-arching research question motivating the chapters of this dissertation is **"How is local to regional air quality impacted by anthropogenic emissions changes occurring at a wide range of spatial and temporal scales?"** Figure 1.1 shows a schematic of factors driving air pollution, as described in the previous section. The figure is not a complete representation of all elements contributing to air pollution and its impacts; rather, it depicts the components that are most relevant for providing context to the research questions addressed in this dissertation. In the figure, the blue dashed line and

black connectors represent the confines of the dissertation within the broader context of factors contributing to air pollution. The research needs under the umbrella of the guiding question are described below.

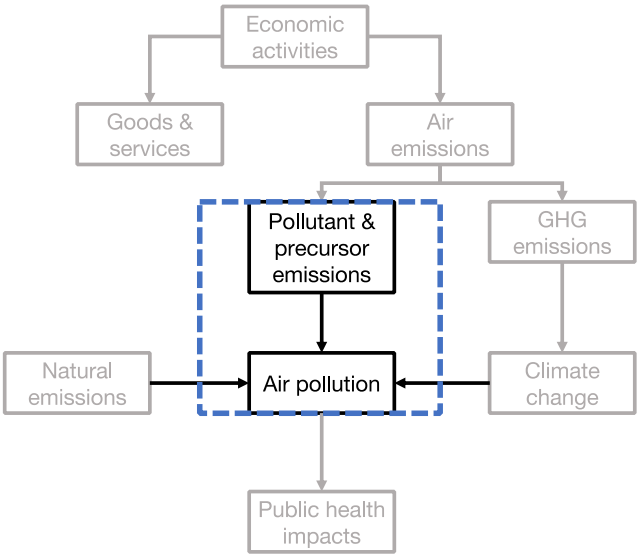

Figure 1.1: Simplified model of factors driving air pollution and the focus of this dissertation. The blue dashed line and black connectors represent the confines of the research questions addressed by this dissertation within the broader context of the many factors contributing to air pollution.

The impacts of ambient air pollution occur disproportionately in low and lower-middle income countries (LMICs) (Brauer et al., 2012; Cohen et al., 2017a). In particular, Latin American cities experience poor air quality leading to severe health and economic impacts (Bell et al., 2011; Franco et al., 2019). In other regions including Europe, China, and North America, air quality modeling has been used as an instrument in the regulatory development and evaluation process to address poor air quality (e.g., Chemel et al., 2014; Craig et al., 2020; Dedoussi et al., 2020; Zhang et al., 2019). However, the application of air quality modeling in Latin America for addressing air quality issues has, until recently (Nedbor-Gross et al., 2018), been more limited. The effective use of models requires resources including air quality observation networks and accurate emissions inventories. In many Latin American cities experiencing poor air quality, significant gaps in observation data and in reliable emissions inventories have constrained the application of air quality models for air quality management strategies (Huneeus et al., 2020; Madrazo et al., 2018; Pinder et al., 2019). However, recent developments in emissions inventories (Pachón et al., 2018), modeling capabilities (de Almeida Albuquerque et al., 2018;

Ferreyra et al., 2016; González et al., 2018; Nedbor-Gross et al., 2017, 2018), and observation networks have made this possible in cities where modeling efforts were previously limited. Despite these recent advances, the contribution of various emissions sources to the air quality burden is not well characterized. In Bogotá, Colombia, a growing megacity where poor air quality leads to severe health impacts (Bell et al., 2011; Cheng et al., 2016; Morales Betancourt et al., 2019; Mura et al., 2020; Rodríguez-Villamizar et al., 2018, 2019), emissions sources have recently been quantified in a new emissions inventory (Pachón et al., 2018), but the contribution of each source to total particulate matter pollution in the city is unclear. ***Therefore, a framework is needed for assessing local emissions source contributions in growing urban centers around the globe.***

In addition to regional formation, pollutants like ozone can be transported on hemispheric scales. As ozone precursor emissions have decreased in the U.S. and other high-income countries, the relative contribution to ozone concentrations from remote anthropogenic and natural sources, as opposed to local sources, has increased (Jaffe et al., 2018; McDuffie et al., 2020; Simon et al., 2015). For the impact of remote contribution to be simulated in these regions adequately, model boundary conditions must be an accurate representation of the incoming atmospheric composition. Typically, boundary conditions are provided by global and hemispheric scale models, which in turn rely on global emissions inventories. As a result, global emissions inventories are a critical component of simulating the formation and intercontinental transport of pollution. However, the development of such emissions inventories can be delayed in time and the inventories themselves uncertain, particularly in low and lower-middle income countries where emissions information may not be readily available, but also high-income countries (Day et al., 2019; Elguindi et al., 2020; McDuffie et al., 2020). Satellite observations of air quality can bridge this gap in time and reduce uncertainty in emissions estimates through inverse modeling (Tong et al., 2016, 2015). Previous inverse approaches directly compared satellite observations and model simulations to estimate surface anthropogenic emissions, but could not distinguish near-surface pollution from that aloft (Cooper et al., 2017; Lamsal et al., 2011). Adjoint model based methods can target surface emissions with higher precision but are heavily computationally burdensome. Accurate and up-to-date international emissions estimates, as well as estimates from natural source such as lightning-NOx emissions, are needed for regional scale modeling with policy relevance, but previous methods either could not distinguish the surface pollution or are

significantly computationally burdensome and relied on an adjoint model. ***Therefore, an updated approach for providing real-time, observationally based, top-down constraints on international and natural emissions is needed for accurate representation of ozone transport in policy-relevant modeling.***

In addition to the contribution of anthropogenic primary and precursor emissions, atmospheric composition is also affected by anthropogenic emissions of GHGs, indirectly through climate change (IPCC, 2021; Jacob and Winner, 2009), as depicted in Figure 1-1. Air pollution is projected to be exacerbated by climate change during the 21st century (Fiore et al., 2012, 2015). However, the magnitude of future climate change and its impacts carry significant uncertainties that have been explored (Fiore et al., 2022; Garcia-Menendez et al., 2017; Manders et al., 2012; Pienkosz et al., 2019; Wild et al., 2012) but have not been systematically quantified for air quality impacts. Previous studies have shown that significant uncertainty in the climate impact on air quality is due to uncertainty in anthropogenic emissions of GHG emissions (Garcia-Menendez et al., 2015) and uncertainty due to climate internal variability (Barnes et al., 2016; Garcia-Menendez et al., 2017; Pienkosz et al., 2019). Climate sensitivity, another key uncertainty in climate projections (Hawkins and Sutton, 2009; Lehner et al., 2020), is also propagated to air quality impact projections but has not been widely explored. Although the climate uncertainties due to GHG emissions scenario, internal variability, and climate sensitivity are well known for climate projections (Deser et al., 2020; Hawkins and Sutton, 2009; Lehner et al., 2020), these climate uncertainties have not been systematically quantified or compared in a unified framework for air quality impacts. ***Therefore, distinct characterization of climate uncertainties in projections of the impact of climate on air quality is needed.***

Beyond the impact to the mean state of air quality, climate is also projected to impact air quality extremes (Fiore et al., 2022; Rieder et al., 2015; Schnell et al., 2016; Shen et al., 2016). However, the impact on extremes has not been previously assessed in the context of uncertainties from GHG emissions, internal variability, and climate sensitivity. Climate impacts on the upper tail of the distribution of future air pollutant concentrations has the potential to aggravate air quality managers' attempts to meet regulatory requirements, despite significant progress in recent decades in decreasing ozone through regulatory actions in the U.S. This can be simulated using coupled climate-chemistry models; however, such models are known to be biased high for ozone, preventing

realistic assessment of future extremes (Liu et al., 2022; Stock et al., 2014; Turnock et al., 2020; Wild and Prather, 2006). ***Therefore, an assessment of the climate impact on future ozone extremes in the context of climate projection uncertainties is needed for scientists and air quality managers.***

1.4 Research objectives

This dissertation targets the overarching question posed by addressing the knowledge gaps identified through the following objectives:

1. Quantify the impact of local anthropogenic emissions sources on urban air quality in a recently enabled air quality modeling system for Bogotá, Colombia.
2. Develop an inverse modeling framework in an air quality modeling platform and provide observationally based constraints on global anthropogenic and natural emissions inventories using satellite observations.
3. Quantify and compare climate uncertainties in projections of the impact of climate change on U.S. air quality.
4. Apply and evaluate bias correction of ozone extremes simulated in a climate-chemistry model and project future ozone extremes using extreme value theory in the context of climate uncertainties due to GHG emissions, internal variability, and climate sensitivity.

Figure 1.2 shows how this dissertation addresses each of these research objectives and the temporal and spatial scales at which each chapter fits in to the overarching research question. While the overarching question has been explored extensively, the individual chapters here each represent a novel analysis. Air quality models are the primary tool used in each of these studies. The motivation, methods, and results of each investigation are presented in Chapters 2-5, along with a brief introduction below.

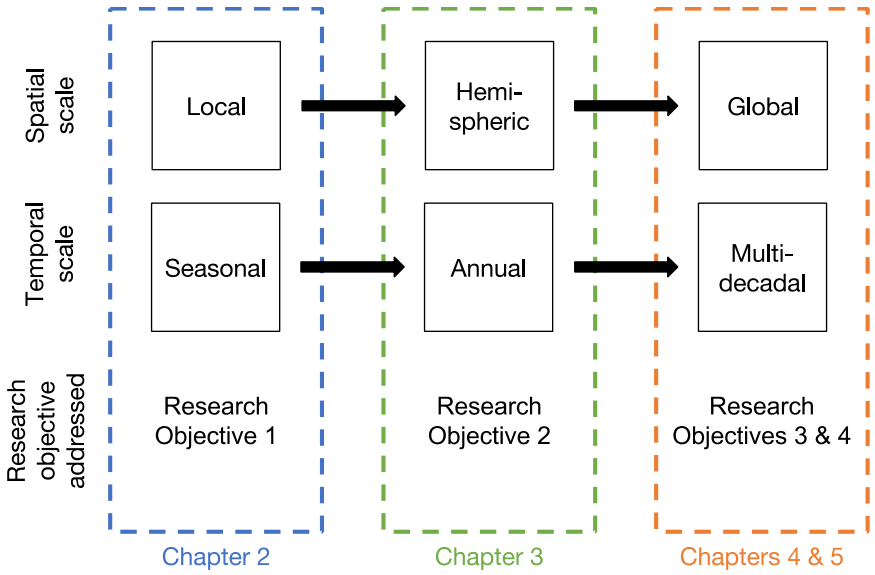

Figure 1.2: Chapters of this dissertation, the research objectives addressed in each chapter, and scales of the modeling systems used.

1.4.1 Chapter 2: Air quality modeling to inform pollution mitigation strategies in a Latin American megacity

This chapter applies sources scaling in a recently developed urban-scale air quality modeling platform for Bogotá, Colombia to identify the impact of various local emissions sources on particulate matter pollution in the city. This chapter advances the application of chemical transport modeling in Latin America where applications were previously limited, an important research need for mitigating poor air quality. Further, new information about the sources of air pollution and their impacts in Bogotá is revealed, which is valuable for policymakers and researchers, and provides a case study for applying such methods in other growing urban centers that have not previously relied on air quality modeling for studying local air quality.

1.4.2 Chapter 3: Inferring and evaluating satellite-based constraints on NOx emissions estimates in air quality simulations

In this chapter, satellite chemical data assimilation is implemented for the first time in the hemispheric scale Community Multiscale Air Quality modeling platform

(CMAQ). 3-D variational data assimilation is coupled to the CMAQ model and a framework for producing inverse estimates of anthropogenic and lightning emissions of NOx using the finite-difference mass-balance approach is developed. In the new framework and data assimilation system developed, near-surface anthropogenic emissions can be distinguished from aloft emissions, a distinction not previously possible in the finite-difference mass balance approach. Natural and anthropogenic emissions are inferred for 2019 using NO_2 retrievals from two satellite instruments. The impact of updated emissions on U.S. and Northern Hemisphere air quality is analyzed.

1.4.3 Chapter 4: Characterizing and quantifying uncertainty in projections of climate change impacts on air quality

Using the framework of climate uncertainty introduced by Hawkins & Sutton (2009), this chapter quantifies and compares climate uncertainties from GHG emissions, internal variability, and climate sensitivity in the impacts of climate on air quality in the U.S. using a single-model ensemble of coupled climate-chemistry simulations. This investigation compares these uncertainties for air quality impacts on ozone and $PM_{2.5}$ for the first time in a unified framework. This chapter sheds new light on air quality projection uncertainties and on the potential for ensemble modeling methods to better project future air quality in the face of climate uncertainty.

1.4.4 Chapter 5: Changes in frequency and magnitude of future extreme ozone events under natural variability and uncertain climate sensitivity

In this chapter, the impact of climate change on extreme ozone episodes in the U.S. is assessed using a single-model ensemble of coupled climate-chemistry simulations along with extreme value theory. First, model simulated ozone is bias corrected using ground-based observations of ozone. Then, simulated and observed extremes are modeled using extreme value theory. The bias correction method and the ability of the bias corrected climate-chemistry model to reproduce observed extremes is evaluated. U.S. ozone extremes in climate-chemistry projections are modeled and the change in future extremes under different climate scenarios is analyzed. Future extremes are

examined in light of climate uncertainties, and the ability of ensemble simulations to better capture natural variability in simulated extremes is evaluated.

1.4.5 Chapter 6: Conclusions

Key insights drawn from each chapter are summarized and potential future research directions are highlighted.

Chapter 2: Air quality modeling to inform pollution mitigation strategies in a Latin American megacity

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Abstract

Poor air quality disproportionally impacts cities in low- and middle-income countries. In Bogotá, Colombia, a metropolitan area with over 10 million inhabitants, fine particulate matter ($PM_{2.5}$) levels regularly exceed air quality guidelines, leading to detrimental effects on health. Although there is public interest to improve the city's air quality, the main sources of $PM_{2.5}$ pollution have not been clearly identified and the use of modeling for policy development in Bogotá has been limited. Here, we apply a modeling framework based on the Community Multiscale Air Quality Modeling System (CMAQ) to conduct seasonal simulations of air pollution in Bogotá and reveal the emissions sectors with the largest contributions to $PM_{2.5}$. Based on these results, we project and compare the air quality benefits of potential pollution mitigation strategies focused on these sources. The analysis finds that resuspended dust from unpaved roads is the largest local source of $PM_{2.5}$ and can contribute over 30% of seasonally-averaged concentration across the city. Vehicles, industrial activity, and unpaved road dust combined are responsible for over 60% of $PM_{2.5}$ pollution in Bogotá. A scenario analysis shows that paving roads can lead to $PM_{2.5}$ decreases of nearly 10 $\mu g/m^3$ by 2030 in some areas of the city, but air quality will deteriorate significantly over others in the absence of additional emissions control measures. Mitigation strategies designed to target the sectors with the largest contributions to $PM_{2.5}$, including road cleaning systems, controls for industrial point sources, cleaner transportation fuels, and updated vehicle fleets, can largely avert projected increases in concentrations, although the impacts of different approaches vary throughout the city. This study is the first to use a comprehensive model to determine sector contributions to air pollution and inform potential emissions control policies in Bogotá, demonstrating an approach to guide pollution management in developing cities facing comparable challenges.

2.1 Introduction

Ambient air pollution causes millions of premature mortalities annually (Burnett et al., 2018; Cohen et al., 2017). These impacts occur disproportionately in low- and middle-income countries (LMICs) (WHO, 2014). Actions that address air pollution challenges at these locations, including aging vehicle fleets, dust resuspension, and industrial coal consumption, can result in large health and environmental benefits. In Latin America, urban centers frequently experience poor air quality and substantial negative impacts associated with air pollution (Bell et al., 2011; Franco et al., 2019). Air quality models provide tools to better understand the sources and processes driving pollutant concentrations, exposure, and public health effects (Burke et al., 2017; NRC, 2007). However, efforts to integrate computational modeling into air quality management in Latin America have been limited. Still, the region offers important opportunities to benefit from the use of air pollution models.

Air quality modeling has been used to study and guide environmental decision-making in North America, Europe, and China for years (Carnell et al., 2019; Cohan et al., 2007; Ou et al., 2020; Palacios et al., 2002; Yang et al., 2019). Chemical transport models (CTMs), which represent the state of the science in air quality modeling, have been used in these countries to evaluate and develop air pollution mitigation policies at the city, regional, and national scales (Chemel et al., 2014; Craig et al., 2020; Dedoussi et al., 2020; Zhang et al., 2019a). In contrast, attempts to apply CTMs in developing areas are often constrained by limitations in the information typically used to support modeling efforts (Pinder et al., 2019). CTMs require significant computational resources and data inputs, and often rely on established monitoring networks and emissions inventories, such as the U.S. Air Quality System (U.S. EPA, 2019) and National Emissions Inventory (Adams, 2020). Global datasets can fill some of the data gaps when local inventories are not available, but do so at a cost to model performance. For Latin American cities, global emissions inventories are inconsistent and inaccurate (Huneeus et al., 2020; Madrazo et al., 2018). Local emissions inventories, when available, have been shown to improve model performance in the region compared to global inventories (González et al., 2018). Although CTM applications focused on developing areas are limited, initial efforts to use them in Latin America are expanding (de Almeida Albuquerque et al., 2018; Ferreyra et al., 2016). Additional research demonstrating the capacity of CTMs to inform air pollution mitigation strategies in the region is needed.

Bogotá, in Colombia, is one of Latin America's six megacities (population >10 million) and among its most polluted (Bell et al., 2011; Cheng et al., 2016; Mura et al., 2020). In 2014, the city experienced annual average $PM_{2.5}$ (particulate matter with aerodynamic diameter ≤ 2.5 μm) concentrations of 21 $\mu g/m^3$, annual averages of over 31 $\mu g/m^3$ at two of the most polluted sites, and 48 days with average $PM_{2.5}$ levels exceeding 50 $\mu g/m^3$ across all sites in Bogota's monitoring network, RMCAB (Red de Monitoreo de Calidad del Aire de Bogotá [RMCAB; (SDA, 2019)]). Ambient particulate matter (PM) in Bogotá has been linked to emergency room visits for respiratory and circulatory diseases (Rodríguez-Villamizar et al., 2019, 2018) and has been estimated to contribute to over 10% of deaths in the city (DNP, 2017). Although air quality modeling has not been an element of Bogotá's environmental policy development process in the past, it has been used in recent research investigating the city's atmospheric composition (Kumar et al., 2016; Nedbor-Gross et al., 2018; Pachón et al., 2018; Reboredo et al., 2015). A few studies have conducted source apportionment of PM_{10} (PM aerodynamic diameter ≤ 10 μm) in Bogotá using receptor modeling and observation-based methods (Ramírez et al., 2018; Vargas et al., 2012), but no efforts to identify the major sources of pollution based on CTMs have been previously performed. Recent studies have identified resuspended particulate matter (RPM) as a major source of PM in the city (Pachón et al., 2018, 2020). Still, the contributions of major emission sectors to total $PM_{2.5}$ in Bogotá remain highly uncertain. New local emissions inventories (Beltrán et al., 2012; Mangones et al., 2019; Pachón et al., 2018; Cardenas-Ruiz, 2012) and the availability of observations from Bogotá´s expanding air quality monitoring network RMCAB, however, have now made it possible to conduct CTM-based source apportionment and evaluate proposed air pollution mitigation strategies.

Here we use a comprehensive CTM to simulate air quality over the Bogotá metropolitan area, identify the main sources of $PM_{2.5}$ pollution in the city, and examine the potential impacts of emissions policies targeting these sources. We rely on a locally developed emissions inventory and apply a source-scaling modeling method to weigh the influence of major emission sectors on $PM_{2.5}$ across the city. Based on insights attained from this analysis, we develop, simulate, and project the air quality benefits of different emissions reduction strategies. This study is the first to use a CTM-based approach to determine sector contributions to air pollution in Bogotá, as well as scenario analysis to evaluate potential mitigation policies. As such, it presents a framework that can guide

the use of air quality modeling in Latin American and LMIC cities facing comparable air quality challenges.

2.2 Data and Methods

To determine emissions source contributions to total $PM_{2.5}$, we model Bogotá air quality with a base case simulation and ten emissions sensitivity simulations. To analyze potential emissions mitigations strategies, we conduct a future base case with emissions projected to 2030 and four future emissions scenario simulations. The following sections describe the modeling system and emissions (Section 2.2.1), model performance evaluation against observations (Section 2.2.2), source apportionment methodology (Section 2.2.3), and future emissions control scenarios (Section 2.2.4).

2.2.1 Air quality modeling

Air quality is simulated with the Community Multiscale Air Quality Modeling System (CMAQ version 5.0.2) (U.S. EPA, 2014) with the Carbon Bond (CB05) chemical mechanism with updated toluene and chlorine chemistry (Sarwar et al., 2012) and AERO6 aerosol module (Appel et al., 2013). The modeling framework including domain design, boundary conditions, and additional refinements are described in Nedbor-Gross et al. (2018). All simulations described here use a 64 km x 64 km grid with 1-km horizontal resolution centered over the Bogotá metropolitan area (Figure A.1) nested within 3 outer domains. Initial and boundary conditions are provided by the outer domains of the nested simulation with outermost boundary conditions derived from downscaled GEOS-Chem global air quality simulations. Meteorology is simulated using the Weather Research and Forecasting Model (WRF version 3.5.1) (Nedbor-Gross et al., 2017). Three-month simulations are used to represent the two seasons characteristic of Bogotá's weather: January-March 2014 (dry season; JFM) and October-December 2014 (rainy season; OND).

A local inventory developed by Pachón et al., (2018) provides emissions for the base-case simulation. The emissions were compiled using top-down estimates and bottom-up approaches aided by field sampling and are the first inventory for Bogotá to be evaluated with a CTM. The inventory includes hourly, 1-km^2 emissions of CO_2, CO, nitrogen oxides (NO_x = NO + NO_2), SO_2, speciated volatile organic compounds (VOCs),

and speciated particulate matter from industrial, mobile, commercial, biogenic, and multiple RPM sources. Pachón et al., (2018) provides a detailed breakdown of the emissions by sector. Originally developed for 2012, it has been updated to 2014 using new vehicle fleet data, mobile source emission factors, and industry growth rates. Additionally, the inventory includes improved RPM emissions for unpaved roads estimated with a natural dust mitigation factor based on local weather conditions following Pérez-Peña et al. (2017), enhanced paved road RPM estimates applying the methodology of (Amato et al., 2011, 2009), and new emissions factors for mobile sources derived with Motor Vehicle Emission Simulator (MOVES; U.S. EPA, 2020) for Bogotá (Ramirez et al., 2019). The inventory, developed in collaboration with the District Secretariat of Environment (Secretaría Distrital de Ambiente, SDA), is the most complete and current compilation of emissions in Bogotá to date.

2.2.2 Observational data and model evaluation

Observed 24-hour average $PM_{2.5}$ concentrations were obtained from the RMCAB network (SDA, 2019). Figure 2.1 shows the locations of the 13 monitors comprising the network in the Bogotá metropolitan area (2.1a) and observed $PM_{2.5}$ concentrations in 2014 (1b). In the discussion of results, we refer to $PM_{2.5}$ at urban/suburban and traffic/industrial sites, which refer to the monitoring site characteristics (SDA, 2019). Urban/suburban sites refer to those representing local mean pollution levels in urban or suburban areas, while traffic/industrial sites refer to those near urban industrial or traffic sources, as labeled on Figure 2.1a. Both the World Health Organization's (WHO) guideline (25 $\mu g/m^3$) and Bogotá's municipal standard (37 $\mu g/m^3$) for 24-hour $PM_{2.5}$ were frequently exceeded in 2014 (Figure 2.1c). Modeled values at observation sites are taken as the simulated concentration in the grid cell containing the site. City-mean model results are estimated as the mean $PM_{2.5}$ concentration across all cells encompassed by the municipal city boundaries shown in Figure 2.1b.

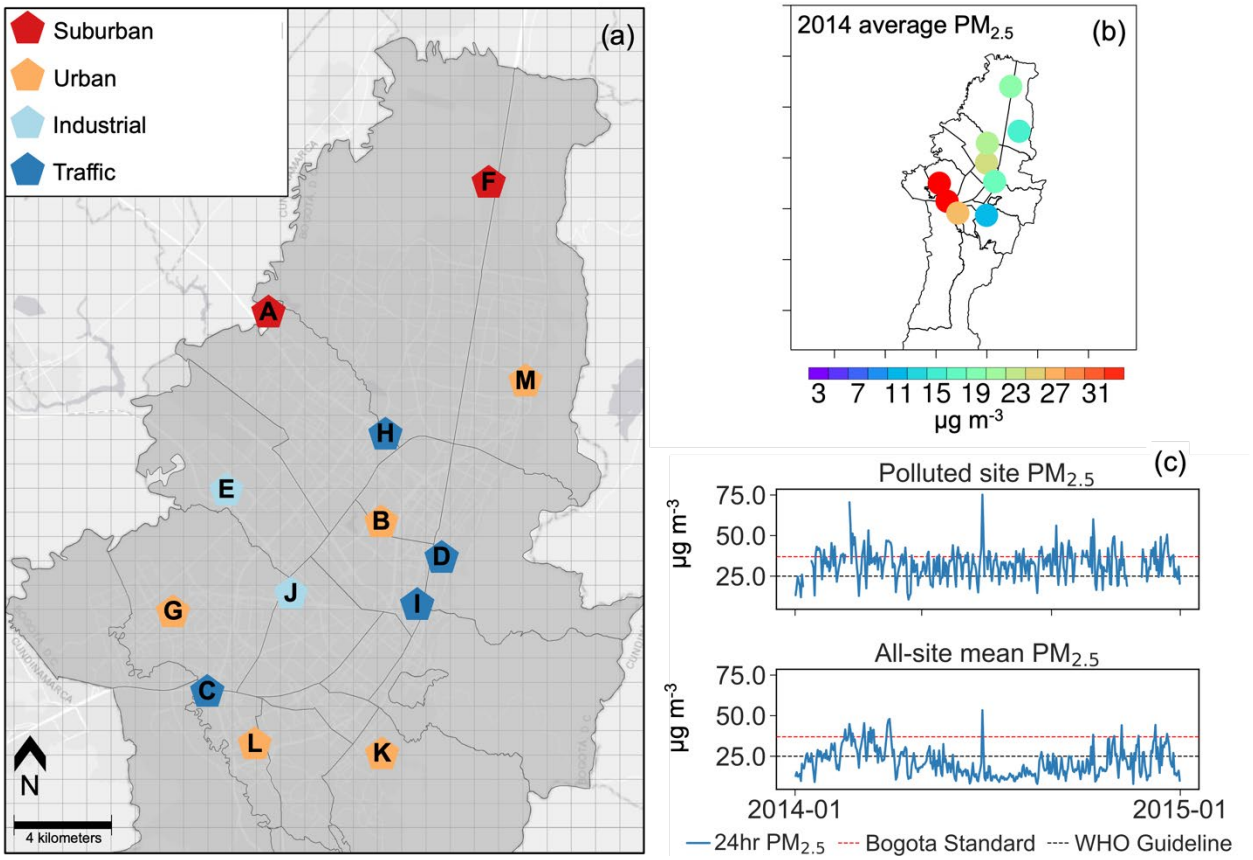

Figure 2.1: RMCAB monitoring network and observed PM2.5 concentrations. (a) Monitor locations and categories. Site names are A: Bolivia, B: Centro de Alto Rendimiento (CDAR), C: Carvajal, D: Convenio 176 10, E: Fontibon, F: Guaymaral, G: Kennedy, H: Las Ferias, I: Min. Ambiente, J: Puente Aranda, K: San Cristobal, L: Tunal, and M: Usaquen. Dark grey area on map delineates Bogotá city limits. (b) Annual-average $PM_{2.5}$ observed at monitoring sites in 2014. (c) 24-hour $PM_{2.5}$ observed at a polluted site (Carvajal) and mean for all monitoring sites in 2014. WHO guideline and Bogotá standard for 24-hour $PM_{2.5}$ are also shown.

We evaluate modeled concentrations against observations using normalized mean bias (NMB) and normalized mean error (NME). Figure A.2 shows that model performance falls within the benchmark NMB and NME criteria for photochemical grid models (Emery et al., 2017) at the majority of the sites. Compared against all monitoring sites, the NME, NMB, and correlation coefficient (R) for 24-hour $PM_{2.5}$ are 44%, -7%, and 0.48, respectively. Sites outside the criteria are those with the lowest observed $PM_{2.5}$ concentrations during the simulated periods. It is important to note that these performance guidelines are specific to model performance in North America, where photochemical modeling has an extensive history and the uncertainties associated with

model inputs (e.g., emissions) are significantly lower. The modeling system for Bogotá is largely able to meet these criteria, lending confidence to its fitness for this analysis.

2.2.3 Source apportionment

We use brute-force source-scaling to apportion $PM_{2.5}$ among local emissions sectors with an annual mean primary $PM_{2.5}$ emissions rate greater than 100 g/s. Sectors with emissions above this threshold are scaled down individually by a scaling factor δ. Sectors below this emissions rate are grouped and scaled as an aggregate source by δ. For each emissions sector, we simulate both seasons with the scaled sector emissions. To determine C_i, the contribution to total $PM_{2.5}$ concentration by emissions sector i, we apply Equation 2.1:

$$C_i = \frac{C_E - C_{E_i(1-\delta)}}{\delta} \qquad \text{Equation 2.1}$$

where C_E is total $PM_{2.5}$ concentration simulated with base-case emissions, $C_{E_i(1-\delta)}$ is total $PM_{2.5}$ concentration simulated with emissions sector i perturbed by 1 – δ, and δ is the emissions scaling factor. To quantify $PM_{2.5}$ from domain boundaries, we apply the same technique, scaling down all boundary conditions. During the JFM and OND seasons, we model the base case, sensitivity simulations for each of eight emissions sectors with emissions rates greater than 100 g/s, a single sensitivity simulation for all emissions sources less than 100 g/s, and a sensitivity simulation for boundary conditions. In total, we conduct 11 simulations for each season for the source apportionment portion of the study. The sum of the contributions from all sectors and boundary (*N*) is compared to the total base-case $PM_{2.5}$, C_E, to assess rescaling accuracy in Equation 2.2:

$$x = \frac{\sum_{i=1}^{N} C_i}{C_E} \qquad \text{Equation 2.2}$$

Here, x is the rescaling fraction and $1 - x$ is the rescaling error. Emissions perturbations have been used extensively as the basis of sensitivity analyses (Cohan et al., 2005; Hakami et al., 2004) and global and regional source apportionment studies (Janssens-Maenhout et al., 2015). Although zero-out source scaling has been applied for source apportionment (e.g., Caiazzo et al., 2013; Lelieveld et al., 2015; Silva et al., 2016; Weagle et al., 2018), brute-force methods can cause rescaling errors if emissions perturbations are large enough to change chemical regimes (Thunis et al., 2019). We choose a 20% scaling factor in order to

produce a measurable response but constrain the potential for significant nonlinearity errors (Fiore et al., 2009; Wild and Akimoto, 2001).

2.2.4 Emissions projections and pollution mitigation strategies

To evaluate the effectiveness of emissions control strategies, we project air quality to 2030 under a business as usual (BAU) scenario based on projections of economic growth, current development plans, and assuming implementation of no new emissions control policies. We then simulate four emissions reduction strategies with BAU as the base emissions. The emissions control measures considered target RPM from roads, industrial point sources, cargo trucks, and passenger vehicles. Table 2.1 summarizes emissions changes under these scenarios and each projection is described below.

Table 2.1: Total emissions and emission changes for base-case simulation and 2030 projections

Base simulations		**PM_{10}**	**$PM_{2.5}$**	**NOx**	**SO_2**	**CO**
2014 base case	Emissions (tonne/day)	66	14	158	4	846
BAU 2030	Emissions (tonne/ day)	24	15	238	7	2,492
	Change*	-63.9%	+5.8%	+50.8%	+92.7%	+194.5%
Emissions reduction strategies						
1. Paved roads cleaning	Emissions (tonne/ day)	21	14	238	7	2,492
	Change**	-11.4%	-4.4%	-	-	-
2. Industrial point sources	Emissions (tonne/ day)	23	15	238	7	2,492
	Change**	-1.9%	-0.7%	-	-	-
3. Heavy-duty vehicle fleet	Emissions (tonne/ day)	23	14	229	7	2,481
	Change**	-3.6%	-5.1%	-3.9%	-1.7%	-0.4%
4. Light-duty vehicle fleet	Emissions (tonne/ day)	21	12	193	3	1,356
	Change**	-12.5%	-19.5%	-18.7%	-64.5%	-45.6%

**Changes for BAU 2030 projections are relative to 2014 emissions*
***Changes for emissions reduction strategies are relative to BAU 2030*

2.2.4.1 Business as usual projection

BAU emissions are projected to 2030 based on anticipated economic and fuel use growth rates. The number of vehicles in Bogotá, with the exception of motorcycles, is projected to increase at an average rate of 2.5% per year, following fuel demand

expectations from the Ministry of Mines and Energy (UPME, 2015). This increases the number of passenger vehicles and cargo trucks in 2030 by approximately 50%. To reflect the rapid growth of motorcycle use recently experienced in the city, the number of motorcycles is projected to rise by 10% per year based on the increase in SDA registration records from 2008 to 2014. In line with the SDA's current targets and 10-year air pollution mitigation plan, the size of Bogotá's public transportation fleet is projected to remain unchanged from 2014 (SDA, 2010). Under the BAU scenario, vehicle emissions factors are held at 2014 levels. Point source emissions are projected based on fuel consumption forecasts derived from industrial and commercial gross domestic product projections for Bogotá (Ecopetrol and ULS, 2016). Average annual growth rates of 3.9% and 2.3% are applied to industrial and commercial point sources, respectively. In accordance with urban development targets (SDA, 2010), the BAU scenario additionally projects paving of all unpaved roads in the city by 2030 at a rate of 5.5 km per month. Road paving decreases RPM, with emissions estimated using the same methods reliant on-road dust loadings applied to create the 2014 base case local emissions inventory (Pachón et al., 2018). Four pollution control strategies focused on major sources of $PM_{2.5}$ are projected by adding emissions controls to 2030 base case emissions.

2.2.4.2 Emissions reduction projections

The air pollution mitigation strategy focused on road RPM considers the implementation of a cleaning system of street sweeper trucks with water sprinklers collecting dust from all main and secondary roads. The road cleaning system is assumed to reduce RPM emissions from roads by 44% based on the observed impacts of a similar implementation in Santiago, Chile since 1998 (Min. Interior Chile, 2005). Other studies have shown dust removal efficiencies between 15% and 60% depending on particle size and type of cleaning device (Querol et al., 2018), with an average removal efficiency of 41% total sediment removal (Amato et al., 2010). Relative to BAU, the control strategy reduces resuspended PM_{10} and $PM_{2.5}$ emissions from all RPM sources in 2030 by approximately 35%.

The emissions control strategy for industrial point sources targets 26 coal-burning industrial facilities which combined represent 60% of Bogotá's BAU industrial PM_{10} emissions. These sources include brick, ceramic, and textile manufacturing facilities. Emissions reductions assume installation of particle capture cyclones on the facilities'

exhaust stacks with a conservative collection efficiency of 70% (Dirgo and Leith, 1985). The industrial controls result in decreases of 18% and 26% for PM_{10} and $PM_{2.5}$ point source emissions respectively, relative to BAU projections.

The pollution mitigation strategy focused on cargo vehicles combines two emissions reduction approaches: fleet upgrades and diesel fuel reformulation. All heavy-duty vehicles added to the city fleet since 2014 are assumed to meet EURO IV emissions standards, as defined by the European Union's (EU) regulations for emissions from the heavy-duty vehicle fleet (EU, 2005). In line with the Ministry of Environment and Sustainable Development's plans to encourage cargo fleet modernization, 10,000 older heavy-duty vehicles with PRE-EURO or EURO I emission controls are removed from the cargo fleet by 2030. An additional 10,000 heavy-duty vehicles are upgraded to EURO IV technology with improved fuel combustion efficiency and selective catalytic reduction. Under the strategy, emissions are further reduced by switching to reformulated diesel fuel with low sulfur content. The decrease in sulfur concentration, from 50 to 10 ppm, is based on the 2030 goals of the National Planning Department (DNP, 2013) and reduces diesel vehicle SO_2 emissions factors by 80%. Compared to the BAU scenario, fleet upgrades and diesel reformulation decrease total $PM_{2.5}$ and PM_{10} mobile source emissions by approximately 20%.

The emissions reduction strategy for light-duty passenger vehicles assumes upgrading control technologies and cleaner gasoline fuel. Under this strategy, 15% of the private light-duty vehicle fleet is changed from TIER 0 standards, with no emissions controls, to technology consistent with TIER 2 standards, which set targets for fleet-average emissions of NO_X, non-methane organic gases (NMOG), CO, formaldehyde, and PM, as defined by the U.S. EPA (U.S. EPA, 2016), by 2030. The upgrades reduce PM_{10} and $PM_{2.5}$ emissions factors of over 300,000 gasoline vehicles in Bogotá by close to 90%. The passenger vehicle strategy additionally assumes that cleaner gasoline fuel with sulfur content reduced from 270 ppm to 10 ppm is introduced in the city, in line with the National Planning Department's 2030 goals (DNP, 2013). The fuel change decreases the SO_2 emission factors for gasoline cars, light-duty trucks, and motorcycles by over 95%. Compared to the BAU scenario, the combined impact of passenger vehicle upgrades and gasoline reformulation reduces annual mobile source emissions by 11%, 18%, 48%, and 93% for $PM_{2.5}$, NO_X, volatile organic carbons (VOCs), and SO_2, respectively, in 2030.

2.3 Results and Discussion

2.3.1 Major sources of $PM_{2.5}$ pollution in Bogotá

Vehicle emissions, unpaved road RPM, and industrial emissions are the largest local sources of $PM_{2.5}$ pollution in Bogotá, summarized with city-average contributions in Figure 2.2. Over 70% of $PM_{2.5}$ pollution is attributed to nine emission sectors: on-road combustion, industrial point sources, and several sources of RPM, including construction, quarrying, main paved roads, secondary paved roads, unpaved industrial roads, and unpaved public roads. Sources outside of Bogotá also contribute significantly to local pollution, with aggregated sources beyond the modeling domain (labeled boundary) dominating among all sources in JFM (33%) and contributing less in OND (18%). RPM from unpaved roads, including industrial and public unpaved roads, is the largest local contributor in both the dry (JFM; 27%) and rainy (OND; 31%) seasons. On-road combustion emissions are the second largest local source, contributing close to 20% of $PM_{2.5}$, followed by industrial point sources. Other sectors emitting RPM combined contribute over 10% of the $PM_{2.5}$ in the city. Disaggregated modeled impacts at urban/suburban and traffic/industrial monitoring sites are shown in Table 2.2. Our finding that unpaved road RPM and on-road emissions contribute most to $PM_{2.5}$ agrees with results of prior receptor modeling studies of PM_{10} in Bogotá, which suggest that road dust and traffic exhaust are the largest sources in the city, with dust sources accounting for between 9% and 51% of PM_{10} depending on season and location (Ramírez et al., 2018; Vargas et al., 2012). A recent measurement campaign also found RPM to be a major source of pollution, contributing 21% of $PM_{2.5}$ (15% from road dust and 6% from construction dust) (Ecopetrol and ULS, 2016).

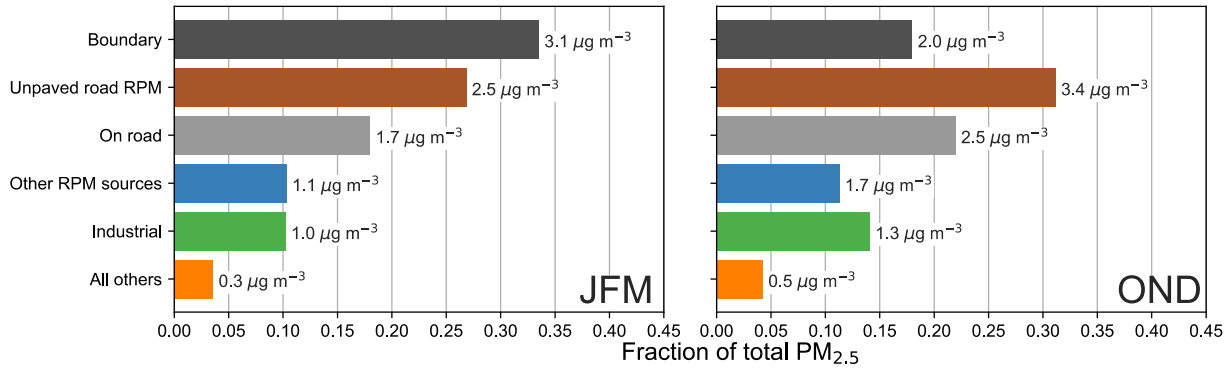

Figure 2.2: City-mean PM2.5 impacts by emissions sector for JFM and OND seasons determined with the source scaling method. Bars indicate fractional contribution relative to total PM2.5, while labels show season-average mass contribution. Unpaved road RPM is the sum of unpaved industrial and unpaved public road RPM. Other RPM sources is the sum of construction, quarrying, main road, and secondary road RPM.

Table 2.2: Modeled $PM_{2.5}$ sector contributions at urban/suburban and traffic/industrial sites

Category	Urban/suburban sites				Traffic/industrial sites			
	JFM		OND		JFM		OND	
	%	μg m^{-3}	%	μg m^{-3}	%	μg m^{-3}	%	μg m^{-3}
Modeled $PM_{2.5}$	-	12.6	-	16.3	-	18.3	-	23.5
Boundary	34.7	4.4	19.4	3.2	19.4	3.5	11.1	2.6
On road	20.2	2.6	24.8	4.0	22.9	4.2	24.5	5.8
Unpaved public road RPM	12.3	1.5	15.5	2.5	15.4	2.8	16.8	3.9
Unpaved industrial road RPM	10.3	1.3	12.2	2.0	11.7	2.1	11.6	2.7
Industrial	9.0	1.1	11.7	1.9	13.5	2.5	17.1	4.0
All others	3.8	0.5	4.4	0.7	4.9	0.9	5.3	1.2
Secondary road RPM	3.5	0.4	4.6	0.7	5.5	1.0	6.6	1.5
Construction RPM	2.4	0.3	3.2	0.5	2.9	0.5	3.0	0.7
Main road RPM	2.2	0.3	2.9	0.5	3.0	0.6	3.5	0.8
Quarrying RPM	1.6	0.2	1.3	0.2	0.9	0.2	0.4	0.1

The spatial distributions of contributions from each emissions sector vary widely, with implications for air quality management. Figure 2.3 maps seasonal contributions to $PM_{2.5}$ by sector. Industrial emissions impacts are localized and affect relatively small areas, contributing most among all sectors to $PM_{2.5}$ at the monitoring site closest to the city center (urban industrial site J in Figure 2.1a). Industrial unpaved road RPM has the most widespread impact, while on-road emissions have the largest impact in the residential north of the city and dominate $PM_{2.5}$ pollution at urban/suburban sites. The highest contribution from any sector is from unpaved public road RPM in the industrial area of the city, due to poor road conditions and heavy traffic. The fractional contributions of on-road, unpaved industrial road RPM, and unpaved public road RPM

mass are similar across urban/suburban and traffic/industrial sites. Modeled contributions at all sites for both seasons are reported in Tables A.1 and A.2. These differences in spatial distribution show that controls aimed at on-road emissions may provide benefits to a higher percent of Bogotá residents due to the larger spatial extent of their impacts, but may less effectively attenuate $PM_{2.5}$ at the locations with the highest concentrations caused by more spatially heterogeneous sectors, such as RPM from unpaved roads and industrial point sources.

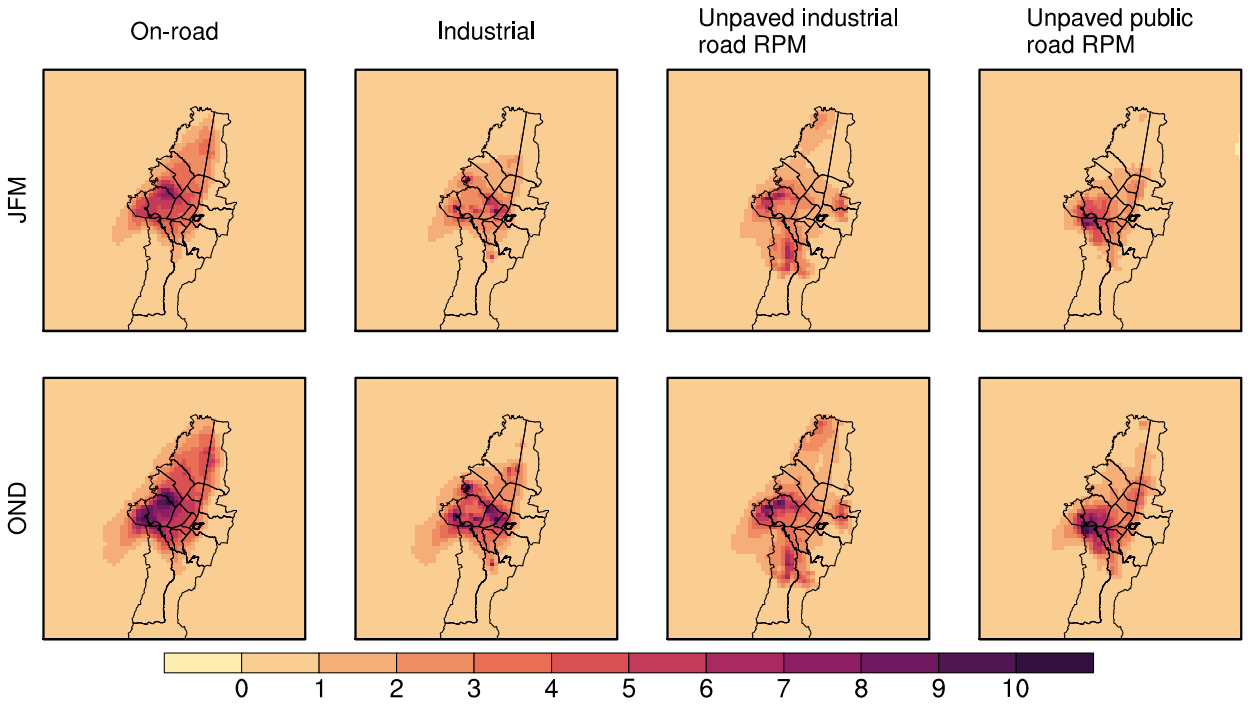

Figure 2.3: Contributions (μg/m³) to Bogotá $PM_{2.5}$ from largest emission sectors during the JFM (top row) and OND (bottom row) seasons. Sectors shown are on-road combustion, industrial point sources, unpaved industrial road RPM, and unpaved public road RPM.

Relative contributions of major emission sectors to city-average $PM_{2.5}$, shown for each season in Figure 2.4, are similar during the dry and rainy seasons despite daily variations with meteorology. Total concentrations are higher during the OND period, compared to JFM, at both urban/suburban (3.7 μg/m³ higher) and traffic/industrial (5.2 μg/m³ higher) sites. The larger relative boundary contribution in JFM is due in part to lower total $PM_{2.5}$ during JFM (9.5 μg/m³) compared to OND (11.3 μg/m³), and an increase in the boundary contribution during JFM (3.2 μg/m³) compared to OND (2.0 μg/m³). This results in an increase across nearly all sectors in OND relative to JFM for local sources. Although lower $PM_{2.5}$ during the rainy season may be expected due to increased wet

deposition, lower average planetary boundary layer (PBL) height and greater atmospheric stability during the rainy season, which favors higher aerosol concentrations, has been observed in the Colombian Andean region where Bogotá is located (Herrera-Mejía and Hoyos, 2019). Figure A.4 shows that PBL height is lower in rainy seasons months, on average, than during dry season months in our simulations. Although sources of RPM from unpaved roads (including public and industrial roads) contribute the most to $PM_{2.5}$ pollution on average, Figure 2.4 shows significant temporal variations in the sectors' impacts. Variations are more pronounced at individual sites, shown in Figure A.3. These daily fluctuations show our approach effectively captures temporal variability in the source breakdown and suggest that addressing $PM_{2.5}$ spikes at specific sites may require plans targeting emission sources that reflect the high variability observed at the sites. The consistency of average source contributions across seasons indicates that mitigation actions targeting the nine sectors identified here would be effective during both rainy and dry seasons.

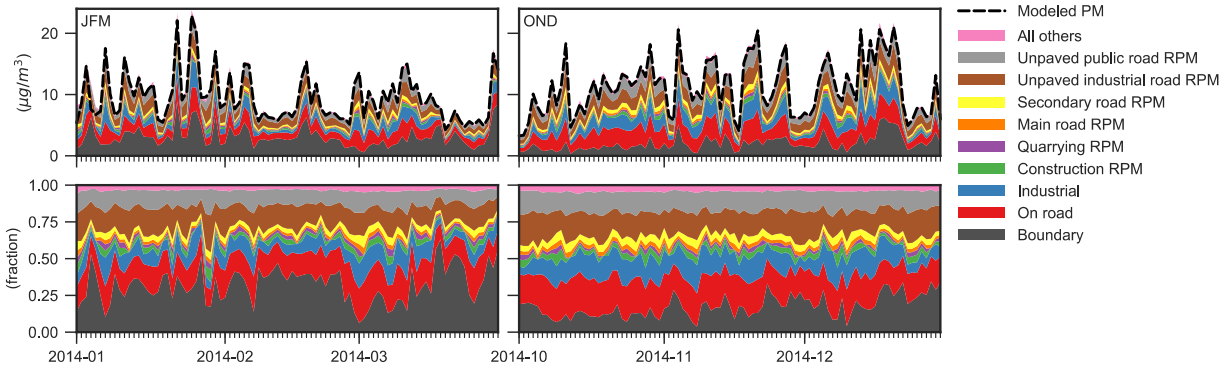

Figure 2.4: Modeled city-average $PM_{2.5}$ and sector contributions in Bogotá for the JFM and OND periods, expressed as $PM_{2.5}$ mass impacts (top row) and fractional contributions (bottom row). The black dashed line represents the total base case modeled $PM_{2.5}$, and not the sum of sector contributions.

Among common source apportionment methods used for regional air quality, source scaling is well suited to inform management strategy (Thunis et al., 2019). A risk of the approach is that the accuracy with which nonlinear processes are captured depends on the dominant chemical regimes and emissions perturbation size. Error is measured by the difference between modeled and rescaled PM, which here is smaller than the contributions from most local emissions sources: 3.4% during JFM and 2.3% during OND (Tables A.1 and A.2). Larger secondary aerosol formation in the JFM period was observed at a background site (Ramírez et al., 2018), indicating the possibility of larger

model rescaling error during JFM. However, a large seasonal difference was not detected in the secondary concentrations modeled in this study. Imperfect scaling is partially associated with secondary inorganic aerosol formation. Scaling emissions down increases sulfate relative to total aerosol mass, leading to relative decreases in NO_3^- and relative increases in SO_4^{2-} and NH_4^+, and ultimately an overestimation of total secondary inorganic mass. The magnitude of this impact is small across the domain. Additionally, nonlinear interactions between meteorology and emissions may add to the $PM_{2.5}$ rescaling error. Despite imperfect rescaling, the source-scaling method captures the relative contributions of emissions sectors to Bogotá's $PM_{2.5}$ pollution with small differences between CMAQ-modeled concentrations and those calculated from source impacts (Figure 2.4). The analysis also demonstrates that the approach may be especially useful in the polluted urban atmospheres of LMICs, where air quality is often heavily influenced by emissions of primary and coarser particulate matter, and the main sources of pollution have not yet been identified.

2.3.2 Future emissions control scenarios

The source apportionment analysis identifies the highest polluting sectors and can guide mitigation actions. To connect the sector impact estimates to potential policy, we analyze four emissions mitigation strategies targeting the largest sources of $PM_{2.5}$. The design of the air pollution reduction strategies, described in Section 2.4, is motivated by the findings discussed in Section 3.1. The large contributions of RPM from unpaved roads highlight an important pollution mitigation opportunity and affirm Bogotá's goal of paving all roads by 2030. Scenarios focused on vehicle and industrial emissions are also examined. Air quality benefits in 2030 are projected under four scenarios: (1) cleaning of paved roads, (2) industrial emissions controls, (3) reduction of cargo fleet emissions, and (4) reduction of light-duty passenger vehicle emissions. Emissions for each scenario simulation are listed in Table 2.1. Figure A.5 shows the distribution of $PM_{2.5}$ emissions projected in 2030 under a BAU and emissions reductions associated with each mitigation scenario. In the 2030 BAU projection, $PM_{2.5}$ increases over central Bogotá by as much as 11 $\mu g/m^3$ and 14 $\mu g/m^3$ in the JFM and OND periods, respectively, relative to the 2014 base case simulation (Figure 2.5). $PM_{2.5}$ decreases are projected along the outskirts of the city, resulting from the assumption that all unpaved roads in Bogotá will be paved by 2030. Despite the considerable emissions reductions realized by paving roads, $PM_{2.5}$

increases driven by industrial activity and traffic reveal the potential impacts of uncontrolled growth in these sectors.

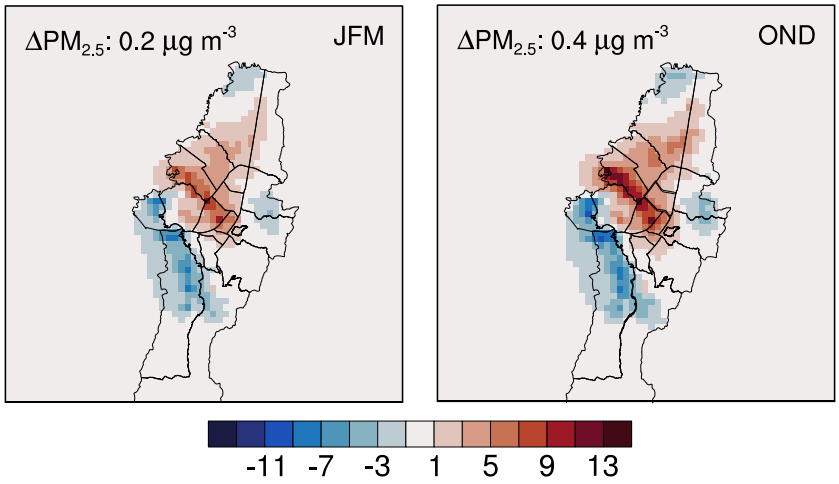

Figure 2.5: $PM_{2.5}$ ($\mu g/m^3$) changes under 2030 BAU scenario relative to 2014 base case simulation, during JFM and OND periods. City-average changes ($\Delta PM_{2.5}$) are also shown on each map.

Emissions control strategies can mitigate projected increases in air pollution. $PM_{2.5}$ reductions achieved by each strategy in 2030, relative to BAU, are shown in Figure 2.6. The strategy focused on light-duty vehicle emissions is projected to have the largest effect on reducing $PM_{2.5}$ across the city. The ubiquity of passenger vehicles leads to widespread impacts across Bogotá in scenario 4, decreasing city-average $PM_{2.5}$ by close to 1 $\mu g/m^3$ during the periods modeled, equivalent to 6% and 9% of BAU $PM_{2.5}$ concentrations in the JFM and OND seasons. Reductions as large as 2.4 $\mu g/m^3$ (JFM) and 4.1 $\mu g/m^3$ (OND) are projected at locations in the southwest of the city. However, the largest decreases relative to total $PM_{2.5}$, up to 15%, are simulated in Bogotá's northwest residential neighborhoods where modeled $PM_{2.5}$ is lower and dominated by vehicle emissions. Urban/suburban monitoring sites (A and F in Figure 2.1), near residential areas and a major highway, benefit most from the light-duty vehicle emissions control strategy with regard to fractional $PM_{2.5}$ decreases.

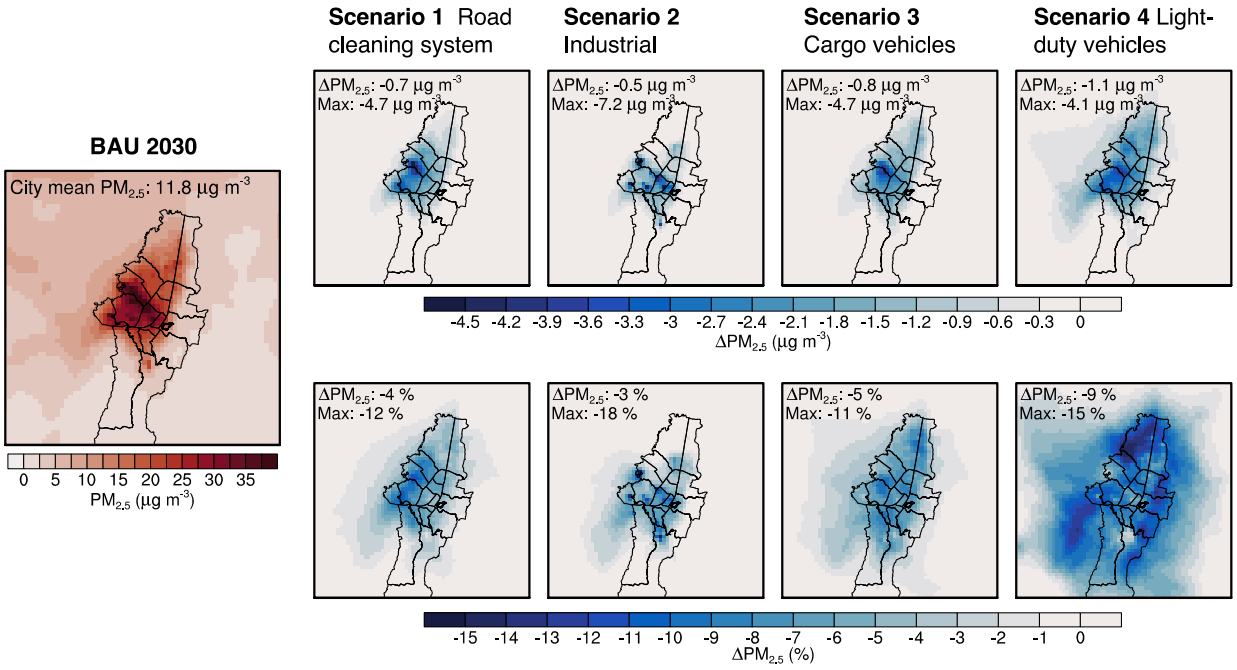

Figure 2.6: 2030 BAU average $PM_{2.5}$ concentrations during OND season (left), with absolute (top) and relative (bottom) changes for each emissions control scenario. City-average ($\Delta PM_{2.5}$) and maximum changes are listed for each mitigation strategy. Analogous results for the JFM season are shown in Figure A.6.

The control strategy targeting heavy-duty cargo vehicle emissions leads to $PM_{2.5}$ reductions with a spatial distribution similar to that of scenario 4, but smaller impacts. Under this scenario, $PM_{2.5}$ decreases by as much as 3.9 µg/m^3 (JFM) and 4.7 µg/m^3 (OND) at specific locations, and by 0.7 µg/m^3 on average across Bogotá, relative to the BAU projection. The reductions are highest in the southwest urban industrial area of the city. Monitoring sites in the southwest of the city are projected to experience the greatest decreases relative to total $PM_{2.5}$ derived from reduced cargo fleet emissions. The largest impact, a 9% drop across both seasons modeled, is projected at the Tunal urban/suburban site (J in Figure 2.1).

Despite differences in emissions reductions (Figure A.5), $PM_{2.5}$ decreases under the control strategies focused on cargo vehicle emissions and paved road RPM are similar. Implementation of a paved road cleaning system (scenario 1), results in city-average decreases of 0.4 µg/m^3 (JFM) and 0.7 µg/m^3 (OND), relative to BAU concentrations. Projected pollution reductions in southwest Bogotá are as large as 4.7 µg/m^3 or 12% of total $PM_{2.5}$, with the Fontibon urban industrial monitoring site (E in Figure 2.1) benefiting most from road cleaning. In addition to the effects on $PM_{2.5}$, the road cleaning system has

a large impact on coarse PM emissions and results in significant decreases in PM_{10} concentrations.

Industrial emissions controls (scenario 2) affect the smallest area among the strategies considered but lead to the largest localized decreases in $PM_{2.5}$ pollution. The reductions projected under the industrial emissions control strategy are as large as 4.9 $\mu g/m^3$ (JFM) and 7.2 $\mu g/m^3$ (OND), relative to BAU concentrations, near targeted facilities. Although air quality benefits are less widespread under this scenario, reductions close to 20% of total $PM_{2.5}$ are attained at specific locations. As expected, urban industrial monitoring sites, such as Puente Aranda (site J in Figure 2.1), are projected to experience the largest impacts from these emission controls.

The scenario analysis shows the impacts diverse emissions control strategies can have on Bogotá's air quality. The results of the source apportionment analysis, revealing large contributions to $PM_{2.5}$ from unpaved roads and on-road sources, support the city's plans to pave all roads by 2030 and maintain the public transportation fleet at its current size. However, in the absence of additional pollution mitigation actions, $PM_{2.5}$ concentrations are projected to rise over much of the city and increase by over 10 $\mu g/m^3$ at some locations over the next 10 years. The emissions control strategies explored here can largely mitigate this anticipated deterioration in air quality, constraining the highest localized $PM_{2.5}$ increments under the BAU projection. Further, under all mitigation scenarios city-average $PM_{2.5}$ concentration is lower than that of the 2014 base case simulation. Although avoided emissions have a larger impact on $PM_{2.5}$ during the OND period, the pollution mitigation actions considered are effective in both the wet and dry seasons.

Air quality benefits can be realized by implementing the strategies individually or in combination. The scenarios analysis suggests that the effects of different measures on $PM_{2.5}$ levels may be comparable. For example, implementation of a road cleaning system and reduction of cargo fleet emissions lead to similar air quality benefits due to their common link to roadways. Still, not all neighborhoods in Bogotá benefit equally from each pollution mitigation strategy. Combined approaches may be needed to prevent $PM_{2.5}$ increases throughout the city. Projecting spatially-resolved benefits of mitigation strategies can help identify those that most impact densely populated areas. Further, different air quality goals may require distinct strategies. Emissions controls targeting the industrial sector may be required to address peak $PM_{2.5}$ concentrations and prevent the

largest exceedances of air quality standards. Reduced emissions from light-duty passenger vehicles may impact the largest population and thus lead to larger public health benefits. While estimates of the air quality effects of possible emissions control strategies can inform policy, further projecting the health and economic impacts of these approaches can provide additional insights.

2.4 Conclusions

This study is the first to use chemical-transport modeling for source apportionment and emissions scenario analysis in Bogotá, Colombia. We apply a comprehensive open-source air quality model to identify the sectors with the largest contributions to $PM_{2.5}$ in the city, where air pollution poses a significant threat to public health, and assess the potential benefits of different mitigation strategies. We find that the largest sources of $PM_{2.5}$ are RPM, industrial emissions, and on-road combustion emissions, despite recent measures to control traffic congestion. The study demonstrates the usefulness of sensitivity-based source apportionment in providing policy-relevant findings by connecting results to actionable air pollution mitigation strategies. Our projections suggest that $PM_{2.5}$ will increase across a large fraction of the city by 2030, unless additional actions are undertaken to control emissions. In the scenario analysis conducted, reducing RPM from unpaved roads is anticipated to decrease $PM_{2.5}$ over sections of Bogotá, while growth in different emissions sectors is anticipated to deteriorate air quality in other areas. The largest localized $PM_{2.5}$ decreases would result from implementing additional controls on industrial facilities, while the most widespread $PM_{2.5}$ reductions would be produced by upgrading the city's light-duty passenger vehicle fleet.

Aspects of the study design and uncertainties in the model results limit the applicability of these results and present additional research opportunities. The contribution of regional biomass burning to $PM_{2.5}$ was not assessed and may have significant impacts in Bogotá (Ballesteros-González et al., 2020). The source apportionment results generally reflect the source strengths in the emissions inventory used. The scenario analysis shows that modeled $PM_{2.5}$ responds to changes in the inventory and is sensitive in particular to resuspended dust emissions, highlighting the importance of accurate local emissions estimates. Bogotá's complex topography introduces uncertainties to the meteorological inputs driving the air quality simulations.

The source-scaling method applied to weigh the influence of emissions sectors, based on first-order sensitivities, may not capture nonlinearities associated with the formation of secondary $PM_{2.5}$. While the study focuses on $PM_{2.5}$ pollution, differences in the chemical composition of PM from various emission sources are not considered. Pollutants other than $PM_{2.5}$, strategies optimized for multi-pollutant control, and the potential for emissions controls to alter prevailing atmospheric chemical regimes were not included in the scenario analysis. Future research can improve this modeling framework. Speciated $PM_{2.5}$ observations would allow model performance and source apportionment results to be better evaluated. Further refinement of emissions inventories, including enhanced estimates for additional sources (e.g., cooking emissions), can support modeling efforts. Simulations spanning multiple years would better capture meteorological variability. Despite these limitations, the modeling framework is able to adequately reproduce observed concentrations, weigh sector contributions based on source scaling with relatively small rescaling errors, and project the air quality benefits of potential pollution mitigation approaches.

The findings of this study can clarify the roles of emissions sources and help prioritize air pollution control strategies in the city. The paired source apportionment and scenario analysis results provide important information to stakeholders interested in attaining cleaner air in Bogotá or similar megacities. The analysis serves as a framework for other large Latin American or LMIC cities with comparable air quality issues, including resuspended dust pollution and high transportation emissions. This study describes a pathway that can be followed to inform air pollution mitigation efforts with chemical-transport modeling at locations where air quality modeling platforms are new. The combined use of local emissions inventories, source-scaling sensitivity modeling, and scenario analysis offers an approach to address $PM_{2.5}$ in heavily polluted areas where primary PM is dominant and the largest contributors to $PM_{2.5}$ are still unclear. Developing modeling frameworks that can inform air quality decision-making in LMICs is an important research need. Cities in these regions experience the greatest detrimental air pollution impacts and offer some of the best opportunities to benefit from such tools.

2.5 Acknowledgments

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Chapter 3: Inferring and evaluating satellite-based constraints on NOx emissions estimates in air quality simulations

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Abstract

Satellite observations of tropospheric NO_2 columns can provide top-down observational constraints on emissions estimates of nitrogen oxides (NO_x). Mass-balance based methods are often applied for this purpose, but do not isolate near-surface emissions from those aloft, such as lightning emissions. Here, we introduce an inverse modeling framework that couples satellite chemical data assimilation to a chemical transport model. In the framework, satellite-constrained emissions totals are inferred using model simulations with and without data assimilation in the iterative finite-difference mass-balance method. The approach improves the finite-difference mass-balance inversion by isolating the near-surface emissions increment. We apply the framework to separately estimate lightning and anthropogenic NO_x emissions over the Northern Hemisphere for 2019. Using overlapping observations from the Ozone Monitoring Instrument (OMI) and the Tropospheric Monitoring Instrument (TROPOMI), we compare separate NO_x emissions inferences from these satellite instruments, as well as the impacts of emissions changes on modeled NO_2 and O_3. OMI inferences of anthropogenic emissions consistently lead to larger emissions than TROPOMI inferences, attributed to a low bias in TROPOMI NO_2 retrievals. Updated lightning NO_x emissions from either satellite improve the chemical transport model's low tropospheric O_3 bias. The combined lighting and anthropogenic emissions updates improve the model's ability to reproduce measured ozone by adjusting natural, long-range, and local pollution contributions. Thus, the framework informs and supports the design of domestic and international control strategies.

3.1 Introduction

Tropospheric nitrogen oxides (NO_x), nitric oxide (NO) and nitrogen dioxide (NO_2), harm human health (Anenberg et al., 2018; Murray et al., 2020) and play a key role in the formation of important secondary atmospheric pollutants, such as O_3 (Jacob, 2000). NO_x is emitted to the troposphere primarily by anthropogenic combustion processes, but natural sources, including soil, lightning, and wildfires, also contribute to the atmospheric NO_x budget (Jacob, 1999). Accurate NO_x emissions are a critical component of local- to global-scale atmospheric chemistry simulations. On hemispheric scales, realistically representing the formation and intercontinental transport of O_3 with models requires adequate global emissions inventories (Itahashi et al., 2020; Zhang et al., 2016, 2008; Verstraeten et al., 2015; Mathur et al., 2017). In regional air quality simulations, which commonly rely on hemispheric or global models for chemical boundary conditions, the relative contribution of long-range pollutant transport to ground-level O_3 concentrations has grown in many areas as O_3 precursor emissions have decreased in the U.S. and other high income countries (HICs) (McDuffie et al., 2020; Jaffe et al., 2018; Simon et al., 2015). As a result, air quality management policies, often informed by regional modeling, are strengthened by accurate and up-to-date global NO_x emissions inventories. However, compilation of bottom-up regional and global emissions inventories, developed from source- and location-specific emissions factors and activity data, is time- and labor-intensive, and can be hindered by limited data. As a result, bottom-up inventories lag emissions changes and are often incomplete. Uncertainties in bottom-up emissions estimates are particularly large for lower-middle income countries (McDuffie et al., 2020; Elguindi et al., 2020) and remain significant for HICs (Day et al., 2019).

Satellite observations of NO_2 can bridge temporal gaps in emissions estimates (Tong et al., 2015, 2016) and constrain uncertainty in emissions inventories through inverse modeling (Lamsal et al., 2011; Goldberg et al., 2021; de Foy and Schauer, 2022). Several methods have been applied to develop top-down emissions estimates using satellite observations and atmospheric models, each carrying advantages and limitations (Elguindi et al., 2020). Adjoint-based methods can provide precise emissions updates, but require significant computational resources (e.g. Qu et al., 2017, 2019; Müller and Stavrakou, 2005; Kurokawa et al., 2009; Cooper et al., 2017; Zhang et al., 2019b; Wang et al., 2020b). Similarly, Kalman filtering and related approaches have been used but are computationally-intensive (e.g. Napelenok et al., 2008; Ding et al., 2020, 2015; Mijling and

van der A, 2012; Miyazaki and Eskes, 2013; Miyazaki et al., 2017, 2012a, b; Sekiya et al., 2021). Mass balance inversion approaches, which scale model emissions by directly comparing model estimates and satellite observations, were introduced by Martin et al. (2003), updated by Lamsal et al. (2011), and have been widely used in research and forecasting (e.g. Boersma et al., 2015; Itahashi et al., 2019; Li et al., 2018; Visser et al., 2019; Zhu et al., 2021; Cooper et al., 2017). Although lower computational costs allow the finite-difference mass-balance (FDMB) approach (Lamsal et al., 2011) to readily update emissions, the method is subject to an emissions smearing effect (Cooper et al., 2017), which can cause emissions updates to be spatially misallocated. Since FDMB uses satellite observations directly, near-surface NO_2 bias cannot be isolated from biases in the middle and upper troposphere, which obscures the surface emissions inference. Further, applications often rely on a single inversion from a single satellite, although available satellite products have been shown to have significant biases. For example, early versions of the Tropospheric Monitoring Instrument (TROPOMI) NO_2 product showed a low bias in urban areas when compared against ground and airborne spectrometer measurements (Judd et al., 2020; Verhoelst et al., 2021) and the Ozone Monitoring Instrument (OMI) NO_2 product has been reported to differ with measurements by ±20% (Lamsal et al., 2014). The impact of biases in satellite-based NO_2 data on mass-balance inversions has not been fully explored despite the wide used of the method to scale NO_x emissions. Minimizing bias in anthropogenic emissions inferences and understanding the potential for them to propagate to emissions updates are needed to improve mass-balance-based inversions.

Here, we introduce a modeling framework that couples satellite chemical data assimilation to the Community Multiscale Air Quality model (CMAQ) and applies an iterative FDMB inversion to estimate NO_x emissions in the Northern Hemisphere. The framework provides observational constraints to improve emissions estimates in areas where emissions are highly uncertain, at a lower computational cost relative to adjoint- and Kalman-filter-based approaches. We apply the framework in an iterative assimilation to infer 2019 NO_x emissions, the first complete year in which OMI and TROPOMI records overlap. In contrast to traditional FDMB, which directly compares modeled and observed columns, our framework improves the FDMB method by first assimilating satellite-retrieved NO_2, and then performing the inversion by comparing model simulations with and without assimilation. In the assimilation step, updates to model concentrations are vertically allocated to model layers. As a result, assimilating the

observed column allows the inversion framework to use only the near-surface portion of the model column in the FDMB inversion, minimizing influences from the upper troposphere and extending the framework proposed by Lamsal et al. (2011). In addition, our analysis compares independent inversions which separately use OMI or TROPOMI NO_2 data. We show that the inverse emissions produced by this framework influence representation of intercontinental O_3 transport to the U.S., offering an opportunity to improve chemical boundary conditions in policy-relevant regional-scale air quality simulations.

3.2 Methods

We develop a framework to update NO_x emissions estimates using the CMAQ chemical transport air quality model (Byun and Schere, 2006), 3D-variational (3DVAR) chemical data assimilation (Sandu and Chai, 2011), and space-based NO_2 observations. We apply the framework to estimate 2019 lightning and anthropogenic NO_x emissions, and compare ground- and space-based NO_2 observations to model simulations using the prior emission (inventory before the framework is applied) and posterior emissions (inventory after the framework is applied) to assess the impact of the updates. Figure 3.1 provides an overview of the framework, in which lightning NO_x (LNO_x) emissions and anthropogenic NO_x (ANO_x) emissions are updated separately.

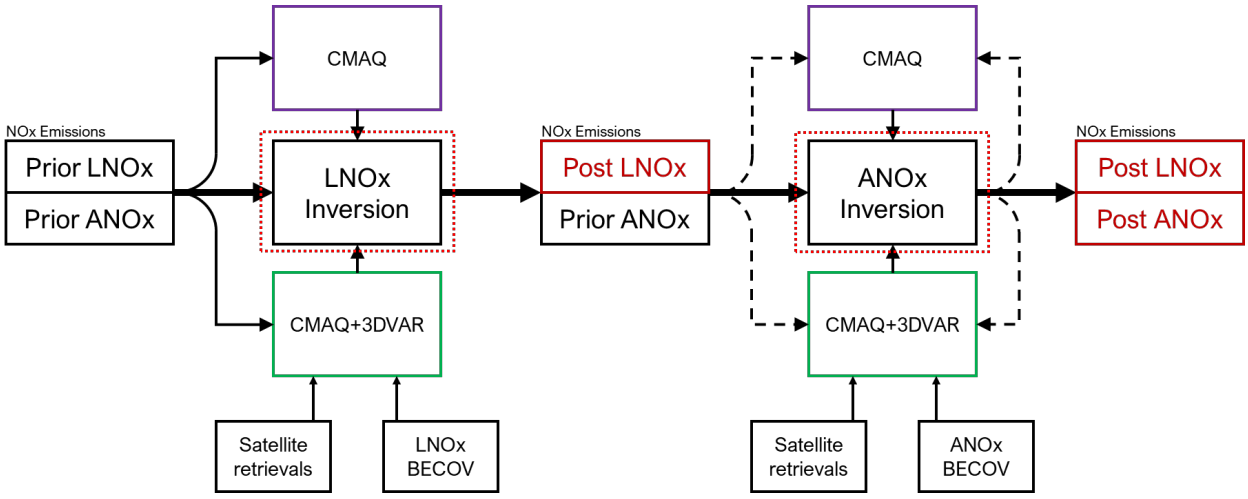

Figure 3.1: NO_x emissions inversion framework. Lightning NOx (LNO_x) emissions are updated in the first step. Then, anthropogenic NO_x (ANO_x) emissions are updated iteratively. CMAQ boxes represent air quality simulations without chemical data assimilation, and CMAQ+3DVAR boxes represents air quality model simulations with chemical data assimilation. Satellite NO_2 retrievals and a background error covariance (BECOV) are inputs to the chemical data assimilation, described in Sect. 3.2.3. Red dotted lines around the inversion boxes correspond to the red dotted lines in Fig. 3.3, which details the inversion algorithm. Dashed black emissions input lines around the ANO_x inversion represent the iterative process. Iteration and convergence criteria are described in Sect. 3.2.6.

3.2.1 Satellite data

We use NO_2 tropospheric column observations from the National Aeronautics and Space Administration's (NASA's) OMI and the Royal Netherlands Meteorological Institute's (KNMI's) TROPOMI instruments in the inversion framework. TROPOMI was launched in October 2017 and provides 7.2×3.6 km^2 resolution NO_2 retrievals, upgraded to 5.6×3.6 km^2 resolution in August 2019 (Van Geffen et al., 2020; Veefkind et al., 2012). TROPOMI's sun-synchronous polar orbit crosses the equator at approximately 1:30 pm local time, allowing the instrument to achieve global coverage in one day. We assimilate the Level-2 tropospheric slant column retrieved from NASA's Earth Science Data Systems Program (https://earthdata.nasa.gov/). The data product is described in the Algorithm and Theoretical Basis Document (ATBD) for TROPOMI NO_2 (van Geffen et al., 2019). We only consider TROPOMI observations with a quality flag greater than 0.5 and cloud fraction lower than 30% in the assimilation, following data product recommendations (van Geffen et al., 2019). We use the latest publicly available versions of the TROPOMI retrieval for 2019 (versions 1.2.2 to 1.3.2) at the time of the analysis. Version 1.3 introduced

updates to cloud processing that decrease noisy hotspots and broadened the range of acceptable air mass factors (Eskes et al., 2021). Information about the updates applied in each version and the dates on which updates were applied is given in Eskes et al. (2021). A research version with an updated retrieval applied to 2019 observations has been developed (van Geffen et al., 2021), but was not yet standard and was not available at the time of this analysis. We discuss the impact of these latest updates in Sect. 3.3.3.

OMI, onboard the Aura satellite launched in 2004, provides tropospheric NO_2 vertical and slant column retrievals with a resolution of 13 × 24 km^2 near nadir in a sun-synchronous polar orbit with a local equator crossing time of 1:45 pm. Global coverage is achieved in two days. We use the NASA Goddard Space Flight Center (GSFC) Level-2 NO_2 product (Krotkov et al., 2019). OMI was impacted by a row anomaly beginning in 2008, reducing the number of usable pixels in the OMI retrieval (Boersma et al., 2018). We include only pixels with cloud fraction lower than 30% and a summary quality flag of 0. Detailed information about the NO_2 data product is included in the OMI ATBD (Chance, 2002) and in (Krotkov et al., 2019).

A low bias has been noted in the versions of TROPOMI NO_2 used for this study (Judd et al., 2020; Verhoelst et al., 2021). Although TROPOMI NO_2 in 2019 has been reprocessed with retrieval version 2.3.1, resulting in an improvement of the bias (Eskes et al., 2021), these reprocessed datasets were not yet available at the time this analysis was conducted. Figure 3.2 compares TROPOMI and OMI tropospheric vertical column density (VCD) for 2019, regridded to the CMAQ grid used. For the VCDs shown in the figure, we remove the effect of the assumed vertical profile of NO_2 from the original satellite product by recalculating the VCDs with the NO_2 vertical profile simulated by CMAQ. In the results, we discuss the low bias in TROPOMI data and explore its impact on emissions inversions.

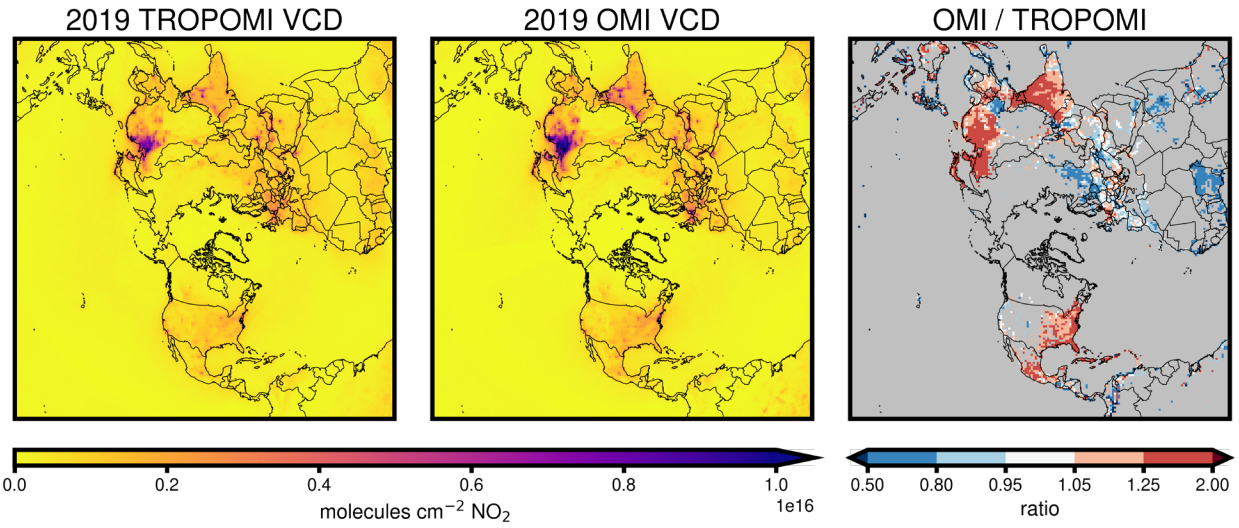

Figure 3.2: 2019 annual average TROPOMI and OMI vertical NO_2 vertical column densities, with CMAQ NO_2 profiles applied, and the ratio between them. Column densities ratios are only shown for the grid cells where NOx emissions updates are applied in the emissions inversion.

3.2.2 Hemispheric air quality modeling

Model simulations in the inversion framework were completed for January–December 2019 using CMAQ v5.3.2 (Appel et al., 2021; U.S. EPA, 2020). CMAQ has been used to simulated air quality over the Northern Hemisphere and shown to adequately capture chemical composition against observations (Mathur et al., 2017). Model inputs and satellite observations are summarized in Table 3.1. Simulations, designed to capture continental-scale pollutant transport, cover the Northern Hemisphere with 108 km horizontal grid spacing and a 44-layer vertical structure reaching 50 hPa (Mathur et al., 2017). The simulations use version CB6r3 of the Carbon Bond 6 chemical mechanism (Luecken et al., 2019), the AERO7 aerosol module (Xu et al., 2018), and updated halogen chemistry (Kang et al., 2021). Anthropogenic emissions are modeled using representative day-of-week emissions that change month-to-month. Representative-day emissions are created by averaging data from the prior emissions inventory on a day-of-week basis by month. For each day of the week and each month, there is a unique hourly emissions file that is used for every matching day of the week in that month. As a result, diurnal and weekly patterns are captured in the emissions, while daily variations that are specific to the prior emissions inventory year are averaged. The prior emissions inventory relies on the best available emissions data at the time of the study. Anthropogenic emissions for North America are from the U.S. Environmental Protection Agency's (EPA) 2017

National Emissions Inventory (NEI) modeling platform (Adams, 2020). Emissions in China are for the year 2015 (Zhao et al., 2018) and emissions for the rest of the hemisphere are based on the Hemispheric Transport of Air Pollution (HTAP) version 2, projected from their original 2010 date to 2014 with scaling factors from the Community Emissions Data System (CEDS). To initialize the 2019 prior and posterior simulations and reduce the impact of chemical initial conditions on the results, we use a 1-year spin-up period not considered for the analyses. CMAQ model runs are driven by meteorology from a retrospective hemispheric simulation using the Weather Research and Forecasting (WRF) model (Skamarock et al., 2008) version 4.1.1 configured following Mathur et al. (2017) and (Xing et al., 2015).

Table 3.1: Prior emissions and model inputs

Data	Year	Source
Prior emissions (North America)	**2017** EPA platform (v7.1)	(Adams, 2020)
Prior emissions (China)	**2015** Tsinghua University	(Zhao et al., 2018)
Prior emissions (Rest of hemisphere)	HTAPv2 (2010) projected to **2014** using CEDS scaling factors	(Janssens-Maenhout et al., 2015; Hoesly et al., 2018)
Prior emissions (LNO_x)	**2017** GEIA*	(Price et al., 1997)
Biomass burning emissions	**2019** FINN*	(Wiedinmyer et al., 2011)
Soil NO_x emissions	**2018** CAMS* v2.1 with canopy reduction factor	(Granier et al., 2019)
Biogenic emissions	**2018** MEGAN*	(Guenther et al., 2006)
Meteorology	**2019** WRF v4.1.1	(Powers et al., 2017)
Satellite observation year	**2019** NO_2 retrievals from OMI and TROPOMI	

* GEIA = Global Emissions Initiative; FINN = Fire Inventory from NCAR; CAMS = Community Atmosphere Modeling System; MEGAN = Model of Emissions of Gases and Aerosols from Nature

3.2.3 Chemical data assimilation in CMAQ

We adjust modeled NO_2 concentrations using satellite observations by coupling the CMAQ model to a data assimilation model, the National Centers for Environmental Prediction (NCEP) Gridpoint Statistical Interpolation (GSI) program version 3.3 (Shao et al., 2016). GSI performs three-dimensional variational (3DVAR) data assimilation by minimizing the cost function, *J*:

$$J = \frac{1}{2}[x^T \mathbf{B}^{-1} x + (H(x) - y)^T \mathbf{R}^{-1} (H(x) - y)] \,, \quad (1)$$

where y is the observation innovation $y = y_o - H(x_b)$, x is the analysis increment $x = x_a - x_b$, $x_a$is the analysis field (NO_2 concentration after application of chemical data assimilation), x_b is the model background (the simulated NO_2 concentration before application of chemical data assimilation), y_o is the satellite observations, $\mathbf{B}$ is the background error covariance matrix, $\mathbf{R}$ is the observation error matrix, and H is the observation operator. To compute the difference between the model column (x_b) and the satellite column (y_o), the observation operator H is applied, which transforms the model background to the form of the satellite observations. For TROPOMI data, the averaging kernel is first converted to scattering weights as

$$w(z) = \mathbf{A}(z) \times \mathbf{M}_{total} \,, \quad (2)$$

where $\mathbf{A}(z)$ is the vertically-resolved TROPOMI averaging kernel for level z, $\mathbf{M}_{total}$ is the air mass factor provided with the satellite data, and $w(z)$ are the vertically-resolved scattering weights. Scattering weights accompany the OMI NO_2 data product, so this step is not needed to assimilate OMI data. Scattering weights are then applied to compute the model slant column as

$$\Omega_s^m = \sum_z \Omega_v^m(z) w(z), \;\; z \le z_{tropopause}, \quad (3)$$

where $\Omega_v^m(z)$ is the model partial vertical column in the troposphere, interpolated to the satellite grid, and Ω_s^m is the model tropospheric slant column density (SCD). The difference between the modeled and observed slant columns, or the observation innovation y in Equation 1, is estimated as

$$\Omega_s' = \Omega_v^o \mathbf{M}_{\text{trop}} - \Omega_s^m, \quad (4)$$

where Ω_s' is the analysis increment, Ω_v^o is the satellite tropospheric VCD, and $\mathbf{M}_{\text{trop}}$ is the tropospheric air mass factor, distributed with the satellite data. We eliminate the influence of the *a priori* satellite vertical profile by computing the analysis increment with

the model and observed SCD, which, unlike the VCD, does not rely on the *a priori* vertical NO_2 profile assumed by the satellite.

We compute **B** using the Generalized background error covariance matrix model (GENBE v2.0) (Descombes et al., 2015), which models background errors by comparing a free-running simulation and a simulation with either lightning or anthropogenic NO_x emissions perturbed. We use GENBE with the prior simulation and a simulation with a uniform -15% perturbation to LNO_x to create 3-dimensional background errors in the upper troposphere for the LNO_x assimilation. After updating LNO_x emissions (as described in Sect. 3.2.5), we create 3-dimensional background errors in the boundary layer for the anthropogenic NO_x assimilation by using GENBE with the LNO_x posterior simulation and a simulation with a -15% perturbation to surface anthropogenic NO_x emissions. Observation error **R** is provided with the satellite data.

Online coupling between GSI and CMAQ was developed in this study to perform the assimilation. At each model timestep in which a satellite observation is available, the CMAQ model simulation is paused and 3DVAR assimilation is performed. The CMAQ model state at that time step is used as x_b. After assimilation using 3DVAR within GSI, CMAQ returns to a free-running mode and the new model state, x, is updated to more closely match the satellite observation. The difference in the monthly average NO_2 VCDs from the assimilation and no-assimilation runs is used in the inversion as $\Delta\Omega$.

3.2.4 Finite difference mass balance inversion

In the inversion framework developed, we iterate the approach of Lamsal et al. (2011). The FDMB process as applied here is summarized in Fig. 3.3. In the past, this approach has been used by directly comparing model and satellite columns (Itahashi et al., 2019; Cooper et al., 2017; Lamsal et al., 2011). We modify the approach by first updating model concentrations with assimilation of satellite observations, and then updating the emissions using the difference between the modeled VCD with and without assimilated satellite information. All updates are performed on a monthly average basis.

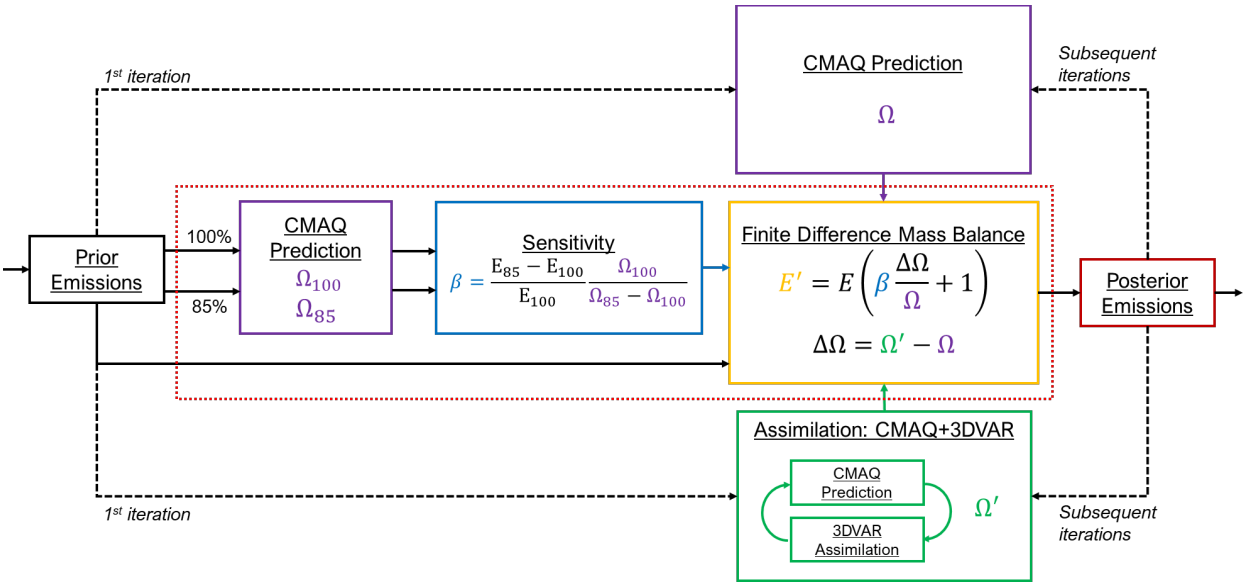

Figure 3.3: FDMB inversion. The red dashed line corresponds to the red dashed lines in Fig. 3.1, and the processes inside show additional details of the FDMB inversion. In this framework, the prior emissions (black box on the far left) are input to the CMAQ model. CMAQ simulations are performed with unperturbed prior emissions (100% arrow and $\boldsymbol{E_{100}}$) and prior emissions with a -15% perturbation (85% arrow and $\boldsymbol{E_{85}}$). The resulting modeled VCDs are $\boldsymbol{\Omega_{100}}$ and $\boldsymbol{\Omega_{85}}$, respectively. These VCDs are used to compute the sensitivity, $\boldsymbol{\beta}$ (blue box). New emissions totals are calculated with FDMB (yellow box), using $\boldsymbol{\beta}$, NO_2 VCD from a CMAQ simulation without assimilation ($\boldsymbol{\Omega}$), and NO_2 VCD from a CMAQ simulation with assimilation ($\boldsymbol{\Omega'}$). When iteration is used, the posterior emissions from the previous iteration are used as input to the CMAQ model to simulate new VCDs, $\boldsymbol{\Omega}$ and $\boldsymbol{\Omega'}$.

In FDMB, following Lamsal et al. (2011), emissions changes are inferred through the relationship

$$\frac{\Delta E}{E} = \beta \frac{\Delta \Omega}{\Omega} \tag{5}$$

where ΔE is the inferred NO_x emissions change, E is the NO_x emissions prior, Ω is the model simulated NO_2 VCD without chemical data assimilation, and $\Delta\Omega = \Omega_{assim} - \Omega$, the monthly average difference between the model simulated tropospheric NO_2 VCD with (Ω_{assim}) and without (Ω) chemical data assimilation. β is a unitless scaling parameter, the Jacobian, that linearly relates NO_2 VCD changes to NO_x emissions changes. β is calculated through finite differencing as

$$\beta = \frac{E' - E}{E} \frac{\Omega_E}{\Omega_{E'} - \Omega_E} \tag{6}$$

where E' is perturbed NOx emissions, Ω_E is the tropospheric NO_2 VCD simulated with model emissions E, and $\Omega_{E'}$ is the tropospheric NO_2 VCD simulated with model emissions E'. To estimate β, we use the same -15% perturbation used to create background errors **B** in the boundary layer. (Cooper et al., 2017) found that using perturbations ranging from 5% to 20% to calculate β changed posterior emissions estimates by less than 2% globally.

3.2.5 Inversion modeling NO_x emissions

In our framework, LNO_x emissions are updated first, separately from anthropogenic emissions. Due to the satellite instruments' sensitivity to NO_2 in the upper atmosphere (e.g. Eskes and Boersma, 2003), small model biases there can influence the total column comparison and adversely impact the anthropogenic emissions adjustment. By updating LNO_x emissions, we aim to decrease this bias and its impact on the ANO_x inversion. We compute the scaling parameter for lightning emissions, β_{LNOx}, using the -15% LNO_x perturbation simulations applied to create background errors for the upper troposphere. We then assimilate satellite NO_2 observations using the background errors for the upper troposphere and apply β_{LNOx} in a single inversion iteration using the full tropospheric VCD to compute spatially-varying LNO_x adjustment factors. Updates to LNO_x are calculated using monthly averages.

After LNO_x emissions are updated, ANO_x emissions are updated by iteratively applying a FDMB inversion independently for each month in 2019. Iterating the FDMB has been shown to improve emissions estimates compared to a single FDMB application (Cooper et al., 2017). In the FDMB iteration, each update to the emissions serves as the prior emissions for the subsequent iteration (represented as black dashed lines in Fig. 3.1). The number of iterations is determined based on the synthetic observation experiment described in Sect. 3.2.6. β is held constant during all ANO_x inversion iterations, and not recalculated each time, to prevent instability in β as changes in the column become smaller with subsequent iterations. In the ANO_x emissions inversion, we only consider grid cells in which local anthropogenic NO_x emissions likely significantly contribute to the satellite-observed NO_2 column by only including grid cells in which anthropogenic NO_x emissions comprise at least 50% of total NO_x emissions following Lamsal et al. (2011), population density is greater than 15,000 people km^{-2} (CIESIN, 2018),

modeled cloud cover is less than 30%, and the local time is 1:00 or 2:00 pm (OMI and TROPOMI overpass times). Only emissions in grid cells meeting these criteria are adjusted. Table B.1 describes each simulation performed for the LNO_x and ANO_x inversions.

The FDMB method assumes emissions impacts are local (i.e., emissions in one grid cell do not affect VCD amounts in neighboring grid cells). This assumption is most valid when NO_x lifetime to chemical losses is shorter than NO_x transport time to neighboring grid cells, which is typical near the surface in coarse resolution models (Martin et al., 2003), such as the one used in this study. However, the assumption is less realistic at finer resolutions and in the upper troposphere, where the lifetime of NO_2 is longer than at the surface and NO_2 concentrations are not directly impacted by coincident near-surface emissions. Even at coarse resolution (e.g., 100-km grid spacing), emissions smearing effects, which occur when the FDMB assumption of local emissions effects is incorrect and emissions are inappropriately adjusted, can appear due to NO_x transport, reservoir species, and chemical feedbacks (Turner et al., 2012; Cooper et al., 2017). Traditional FDMB, which directly compares modeled and remotely sensed columns, cannot address this effect. Assimilating the satellite VCD introduces an additional complication. The horizontal length scales (on the order of several hundreds of kilometers) used in the background error extend beyond the grid cell horizontal dimensions (nominally 108km) in the middle and upper troposphere and, as a result, NO_2 changes introduced by assimilation ($\Delta\Omega$) do not have a local relationship with surface emissions directly below. In our work, assimilating the observed column information, instead of directly comparing modeled and satellite retrieved VCDs, allows the analysis to be restricted to the lower troposphere, mitigating both the misallocation errors of FDMB and the effect of horizontal length scales extending beyond the grid cell dimension. To that end, we limit the anthropogenic emissions analysis to the lowest 20 model layers, which is nominally from the ground to ~720 hPa over non-mountainous terrain in the summer, and use that partial column to calculate $\Delta\Omega$ in the FDMB inversion. $\Delta\Omega$ for a single month above and below the threshold is illustrated in Fig. B.1. By applying this cutoff, we focus the inversion on surface anthropogenic NO_x.

3.2.6 Inversion system testing

We conduct a synthetic observation experiment to evaluate the ability of the inversion system to constrain emissions to a known perturbation. Artificial NO_2 observations were generated from CMAQ simulations with unperturbed emissions and NO_x emissions reduced by 15%. As expected, assimilating the synthetic observations derived from a simulation with unperturbed emissions results in an analysis increment of zero. The results of an iterative emissions inversion based on the synthetic observations derived from the simulation with perturbed emissions are shown in Fig. B.2. Across Northern Hemisphere regions, the normalized mean error (NME) relative to the known perturbed emissions, and the rate at which it changes, decrease with subsequent iterations. The NME is minimized after 7–9 iterations, depending on the region. In all subsequent results, emissions inferences made with 8 iterations of the inversion system are shown and analyzed. Convergence of the inversion in different global regions adds confidence to the system's ability to constrain real-world emissions.

3.3 Results

3.3.1 Lighting NO_x emissions updates

Assimilation of retrievals from either satellite increases LNO_x emissions across all seasons, relative to the prior emissions (monthly climatology from GEIA), with largest changes occurring during the summer (Figs. B.3 and B.4). Applying 2019 OMI data increases total LNO_x emissions in 2019 by 20% over the GEIA climatology, while assimilation of TROPOMI data increases LNO_x emissions by 24%. The emissions increases inferred by both satellite products are driven by NO_2 increases in the mid and upper troposphere due to assimilation, with changes near the surface negligible in comparison. Increases in background areas with small NO_2 column totals and subsequent LNO_x increases in these areas suggest a low bias in modeled background NO_2 relative to observations from both satellites. A low bias agrees with the findings reported by other model-satellite NO_2 comparisons (Silvern et al., 2019; Qu et al., 2021; Goldberg et al., 2017). The LNO_x emissions adjustments inferred here decrease the differences between modeled and satellite-derived NO_2 in the upper troposphere and decrease the bias that differences in the upper troposphere can introduce to the subsequent ANO_x inversion.

3.3.2 Impact of assimilation on modeled NO_2 vertical column density

Figure 3.4 shows the change to CMAQ-modeled tropospheric VCD ($\Delta\Omega$) caused by assimilating NO_2 observations from OMI or TROPOMI with background errors for the boundary layer, before applying any emissions adjustments. In Fig. 3.4, and throughout the results, $\Delta\Omega$ reflects differences near the surface (as described in Sect. 3.2.5). Assimilating OMI NO_2 data generally increases modeled NO_2 columns near populated areas in China, India, and the U.S. In contrast, assimilating TROPOMI NO_2 data decreases modeled NO_2 columns more widely across the Northern Hemisphere. The changes brought about by assimilating satellite data are larger during the winter and fall, and smaller in the spring and summer, when NO_x lifetime is shortest and NO_2 columns are smaller. During the winter in northeast China, where the assimilation impacts are most apparent, the seasonal average change due to assimilation reaches 1.8×10^{15} molecules cm^{-2} for OMI and -2.8×10^{15} molecules cm^{-2} for TROPOMI. The direction of $\Delta\Omega$ after assimilation of OMI data is more heterogeneous and shows a stronger seasonality, while $\Delta\Omega$ based on assimilating TROPOMI data is consistently negative. Over Europe, $\Delta\Omega$ after assimilating OMI observations is close to zero in warm months and negative in colder seasons. Assimilating satellite-observed NO_2 increases the NO_2 levels modeled over the ocean and less-populous areas, such as the Sahara, with low NO_x emissions and small NO_2 column amounts.

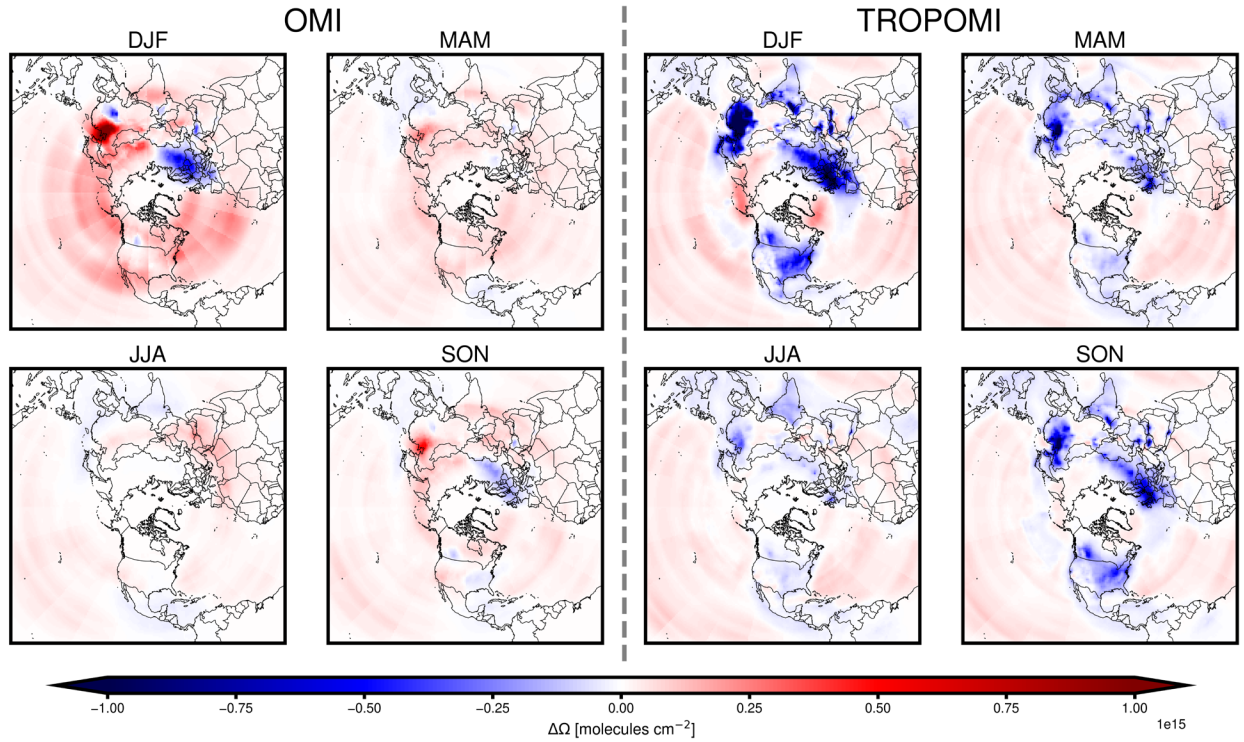

Figure 3.4: Seasonal NO_2 VCD change (**ΔΩ**) from CMAQ simulation using prior emissions after assimilating OMI or TROPOMI tropospheric NO_2 observations and modeling atmospheric composition with prior NO_x emissions. **ΔΩ** shown for winter (DJF), spring (MAM), summer (JJA), and fall (SON).

Over polluted areas, the direction of ΔΩ for the TROPOMI or OMI data assimilations tends to differ. This discrepancy is likely due to the low bias in TROPOMI-derived tropospheric NO_2 columns, which has been reported to be approximately 10% over the U.S., Europe, and India, and greater than 20% over China when compared with the OMI Quality Assurance for Essential Climate Variables (QA4ECV) retrieval (van Geffen et al., 2021; Verhoelst et al., 2021; Wang et al., 2020a; Li et al., 2021a). Over background areas, the analysis increments that results from assimilation of observations from both satellites generally agree. The consistency suggests a low bias in modeled background NO_2 concentrations and also agrees with the low bias in CMAQ-modeled free tropospheric NO_2 reported by (Goldberg et al., 2017). Such a bias can contribute to the positive analysis increment over background areas. However, NO_2 columns observed in these regions may be smaller than the retrieval accuracy of 0.7×10^{15} molecules cm^{-2} (Van Geffen et al., 2020), reducing confidence in the analysis increment at these locations. In the anthropogenic emissions inversion, our filtering criteria exclude from the analysis background areas which are more likely to have low VCD amounts.

3.3.3 Emissions inversion

Season-average β values, relating NO_2 vertical column differences to anthropogenic near-surface NO_x emissions updates, are shown in Fig. B.5. Based on our criteria for gridcell inclusion in the inversion, described in Sect. 3.2.5, we consider 13% of the grid cells in the domain, which represent 88% of prior anthropogenic NO_x emissions. Seasonal domain-average values range from 1.33 to 1.66, and are lower in the winter and higher in the summer. A β value less than 1.0 results in an emissions update that is smaller than the VCD change, while a β greater than 1.0 has the opposite effect. β tends to be less than 1.0 in polluted regions during colder months and larger during warmer months and in less polluted regions, although many grid cells which are less polluted are not considered in the analysis. The scaling factors are smallest over China, and larger over the U.S., India, Mexico, and Europe. The differences among regions stem from local differences in NO_x lifetime and transport. In Indonesia and sub-Saharan Africa, lower emissions and a small response from tropospheric VCD to anthropogenic emissions perturbations can lead to large β values. To prevent overly large or small β values, we constrain the factor to between 0.1 and 10, following (Cooper et al., 2017). Scaling factors estimated here are larger than the 1.16 global-average previously reported by Lamsal et al. (2011). However, in Lamsal et al. (2011) modeled NO_2 vertical columns were sampled at the morning SCanning Imaging Absorption spectroMeter for Atmospheric CHartographY (SCIAMACHY) overpass time, rather than the afternoon OMI or TROPOMI overpass times, and β tends to be closer to 1.0 during the morning in regions with high NO_x emissions (Li and Wang, 2019). Li and Wang (2019) show that over rural regions with lower NO_x concentrations, β is larger at the OMI or TROPOMI overpass window than at the SCIAMACHY overpass window, suggesting a larger overall β for analyses based on OMI or TROPOMI products should be expected. Additionally, NO_x emissions have decreased considerably in several regions of the Northern Hemisphere after the Lamsal study was conducted (2011), including the U.S. (Tong et al., 2015) and China (Miyazaki et al., 2017), which has changed the sensitivity of NO_2 VCDs to NO_x emissions (Silvern et al., 2019; Qu et al., 2021).

Annual bottom-up prior ANO_x emissions estimates are shown in Fig. 3.5. Season-average ANO_x emissions inferences from the inversions based on OMI and TROPOMI observations are shown in Fig. 3.6. The use of OMI observations generally tends to increase emissions in most industrialized nations outside of Europe. NO_x emissions

increases driven by OMI observations are largest in winter and spring and smaller, or slight decreases, in summer and fall. In contrast, the use of TROPOMI retrievals tends to drive a decrease in NO_x emissions across all seasons and continents, with the largest impacts in the summer and smallest in the spring. The largest emissions changes based on both OMI and TROPOMI retrievals are in Northeast China during the winter. Over India, OMI inferred changes are concentrated in Central India where prior emissions are lower, while the largest changes inferred from TROPOMI are in the Northern, Eastern, and Southern zones, where prior emissions are highest. Relative NO_x emissions changes driven by TROPOMI observations tend to be small over dense urban areas, with more uniform decreases over cells with lower emissions

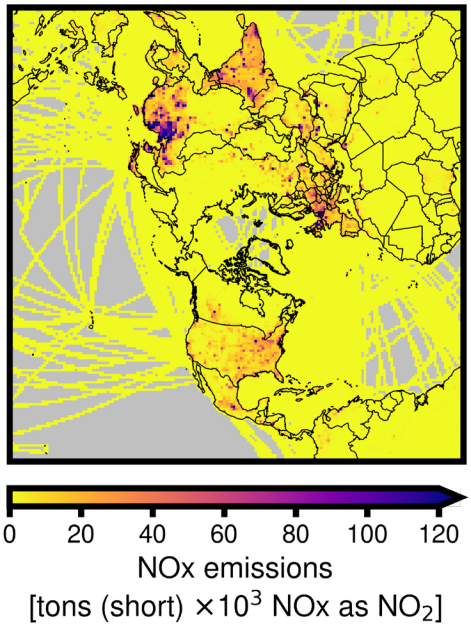

Figure 3.5: 2019 prior anthropogenic NO_x emissions totals. Data sources are described in Table 3.1.

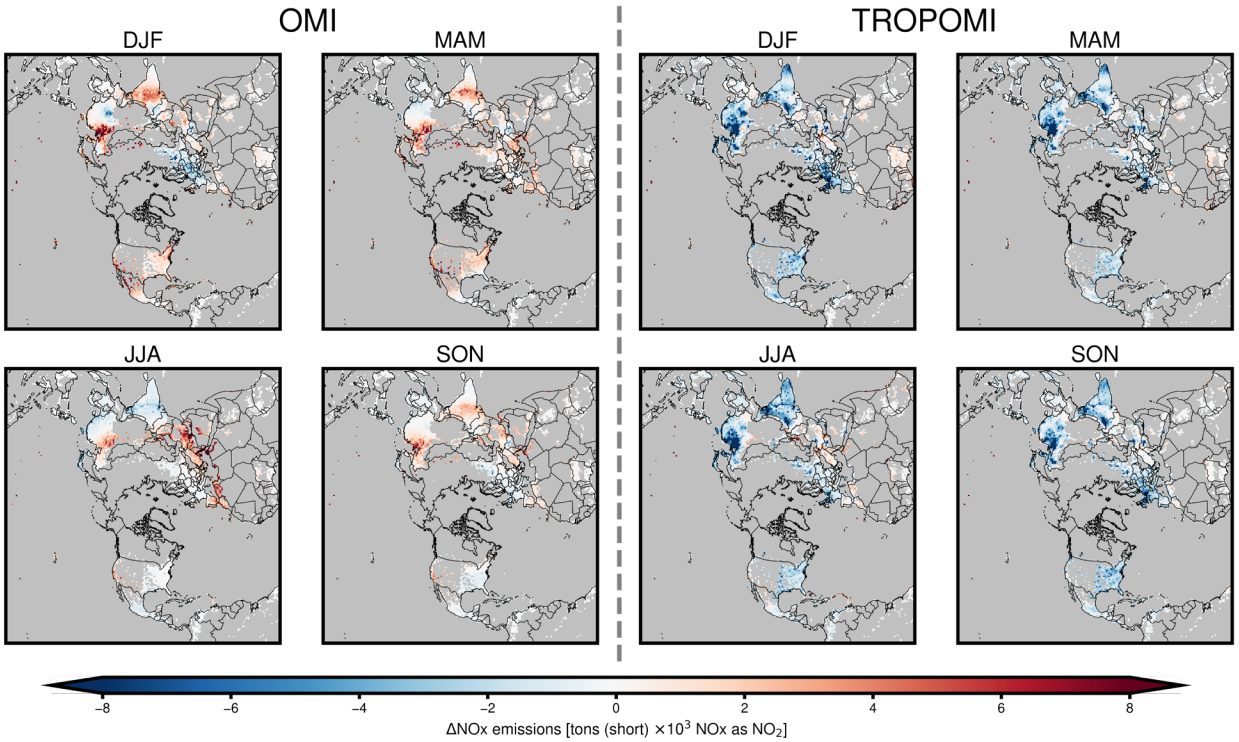

Figure 3.6: Season-average NO_x emissions changes from inverse modeling updates based on OMI and TROPOMI observations. Emissions changes are shown for winter (DJF), spring (MAM), summer (JJA), and fall (SON).

ANOx emissions totals and inferred changes are explored for China, India, Europe, Mexico, and the U.S. (Fig. 3.7). We also show 2019 NO_x emissions totals from the Copernicus Atmosphere Monitoring Service (CAMS) bottom-up emissions inventory (Granier et al., 2019), and the NASA Tropospheric Chemical Reanalysis products 2 (TCR-2) satellite-inferred inventory (Miyazaki et al., 2019, 2020). TCR-2 top-down NOx emissions are constrained using satellite observations of NO_2, CO, O_3, and SO_2 at a resolution of 1.125°× 1.125° and are further described in Miyazaki et al. (2017). CAMS anthropogenic NO_x emissions are based on the Emissions Database for Global Atmospheric Research (EDGAR version 5.3) estimates for 2015 (Crippa et al., 2020) projected to 2019 using CEDS scaling factors, and are provided at 0.1°× 0.1°. Both datasets provide monthly anthropogenic NO_x totals. Except for Europe, assimilation toward OMI retrievals increases annual emissions totals in the regions analyzed, while using TROPOMI retrievals decreases them. TCR-2 NO_x emission estimates are larger than the prior emissions used by our inverse modeling framework, except for India, while CAMS totals are lower than the prior emission estimates and similar to TROPOMI inferred emissions. Across the regions considered, TROPOMI infers an average annual decrease

to NO_x emissions from the regions of -33%, while OMI infers a +9% increase. In Europe, the only region where the sign of the inferred changes match, use of OMI retrievals results in a -1% change, while applying TROPOMI observations leads to a -36% decrease in NO_x emissions. The largest total changes are inferred in the highest emitting region, China, while the greatest relative changes, -41% inferred with TROPOMI, are for India, where emissions are highly uncertain. Changes inferred with OMI observations over the U.S. are greater than $1{,}200 \times 10^3$ short tons NO_x as NO_2 per year, but smaller than the difference between our prior U.S. emissions estimates and TCR-2 or CAMS estimates. A change of -3,000 $\times 10^3$ short tons NO_x as NO_2 emitted annually in the U.S. as inferred by TROPOMI, over 30% of the prior emissions, differs significantly from National Emissions Inventory estimates but leads to a total close to that of the 2019 CAMS inventory.

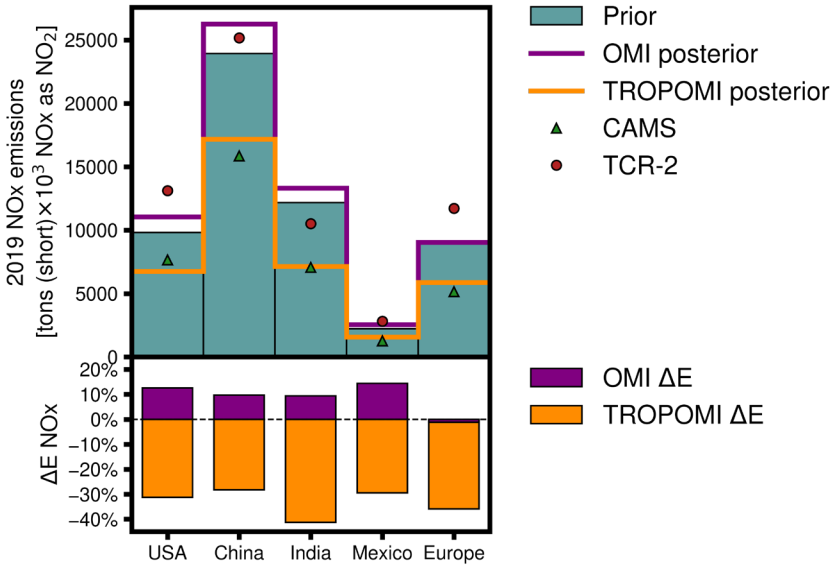

Figure 3.7: Prior and satellite-inferred 2019 anthropogenic NOx emissions in select global regions. Top plot shows total emissions (as NO_2) from prior emissions estimates, inference with OMI or TROPOMI observations (OMI and TROPOMI posterior), and CAMS or TCR-2 inventories in the U.S., China, India, Mexico, and Europe. The bottom plot shows the percent change (ΔE NO_x) inferred with OMI or TROPOMI data, relative to prior emission estimates, for each region.

Across the months simulated, inferences using OMI retrievals consistently lead to higher NO_x emissions than using TROPOMI retrievals. Figure 3.8 shows monthly NO_x emission totals and inferred changes for several global regions. The magnitude of changes is generally smallest in summer months and largest in winter months for both OMI and TROPOMI inferred emissions. Monthly prior emissions totals lay between the OMI and TROPOMI inferences, except for summertime emissions in India and Mexico where both satellite inferences decrease NO_x emissions. Over Europe, both satellite

products infer a decrease during the winter, although fewer valid satellite pixels due to snow cover at high latitudes and longer winter NO_2 atmospheric lifetimes may influence the inference.

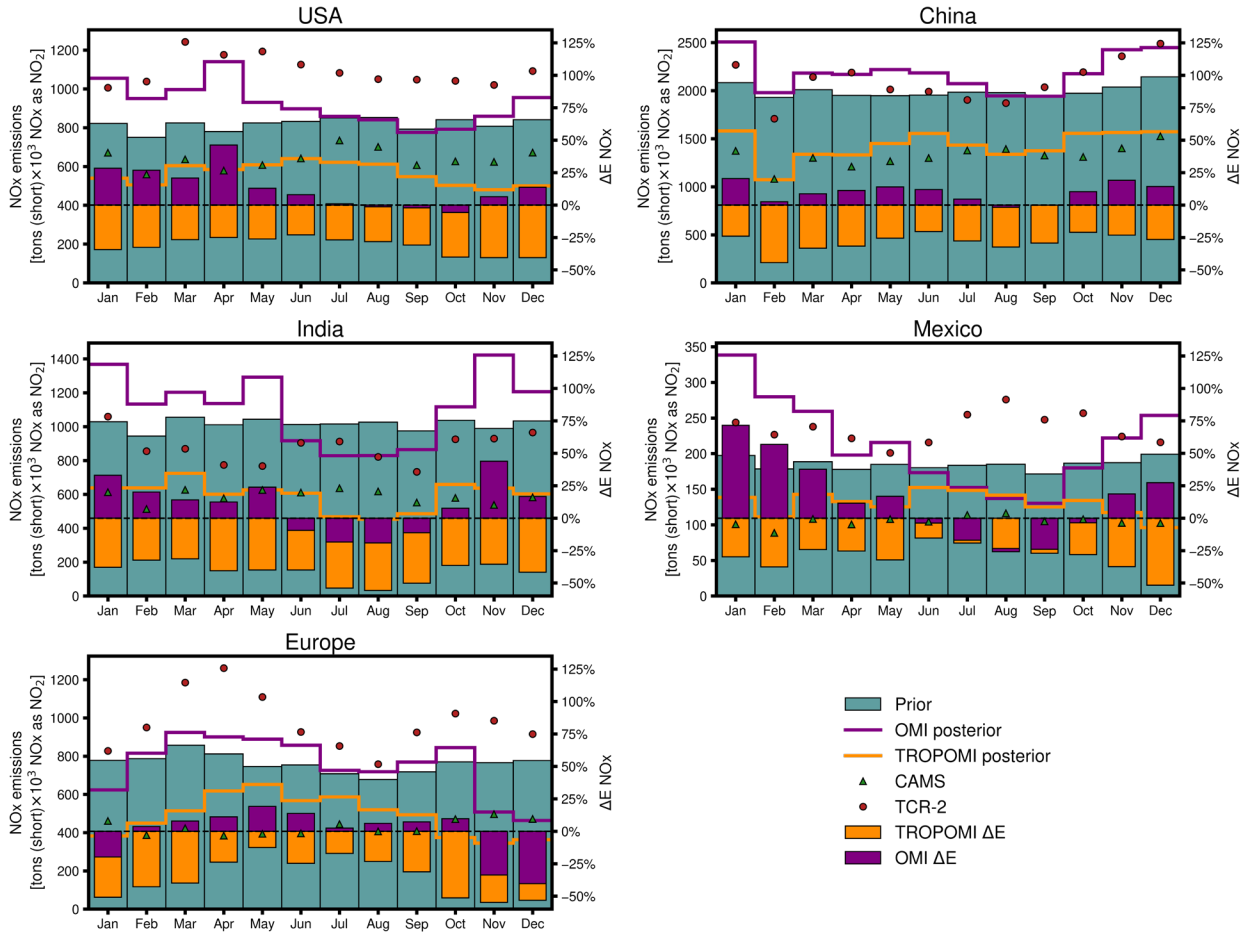

Figure 3.8: Monthly prior and satellite-inferred anthropogenic NO_x emissions in 2019 in select global regions. Total monthly emission (as NO_2) from prior emission estimates, inference with OMI or TROPOMI observations (OMI and TROPOMI posterior), and CAMS or TCR-2 inventories in the U.S., China, India, Mexico, and Europe are shown. Percent change (ΔE NOx) inferred with OMI or TROPOMI data, relative to prior emission estimates, for each region are shown by the purple and orange bars.

Based on reported NO_x emissions trends (McDuffie et al., 2020; U.S. EPA, 2022a), changes from the prior emissions inventory (Table 3.1) to 2019 are expected. Relative to the prior emissions, significant decreases in NO_x emissions in China, estimated for 2015 in the prior inventory, and smaller reductions in Europe and North America, reported for 2014 and 2017 in the prior inventory, respectively, should be anticipated. TROPOMI inferred emissions reflect the direction anticipated for these changes, but with larger than expected magnitudes. For example, the 28% decrease in anthropogenic NO_x emissions

over China between 2015 and 2019 inferred from the TROPOMI observations is substantially larger than the 8% decrease estimated between 2015 and 2017 by the Community Emissions Data System (McDuffie et al., 2020). Bottom-up estimates indicate that anthropogenic NO_x emissions in the U.S. have decreased through 2019 (U.S. EPA, 2022b). Although the direction of the emissions change inferred from TROPOMI agrees with the trend in bottom-up estimates, its magnitude is larger than expected. An underestimate of U.S. emissions in winter in the prior inventory when compared with OMI inferences contrasts with field study results reporting no bias in Northeastern U.S. winter emissions estimates (Jaeglé et al., 2018; Salmon et al., 2018). In India, bottom-up emissions inventories report sustained growth of NO_x emissions (Kurokawa and Ohara, 2020; McDuffie et al., 2020) and NO_2 levels observed by OMI have been increasing since 2005 (Goldberg et al., 2021; Cooper et al., 2022). The decrease in anthropogenic NO_x emissions inferred by TROPOMI observations contrasts with these trends in bottom-up estimates and OMI observations.

The low bias known to affect TROPOMI NO_2 observations influences the results of the emissions inversion, which targets grid cells with high emissions, likely leading to larger than expected inferred emissions decreases. We conduct an inversion using the reprocessed TROPOMI NO_2 version 2.3.1 (van Geffen et al., 2021) to infer NO_x emissions for January 2019, and find that the updated data increases the TROPOMI posterior inference by 17% over the U.S. and 4% in China relative to version 1.2.2, but still differs significantly from that obtained using OMI observations (Figs. B.10 and B.11). The differences between emissions inferred by OMI and TROPOMI observations highlight the importance of ongoing efforts to harmonize OMI and TROPOMI NO_2 retrieval algorithms, such as the NASA Multi-Decadal Nitrogen Dioxide and Derived Products from Satellites (MINDS) (Lamsal et al., 2020) and the QA4ECV (Boersma et al., 2017) datasets.

In addition to smearing effects, coarse resolution models can artificially alter nonlinear NO_2 chemistry, leading to biases in inferences of NO_x emissions from satellite NO_2 columns (Valin et al., 2011; Sekiya et al., 2021; Lamsal et al., 2011). Higher resolution simulations can better resolve β and reduce biases caused by nonlinear chemistry. Additional errors in the emissions estimates may be associated with emissions from non-anthropogenic NO_x sources. Although the emissions inversion targets anthropogenic sources only, changes in NO_2 columns observed by the satellite instruments driven by

natural NO_x emissions processes may not be captured by in the air quality model simulations and subsequently lead to biased anthropogenic emissions inferences (Li et al., 2021a).

The emissions resulting from the inverse modeling framework are comparable to CAMS and TCR-2 2019 emissions estimates in several ways. In the U.S., China, and Europe, the magnitudes of ANO_x emissions from OMI retrievals are comparable to TCR-2 NO_x emissions estimates and exhibit similar monthly patterns. Annual NO_x emissions inferred from OMI observations are also relatively similar to TCR-2 estimates for India and Mexico, although monthly emissions patterns differ. Unlike TCR-2 emissions estimates, which are is also constrained by OMI NO_2 observations, 2019 CAMS emissions estimates are projected from 2015 bottom-up data. However, CAMS estimates provide a representation of anticipated emissions trends. In all regions considered, CAMS NO_x emissions estimates are close to the TROPOMI inference annual totals and lower than the prior emissions, OMI inferences, and TCR-2 estimates, potentially suggesting that global NO_x emissions have not decreased as much as anticipated by the CAMS inventory projections.

3.3.4 Impacts of emissions updates on modeled NO_2 and O_3

We evaluate and compare the CMAQ simulations' ability to reproduce observed pollutant concentrations when driven with NO_x emissions estimates from the prior inventory and those derived by the inverse modeling framework. Figure 3.9 compares 2019 OMI and TROPOMI NO_2 VCD retrievals with modeled NO_2 VCDs using the prior emissions with no updates, LNO_x emissions updates, and LNO_x and ANO_x emissions updates. Satellite-based LNO_x emissions updates improve CMAQ model performance—correlation coefficient (R), normalized mean error (NME), and normalized mean bias (NMB)—when evaluated against tropospheric VCD retrievals, relative to model performance with the prior emissions. OMI-inferred ANO_x emissions updates further improve CMAQ model performance evaluated against VCD retrievals, decreasing NMB from -20% to -5% and NME from 38% to 28%. Model performance is improved by using OMI data in the inverse modeling framework across all seasons (Figs. B.6-B.9). Although LNO_x emissions updates derived from TROPOMI observations improve model bias and error relative to the CMAQ simulation using prior emission estimates, TROPOMI-inferred anthropogenic emissions do not, except during summer months (Figs. 3.9 and

B.6-B.9). The lack of significant improvements in CMAQ-simulated NO_2 VCDs after applying the emissions inversion with TROPOMI NO_2 retrievals prior to the version 2.3.1 update (van Geffen et al., 2021) may be associated with changing chemical regimes that are not captured in the emissions inversion process.

Changes in modeled VCD due to assimilation and the emission inferences calculated in the TROPOMI ANO_x inversion exceed the emissions perturbation and VCD changes used to calculate β. For example, over the eastern U.S., the -15% emissions perturbation used to calculate β leads to VCD changes of -15% on average in winter, but assimilating TROPOMI retrievals leads to VCD changes ($\Delta\Omega$) of -19% on average in the winter, with individual changes exceeding -30%. Modeled NO_x chemistry and NO_2 vertical profiles after assimilating TROPOMI retrievals may be different than those used in the calculation of β. As a result, assimilating TROPOMI retrievals in the ANOx inversion may lead to modeled NO_2 vertical profiles which are inconsistent with the precalculated β used in the FDMB relationship and less reliable subsequent emissions inferences. In contrast, the magnitude of VCD changes due to assimilating OMI retrievals over the eastern U.S. in winter is 8%, well within the magnitude of the VCD changes used to precalculate β. This highlights the importance of applying a β sensitivity valid for the magnitude of anticipated emissions changes in FDMB inversions and the potential consequences of relying on satellite-derived retrievals with pre-existing biases in emissions inversions.

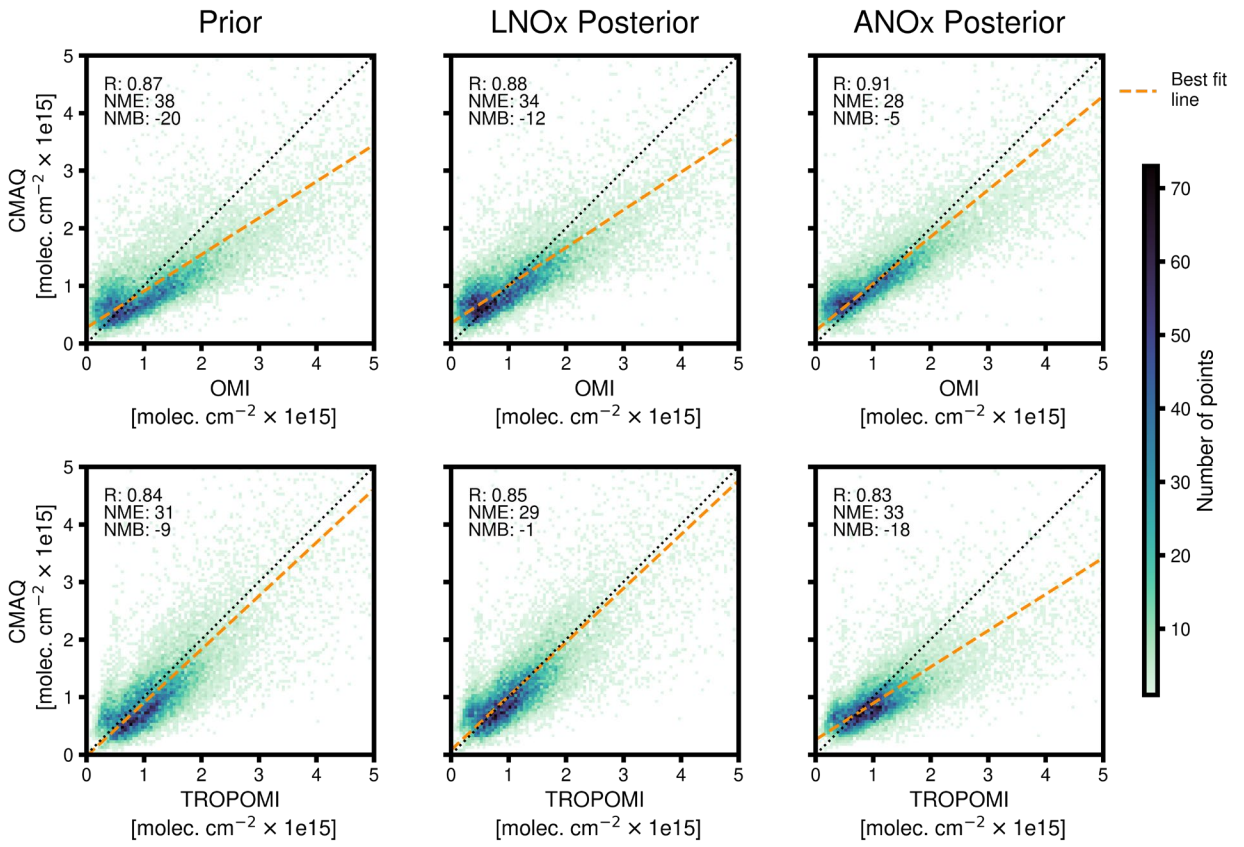

Figure 3.9: Impact of NOx emissions updates on modeled NO_2 VCDs. Plots compare 2019 season-average CMAQ-modeled NO_2 VCD at each model grid cell in which NOx emissions were updated by the inverse modeling framework against OMI and TROPOMI tropospheric NO_2 VCD retrievals averaged in each model grid cell. Modeled NO_2 VCD using prior emissions (Prior), inferred LNOx emissions (LNOx posterior), and inferred lightning and anthropogenic NOx emissions (ANOx posterior) are each compared with NO_2 VCD retrievals. Top row plots compare retrievals and modeled VCD based on OMI observations, while bottom row plots compare retrievals and modeled VCD based on TROPOMI observations. Linear regression line, correlation coefficient (R), normalized mean error (NME), and normalized mean bias (NMB), relative to tropospheric NO_2 VCD retrievals, are shown for each CMAQ simulation.

Comparing CMAQ-modeled O_3 to ozonesonde measurements from the World Ozone and Ultraviolet Radiation Data Centre (WOUDC) network shows the impacts updating LNO_x emissions on simulated tropospheric O_3 (Fig. 3.10). Above 300 hPa, the model is biased low, but neither update has a major impact on this bias. However, within the free troposphere, the effects of LNO_x emissions updates are larger. LNO_x satellite-inferred emissions from both satellites increase O_3 and subsequently improve the model's low O_3 bias across all seasons, with the strongest effect in the summer. This suggests a low background NO_2 in our prior simulation, consistent with several studies

demonstrating that models underestimate background NO_2 (Goldberg et al., 2017; Silvern et al., 2019; Qu et al., 2021).

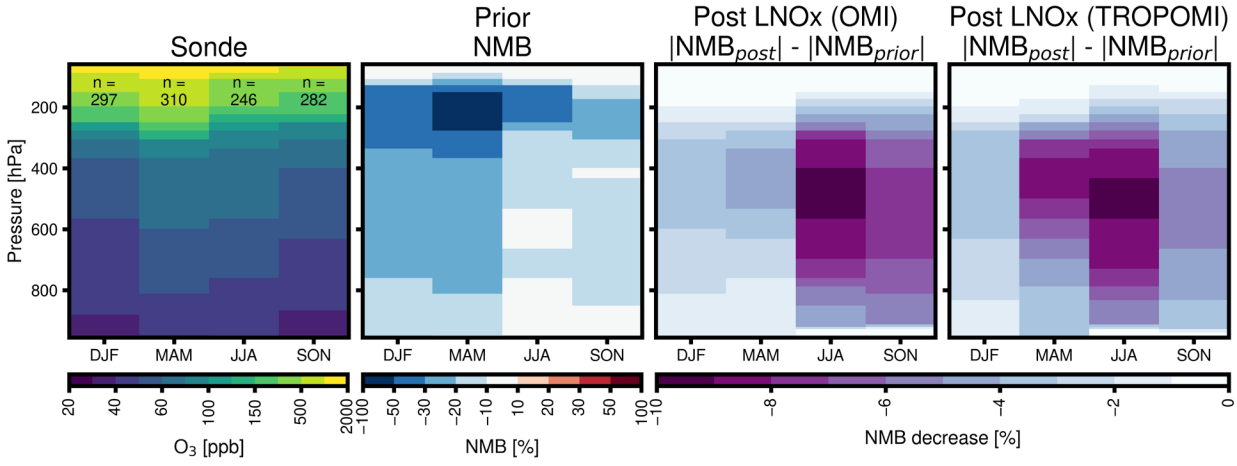

Figure 3.10: Ozonesonde observations from the WOUDC network and impact of lightning emissions inferences on modeled ozone. Left plot shows sonde observations averaged in each season and total number of launches per season. The NMB is shown for the prior emissions simulation. Plots on the right show the decrease in the NMB, relative to the prior simulation, for simulations with LNO_x emissions updated with OMI and TROPOMI data.

Comparisons of CMAQ-modeled NO_2 and O_3 concentrations with ground-level measurements highlight the challenges of reproducing local air quality with a coarse scale model, but suggest potential to improve model performance with satellite-derived NO_x emissions updates. Table 3.2 shows statistics evaluating modeled ground-level daily average NO_2 and maximum 8-hour O_3 concentrations over the U.S. against observations from 1,218 monitoring sites in the Air Quality System (AQS) (U.S. EPA, 2022b), excluding near-road monitors for which the gridded NO_2 fields are not representative of. Statistics for each season are included in Tables B.2 and B.3. There is a significant low bias in CMAQ-predicted ground-level NO_2 concentrations compared with monitoring site measurements, likely due to the model's coarse grid resolution and the aggregation of NO_2 monitors within urban areas with high NO_x emissions and large concentration gradients. CMAQ simulations at higher horizontal resolution do not show the same bias against NO_2 surface observations (Toro et al., 2021). Agreement between modeled and observed ground-level NO_2 concentrations is improved by using OMI-inferred NO_x emissions, compared with the prior emissions simulation, in particular during winter and spring months. Model performance evaluated against ground-level O_3 measurements improves to a smaller extent with OMI-inferred NO_x emissions during winter and spring months. The use of TROPOMI-inferred emissions has mixed impacts on CMAQ

performance against observed ground-level NO_2 and O_3 concentrations, leading to limited gains in seasonal R and some seasonal biases and errors, but also less agreement with observations for other seasonal statistics. In the U.S., the network of ground-based air quality observations is relatively large. However, in some regions where emissions uncertainties are expected to be especially high, ground-based observations are significantly limited and less accessible. Assessing the impact of emissions updated against ground-based observations in these regions, although a challenge, would provide further evaluation of the inversion framework in locations where satellite retrievals have the largest potential to provide important constraints to emissions estimates.

Table 3.2: CMAQ model performance evaluated against daily average NO_2 (DA NO_2) and maximum 8-hour O_3 concentrations (MDA8 O_3) observed in 2019 by AQS monitoring sites in the U.S. Near-road monitors are not considered. Statistics are shown for simulations using prior emissions (Prior), lightning and anthropogenic NOx emissions inferred with OMI data (OMI-inferred), and lightning and anthropogenic NOx emissions inferred with TROPOMI data (TROPOMI-inferred). Coefficient of determination (R), normalized mean error (NME), and normalized mean bias (NMB), relative to AQS observations, are estimated for each CMAQ simulation.

Pollutant	NO_x emissions	R	NME	NMB
MDA8 O_3	Prior	0.65	15.9%	-1.4%
	OMI-inferred	0.68	15.4%	3.4%
	TROPOMI-inferred	0.68	15.3%	-3.3%
DA NO_2	Prior	0.45	62.2%	-56.9%
	OMI-inferred	0.52	57.4%	-49.6%
	TROPOMI-inferred	0.45	71.7%	-69.9%

3.3.5 Impacts of emissions updates on long-range O_3 transport

Global NO_x emissions estimates affect model simulations of long-range air pollution transport. To explore these impacts, we examine the response of CMAQ-modeled transpacific O_3 to the inverse modeling framework's NO_x emissions updates. Figure 3.11 shows season-average changes in simulated free-tropospheric O_3 over the North Pacific Ocean resulting from the use of OMI- and TROPOMI-inferred ANO_x emissions, relative to the emissions simulation with LNO_x emissions updated. As expected, the emissions inversions lead to O_3 variations that follows NO_x emissions changes inferred for each satellite's observations, with OMI inferences resulting in higher

O_3 concentrations and TROPOMI inferences resulting in lower O_3 concentrations over the North Pacific Ocean. Season-average differences with respect to the prior emissions simulation are as large as +1.8 ppb in winter, using OMI-based updates, and -1.9 ppb in spring, using TROPOMI-based updates. Combined with transpacific wind patterns, the effects of the NO_x emissions inversions on modeled O_3 suggest potential implications of uncertain Asian emissions estimates for U.S. air quality management and emphasize the impacts of biases in satellite retrievals in inverse modeling systems.

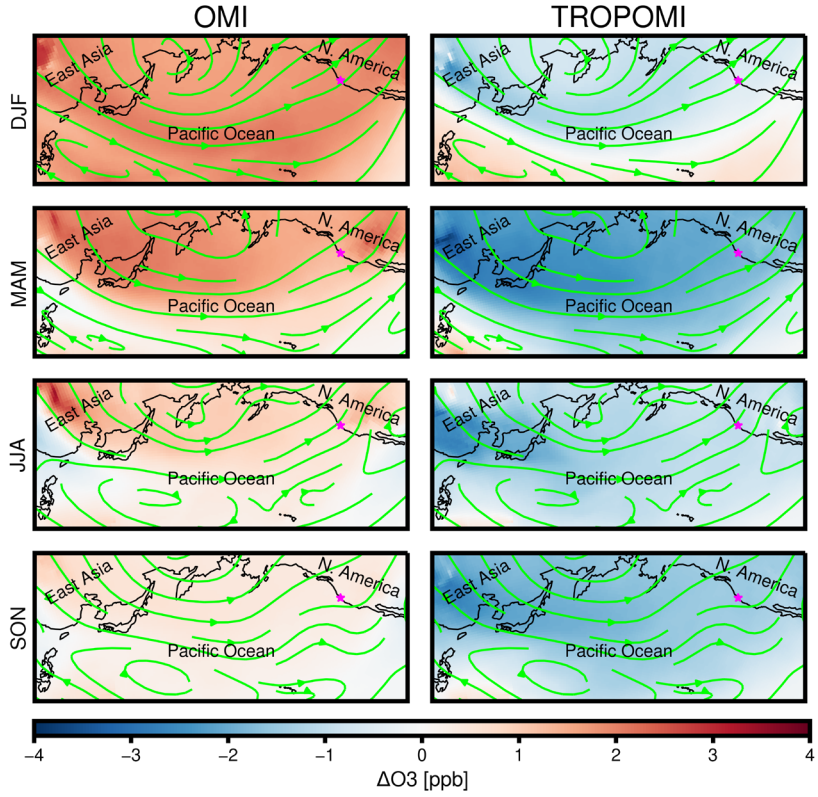

Figure 3.11: Changes 2019 season-average free-tropospheric O_3 concentrations (averaged between 750-250 hPa) simulated over the North Pacific Ocean using lightning and anthropogenic NO_x emissions inferred with OMI or TROPOMI observations, relative to simulation using prior ANO_x emissions and updated LNO_x emissions. Differences are shown for winter (DJF), spring (MAM), summer (JJA), and fall (SON). Arrows depict season-average free-tropospheric winds (750-250 hPa). Star marker indicates location of Trinidad Head, California.

At the Trinidad Head, California, a location where atmospheric composition is relatively unaffected by local emissions sources and responsive to transpacific pollution transport (Fig. 3.11), differences in modeled daily average free-tropospheric O_3 concentrations can reach +5 ppb or -3 ppb. Figure 3.12 compares CMAQ-modeled vertical O_3 profiles to observations from 39 ozonesondes launched in at Trinidad Head in 2019 (WOUDC, 2015). Relative to the CMAQ simulation using prior emissions, NO_x emissions

updates inferred from OMI and TROPOMI data can improve the model's ability to reproduce ozonesonde O_3 distributions measured from the site, in particular during winter and spring when the discrepancies between modeled and observed concentrations are largest. These impacts on modeled vertical O_3 profiles are largely driven by changes the modeling framework's updates to lightning NO_x emissions. The inferred LNO_x increases from each satellite improve O_3 biases, while subsequent anthropogenic updates have smaller impacts, suggesting that biases in O_3 could be driven by background NO_2 composition in the model and not solely by long-range transport resulting from anthropogenic emissions.

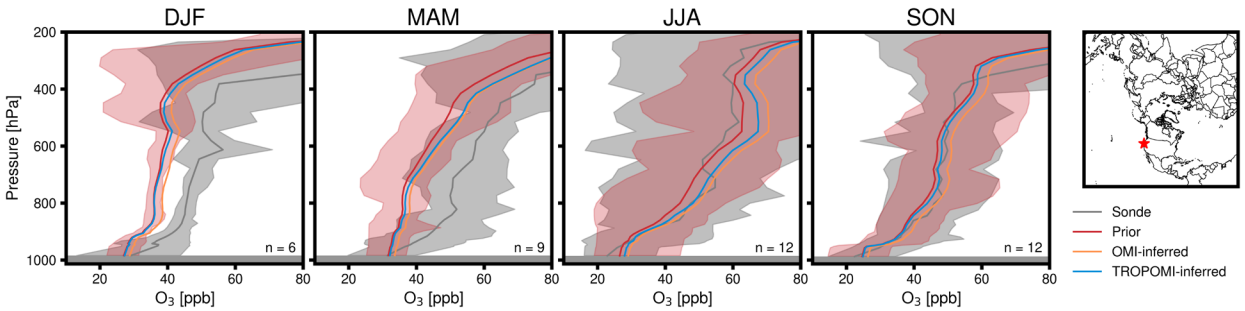

Figure 3.12: Season-average vertical O_3 concentration profiles modeled by CMAQ and measured by ozonesondes launched at Trinidad Head, California, in 2019. Vertical distributions are shown for simulations using prior emissions (Prior), lightning and anthropogenic NOx emissions inferred with OMI data (OMI-inferred), and lightning and anthropogenic NOx emissions inferred with TROPOMI data (TROPOMI-inferred). Modeled season-average profiles are shown during winter (DJF), spring (MAM), summer (JJA), and fall (SON) for days and times matching ozonesonde launches. Shading around sonde and prior emissions profiles represent the maximum and minimum O_3 at each pressure level. Map shows location of the Trinidad Head launch site.

3.4 Conclusions

In this study, we describe a satellite chemical data assimilation and inverse emissions modeling framework based on the CMAQ hemispheric air quality modeling platform. In the framework, data assimilation adjusts modeled NO_2 concentrations online using satellite retrievals of tropospheric NO_2 VCDs. The NO_2 column changes drive the FDMB inversion, resulting in satellite-constrained top-down emissions estimates. Here, we implement the framework in a NO_x emissions inversion to separately update 2019 Northern Hemisphere lightning and anthropogenic NO_x emissions estimates using NO_2 products from the OMI and TROPOMI satellite instruments. Relative to the modeling platform's prior emissions derived from regional and global emissions inventories, updates inferred using OMI and TROPOMI observations change average anthropogenic

NO_x emissions by –41% to +12% in China, the U.S., India, Europe, and Mexico. Evaluated against ground-based NO_2 observations recorded over the U.S. in 2019, the model performs best when using OMI-updated emissions, although a low bias in CMAQ predictions using prior emissions persists into simulations with satellite data assimilation. Compared with U.S ground-based O_3 observations, satellite-inferred emissions have mixed impacts on model performance, improving agreement with the measurements during certain months. LNO_x emissions inferences improve modeled O_3 when compared against ozonesonde observations across the Northern Hemisphere. The framework's NO_x emissions updates also affect model estimates of trans-Pacific O_3 transport, a source of growing concern in the U.S., with changes ranging from -3 ppb to +5 ppb in simulated O_3 at a remote West Coast site resulting from use of satellite-inferred emissions.

The modeling framework presented has several limitations. The computational cost is greater than that of traditional FDMB inversions due to the assimilation step. However, the computational burden is comparable or less than other satellite assimilation methods such as Kalman-Filter and adjoint 4D-variational approaches. In addition, the framework requires minimal code changes to the underlying CTM, so inverse estimates will improve as the underlying air quality model is updated with little additional effort needed to implement this framework. The global coverage of instruments on polar-orbiting satellites, such as Aura and Sentinel-5P, makes the emissions inversions possible but does not allow satellite observations to inform diurnal emissions variations. Upcoming geostationary satellite missions, including GEMS, TEMPO, and Sentinel-4, will provide this capability. Our approach, which balances computational costs and precision in the inversion, is subject to several assumptions. As in all mass-balance based approaches, our method fully attributes the change in the VCD to emissions changes. To the extent that column differences are due to chemistry or transport and not emissions, this assumption introduces error into mass-balance inversions, including the inversion implemented in our framework. Large changes to the model concentrations resulting from the chemical data assimilation may invalidate assumptions in the subsequent FDMB inversion, leading to biases in the inferred emissions. The FDMB inversion treats each grid cell independently and cannot relate NO_2 column changes in one grid cell to emissions in another. Although emissions smearing in the approach is mitigated by only analyzing the lower portion of the model column, our

emissions changes may be less precise than targeted assimilation methods, such as 4DVAR adjoint-based methods. Further, coarse grid resolution exacerbates biases in modeled NO_2 columns (Valin et al., 2011) and inferred NO_x emissions (Sekiya et al., 2021). The air quality model used here does not include stratospheric chemistry, which could affect comparisons against NO_2 retrievals. Nevertheless, the framework shows the potential to improve air quality model predictions using satellite-derived emissions updates, in particular for regions with highly uncertain emissions inventories.

Emissions inversions based on satellite observations can provide valuable information for air quality modeling by addressing the gaps in bottom-up emissions inventories. However, our analysis shows that such inversions and subsequent air quality simulations can be strongly influenced by uncertainties and biases in the satellite data products used. In the analysis conducted, NO_x emissions inferred from TROPOMI observations appear biased low when assessed against those inferred from OMI data and surface and concentration measurements. The bias is consistent with recent research showing a low bias in TROPOMI v1.2 and v1.3 tropospheric columns (Judd et al., 2020; Verhoelst et al., 2021; Li et al., 2021a; van Geffen et al., 2021). The results highlight the importance of efforts to develop robust and consistent satellite data products for use in air quality modeling evaluation, assimilation, and emissions inversions. Ongoing efforts to this end include the MINDS (Lamsal et al., 2020) and the QA4ECV (Boersma et al., 2017) projects. This study also emphasizes the need for longer-term satellite data assimilation and comparisons of established and new satellite data products. The framework introduced here can serve a generalized tool with applications beyond those explored in this study, and allows new satellite data products to be incorporated as they become available. As satellite data products evolve and advance, the emissions inferred by the framework will improve.

Chapter 4: Characterizing and quantifying uncertainty in projections of climate change impacts on air quality

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Abstract

Climate change can aggravate air pollution, with important public health and environmental consequences. While major sources of uncertainty in climate change projections—greenhouse gas emissions scenario, model response, and internal variability—have been investigated extensively, their propagation to estimates of air quality impacts has not been systematically assessed. Here, we compare these uncertainties using a coupled modeling framework that includes a human activity model, an Earth system model of intermediate complexity, and a global atmospheric chemistry model. Uncertainties in projections of U.S. air quality under 21st century climate change are quantified based on a climate-chemistry ensemble that considers multiple initializations, representations of climate sensitivity, and climate policy scenarios, under constant air pollution emissions. We find that climate-related uncertainties are comparable at mid-century, making it difficult to distinguish the impact of variations in greenhouse gas emissions on ozone and particulate matter pollution. While greenhouse gas emissions scenario eventually becomes the dominant uncertainty based on the scenarios considered, all sources of uncertainty are significant through the end of the century. The results provide insights into intrinsically different uncertainties in projections of air pollution impacts and the potential for large ensembles to better capture them.

4.1 Introduction

Climate change is expected to exacerbate air pollution during the 21st century (Jacob and Winner, 2009; Fiore et al., 2012). Under a warmer climate, projected changes in meteorology, atmospheric chemistry, and natural emissions will deteriorate air quality

over many populated areas (Fiore et al., 2015). This climate-induced impact on air quality, commonly termed the climate penalty, is distinct from changes brought about by changing pollutant emissions and can have significant public health and economic consequences (Fann et al., 2021; Hong et al., 2019; Nolte et al., 2018a; Silva et al., 2016a, 2017). The Fourth National Climate Assessment projects that in the absence of climate policy, by 2090 the climate penalty on ozone (O_3) pollution will annually cause 500 additional deaths and cost $8 billion in the U.S., compared to air quality under a climate-mitigation pathway (U.S. EPA, 2017; Nolte et al., 2018a). Although the interactions between particulate matter pollution and climate change are less clear (Fiore et al., 2015; Von Schneidemesser et al., 2015), a climate penalty is also expected for $PM_{2.5}$ (particulate matter with aerodynamic diameter ≤ 2.5 μm) (Shen et al., 2017; Day and Pandis, 2015; Nolte et al., 2018b), the environmental pollutant responsible for the largest public health burden (Cohen et al., 2017). Climate change mitigation policies can reduce health damages by modulating the climate penalty and its impacts (Garcia-Menendez et al., 2015), but estimates of these co-benefits are subject to significant uncertainties.

Characterizing uncertainty is integral to projections of climate change and its impacts (Deser et al., 2012; National Research Council, 2012). Uncertainty in climate projections stems from three major sources: greenhouse gas (GHG) emissions scenario, climate model response, and internal variability (Hawkins and Sutton, 2009). Internal variability arises from natural variations in the climate system and is most effectively captured by initial condition ensembles (Deser et al., 2020; Deser, 2020; Lehner et al., 2020; Mankin et al., 2020; Fiore et al., 2022). Model response uncertainty is the differing response of climate models to the same radiative forcing and has been assessed by quantifying variability in projections across multi-model ensembles (Lehner et al., 2020). In the absence of multi-model ensembles, altering the climate sensitivity in a single-model framework has been shown to approximate model response for uncertainty analyses (Monier et al., 2015). Emissions scenario uncertainty can be addressed by increasing the accuracy of emissions estimates or projecting multiple climate policy futures. Understanding how these drivers of uncertainty propagate to projections of climate impacts is critical for policy analyses considering air quality impacts and co-benefits of mitigation strategies. While climate uncertainties in air quality projections have been explored (Garcia-Menendez et al., 2017; Pienkosz et al., 2019; Manders et al., 2012; Wild

et al., 2012; Fiore et al., 2022), these uncertainties have not been systematically weighed or compared.

Here, we quantify and compare major uncertainties in projections of the climate penalty on U.S. air quality. Uncertainties are derived from an extensive simulation ensemble designed to simulate the climate-induced impacts on air pollution and generated with a modeling framework that includes a human activity model, an Earth system model of intermediate complexity, and a global atmospheric chemistry model. Uncertainties in $PM_{2.5}$ and O_3 impacts due to internal variability, GHG emissions scenario, and model response are characterized in the ensemble by multiple model initializations, representations of climate sensitivity, and climate policy scenarios. We find that these uncertainties are comparable at mid-century and can make it difficult to discern the impact of changes in GHG emissions on U.S. air quality. By the end of the century, however, GHG emissions scenario is the dominant source of uncertainty followed by climate sensitivity, while internal variability remains close to its mid-century levels. We further examine the influence of internal variability on air quality projections ranging from annual to decadal scale. The analysis also examines the influence of climate sensitivity in simulations of air quality under climate change for the first time, providing insights into the weight of model response uncertainty in projections of air quality impacts. The results of this analysis highlight the potential impacts that future large and multi-model climate-chemistry ensembles can have on improving our understanding of climate change effects on air pollution in the face of uncertainty.

4.2 Methods

4.2.1 Modeling framework

We use a modeling framework that couples the Massachusetts Institute of Technology Integrated Global System Model and the National Center for Atmospheric Research Community Atmosphere Model (MIT IGSM-CAM) (Monier et al., 2013) to generate the simulation ensemble used in this study. The MIT IGSM is an integrated assessment model that links the MIT Earth System model (MESM) (Sokolov et al., 2018), an earth system model of intermediate complexity, and Economic Projection and Policy Analysis (EPPA) model (Paltsev et al., 2005a; Chen et al., 2016) to produce internally consistent projections of global climate, policy, and economic activity. The MESM

includes atmosphere, ocean, and land components to represent Earth system processes and interactions (Sokolov et al., 2018). The EPPA model is a computable general equilibrium model of the world economy that projects economic activity and emissions of climate forcers under policy constraints. A feature of the MIT IGSM-CAM framework is the ability to alter climate model response by modifying climate sensitivity with a cloud radiative adjustment method (Sokolov and Monier, 2012) and to conduct simulations under different representations of internal variability generated by perturbing initial atmospheric, land, and ocean conditions in the MESM. MIT IGSM-CAM simulations have been shown to realistically reproduce observations of the historical and present-day climate system (Monier et al., 2013; Sokolov et al., 2018). The framework has been used to project climate change impacts on multiple sectors, including public health, energy use, and infrastructure (Waldhoff et al., 2015). Here, three-dimensional meteorological fields projected by the MIT IGSM-CAM framework are used to simulate global atmospheric chemistry with the Community Atmosphere Model with Chemistry (CAM-Chem) at 1.9° × 2.5° horizontal resolution. Interactive chemistry and two-way feedbacks between the MIT IGSM-CAM and CAM-Chem fields are not modeled. By using one-way coupling and relying on an Earth system model of intermediate complexity with high computational efficiency, the framework allows the simulations required by the study's uncertainty analysis, thousands of years of modeled global air quality, to be completed. Detailed descriptions of the MIT IGSM-CAM modeling framework and approach are included in Monier et al. (2013) and (2018).

Global atmospheric composition is modeled with CAM-Chem (version 1.1.2) (Lamarque et al., 2012), driven by MIT IGSM-CAM meteorological fields. In the configuration used, CAM-Chem simulates atmospheric concentrations of over 100 gas phase and aerosol species from surface level to the lower stratosphere. For this study's projections of climate impacts on $PM_{2.5}$, we consider changes in ground-level concentrations of sulfate (SO_4), ammonium nitrate (NH_4NO_3), organic aerosol (OA), and black carbon (BC), and estimate $PM_{2.5}$ mass following Val Martin et al., (2014). Secondary organic aerosol (SOA) production is simulated with a two-product scheme linking its formation to oxidation of non-methane hydrocarbons (Lamarque et al., 2012). The simulations use an optimized dry deposition scheme (Val Martin et al., 2014) and model the effect of temperature on biogenic isoprene and monoterpene emissions based on the Model of Emissions and Aerosols (Lamarque et al., 2012). The response of other internal

emissions sources to climate change, including wildfires and dust emissions, is not included in the projections. CAM-Chem predictions of O_3 and $PM_{2.5}$ concentrations have been evaluated against observations, showing the model is able to generally reproduce the distribution of ground-level O_3 and aerosol species over the U.S. (Lamarque et al., 2012; Brown-Steiner et al., 2015; Tilmes et al., 2015). When CAM-Chem is coupled to the MIT IGSM-CAM, the frameworks' performance for historical O_3 compared against monitoring site measurements is similar to that reported in CAM-Chem model evaluations (Lamarque et al., 2012), with the simulations reproducing observed spatiotemporal patterns but exhibiting a positive bias against observations, a typical characteristic of chemistry-climate models (Turnock et al., 2020). The framework's predictions of $PM_{2.5}$ reproduce average annual $PM_{2.5}$ concentrations observed at over 400 U.S. EPA monitoring sites during the historical period, although the comparisons indicate a positive bias for SO_4 and a negative bias for EC and OA in the modeled results (Pienkosz et al., 2019). In the CAM-Chem simulations, emissions of air pollutants and precursors are based on the Precursors of Ozone and their Effects in the Troposphere inventory (Lamarque et al., 2012). To isolate the effect of climate on air quality, anthropogenic emissions are fixed at start-of-century levels (2000) in all atmospheric chemistry simulations. Additional description and discussion of the CAM-Chem framework and simulations are included in Garcia-Menendez et al. (2015, 2017), and Pienkosz et al. (2019).

4.2.2 Simulation ensemble of climate change impacts on U.S. air quality

The study's modeling framework was used to generate a simulation ensemble of 21st century climate change impacts on air quality with multiple realizations of internal variability, climate sensitivity, and climate policy and is illustrated in Fig. C.1. Three scenarios of economic activity, climate forcer emissions, and climate change are used to investigate emissions scenario uncertainty. The scenarios, developed for U.S. EPA's Climate Change Impacts and Risk Analysis (CIRA) Program (U.S. EPA, 2017), include a reference scenario (REF) with no global climate policy, unconstrained greenhouse gas emissions, and a total radiative forcing of 10 W/m^2 at the end of the century, and two stabilization scenarios that constrain total radiative forcing in 2100 to 4.5 W/m^2 (P4.5) and 3.7 W/m^2 (P3.7). The REF scenario is comparable to the Representative Concentration Pathway (RCP) 8.5, with atmospheric CO_2 reaching approximately 830 ppm by 2100,

compared with over 900 ppm under RCP 8.5 (Moss et al., 2010). Higher radiative forcing in REF is primarily due to larger anthropogenic and biogenic CH_4 emissions (Paltsev et al., 2015). Climate stabilization scenarios P4.5 and P3.7 lead to end-of-century CO_2 concentrations between those of RCP4.5 and RCP2.6 (Moss et al., 2010), with the implementation of a global tax on carbon emissions constraining them to lower than 500 ppm. Scenario likelihood is not considered in uncertainty estimates. The scenarios cover a broad range of emissions and radiative forcing levels, from very high GHG emissions to ambitious emissions reductions. While recent analyses suggest that high CO_2 emissions scenarios (e.g., RCP 8.5 and Socio-economic Pathway SSP5-8.5) have a low likelihood (Peters and Hausfather, 2020), we include the REF scenario as an informative upper limit scenario without climate change mitigation, in line with the Intergovernmental Panel on Climate Change's (IPCC's) Sixth Assessment Report (AR6) which concludes that such future concentration levels cannot be disregarded (Arias et al., 2021).

To investigate the influence of climate model response in projections of climate change impacts on air quality, simulations are conducted with three climate sensitivities corresponding to the IPCC AR6's best estimate, 3.0 °C (CS30), and values at the lower and upper end of the assessment's very likely range, 2.0 °C (CS20) and 4.5 °C (CS45) (Arias et al., 2021). Under the REF scenario and a 3.0 °C climate sensitivity, U.S. average temperature is projected to increase by 5.4 °C relative to start-of-century, within the National Climate Assessment's projected range for the RCP8.5 scenario (Vose et al., 2017). Temperature changes projected in our ensemble under each climate sensitivity and policy scenario considered are included in Table C.2. While different climate sensitivities among ensemble members are used to represent varying climate model response, they do not fully capture model response uncertainty. Meteorological variables influencing air pollution may not scale directly with climate sensitivity, and multi-model ensembles are better suited to capture differing responses to forcings. This analysis examining climate sensitivity, however, is the first to specifically explore a major component of structural uncertainty in projections of climate change impacts on air quality. To represent a range of internal variability, five realizations of internal variability are included in the ensemble, each generated by applying initial conditions in the atmosphere and land components in the MIT IGSM-CAM simulations taken from a random year of a pre-industrial control simulation (Monier et al., 2013). Detailed descriptions of the scenarios'

economic, emissions, and climate projections are presented in Monier et al. (2015) and Paltsev et al. (2015).

Atmospheric chemistry is modeled with CAM-Chem for 30-year periods at the start (1981-2010), middle (2036-2065), and end (2086-2115) of the 21st century, under each policy scenario, climate sensitivity, and realization of internal variability. The resulting ensemble includes 150 annual simulations of U.S. air quality for each of the three policy scenarios, climate sensitivities, and time-periods considered. In total, over 2,800 annual simulations of atmospheric chemistry were conducted for this analysis. Climate change impacts on O_3 and $PM_{2.5}$ pollution are estimated as the difference between concentrations simulated with REF at the start of the century and the future periods simulation with the corresponding climate sensitivity. Population-weighted estimates are based on U.S. population distribution at the start of the century using the Gridded Population of the World, Version 3 dataset (CIESIN, 2005). Regional averages are based on the regions defined in the U.S. National Climate Assessment (U.S. Global Change Research Program, 2018).

4.2.3 Uncertainty estimates

We weigh climate-related uncertainties in projections of climate change impacts on U.S. air quality by applying the methodology of Monier et al. (2015) to the CAM-Chem simulation ensemble. Here, we define scenario, climate sensitivity, and internal variability uncertainty in air quality projections as the range in simulated pollutant concentration impacts resulting from different GHG emissions scenarios, climate sensitivities, and climate model initializations, respectively. Uncertainties are determined at the middle and end of the 21st century. We begin by computing the climate-induced change (the climate penalty) for each annual projection in the ensemble as the change in annual average ground-level daily maximum 8-hour average O_3 or daily average $PM_{2.5}$, relative to start-of-century concentrations of the corresponding annual simulation in the matching initial condition and climate sensitivity ensemble member.

To determine scenario uncertainty, we first control for internal variability by averaging across 30-year periods and initial condition ensemble members under each emissions scenario and climate sensitivity. This results in projected mean changes in air quality at the middle and end of the 21st century for each emissions scenario and each climate sensitivity. Next, we find the range between the emissions scenarios resulting in

the largest and smallest air quality impact at each climate sensitivity. Finally, we calculate the mean of these ranges to control for climate sensitivity. The resulting range is defined as the uncertainty in the climate impact on air quality due to emissions scenario.

For climate sensitivity uncertainty we repeat the process, instead calculating the range between climate sensitivities resulting in the largest and smallest impacts under each emissions scenario, and then averaging these ranges.

To compute uncertainty due to internal variability, we first average across each 30-year period in the ensemble, each of which is associated with a different climate model initialization. We then calculate the difference between the initial condition ensemble member means with the largest and smallest air quality impacts under each policy scenario and climate sensitivity. The average of these ranges represents a range of internal variability uncertainty in 30-year mean projections. In addition to estimating internal variability uncertainty for 30-year simulations, we explore internal variability for 15-, 10-, 5-, and 1-year time slices. We measure these uncertainties by calculating rolling 15-, 10-, 5-, or 1-year averages within the 30-year period of each initial condition ensemble member, resulting in a series of average air quality impact values for each time slice length, under each climate sensitivity and emissions scenario. We group rolling averages by climate sensitivity and policy scenario, and then calculate the internal variability uncertainty as the average of the ranges within each group.

4.3 Results

4.3.1 Uncertainty in climate change impacts on U.S. air quality

Climate change is projected to negatively impact ground-level O_3 and $PM_{2.5}$ pollution in the U.S. Climate stabilization policies targeting total radiative forcings of 4.5 and 3.7 W m^{-2} (P4.5 and P3.7) by 2100 under a climate sensitivity of 3.0 °C (mean surface temperature change per doubling of atmospheric CO_2) limit projected population-weighted annual impacts at the end of the century to 0.4 and 0.3 μg m^{-3} for $PM_{2.5}$, and 0.3 and 0.6 ppb daily maximum 8-hour O_3 (all O_3 concentrations are reported here as daily maximum 8-hour average). However, under a no-policy reference greenhouse gas emissions scenario (REF; see Methods), a less likely but informative scenario, mean climate penalties on annual U.S. population-weighted $PM_{2.5}$ and O_3 are projected to be 0.6 μg m^{-3} and 0.8 ppb at mid-century, and 1.5 μg m^{-3} and 3.2 ppb by the end of the century.

For O_3-season (May–Sept.) concentrations, the projected mean climate penalty under the REF scenario at century end is 8.1 ppb. Climate change impacts on O_3 levels projected under a 3 °C climate sensitivity agree with previously reported climate penalty estimates over the Northeast, Midwest, Southeast, and California, but show differences in other areas, in particular over the Northwest (Nolte et al., 2021; Schnell et al., 2016; Fiore et al., 2015). There is less consistency among reported simulations of climate change impacts on $PM_{2.5}$ pollution. The projections used here agree with those from specific studies over particular areas (e.g., the Northeast and Southeast), but can differ with others depending on $PM_{2.5}$ component or region (Xu and Lamarque, 2018; Nolte et al., 2018b; Westervelt et al., 2016). Comparisons between this ensemble's projections and others are further discussed in Pienkosz et al., (2019) and Garcia-Menendez et al. (2017). Despite clear mean penalties by the end of the century, a closer look at the ensemble of over 2,800 years of simulated air quality (Fig. 4.1a-b) reveals substantial uncertainty in these estimates of climate change impacts on pollutant concentrations. Internal variability can lead to projections of climate-induced changes on annual U.S. population-weighted $PM_{2.5}$ and O_3 that range from -0.2 to 3.2 μg m^{-3} and -1.5 to 8.0 ppb, respectively, and simulations under the different climate sensitivities considered (2.0, 3.0, and 4.5 °C) differ by up to 1.5 μg m^{-3} and 4.1 ppb in the mean climate impacts projected for a given emissions scenario.

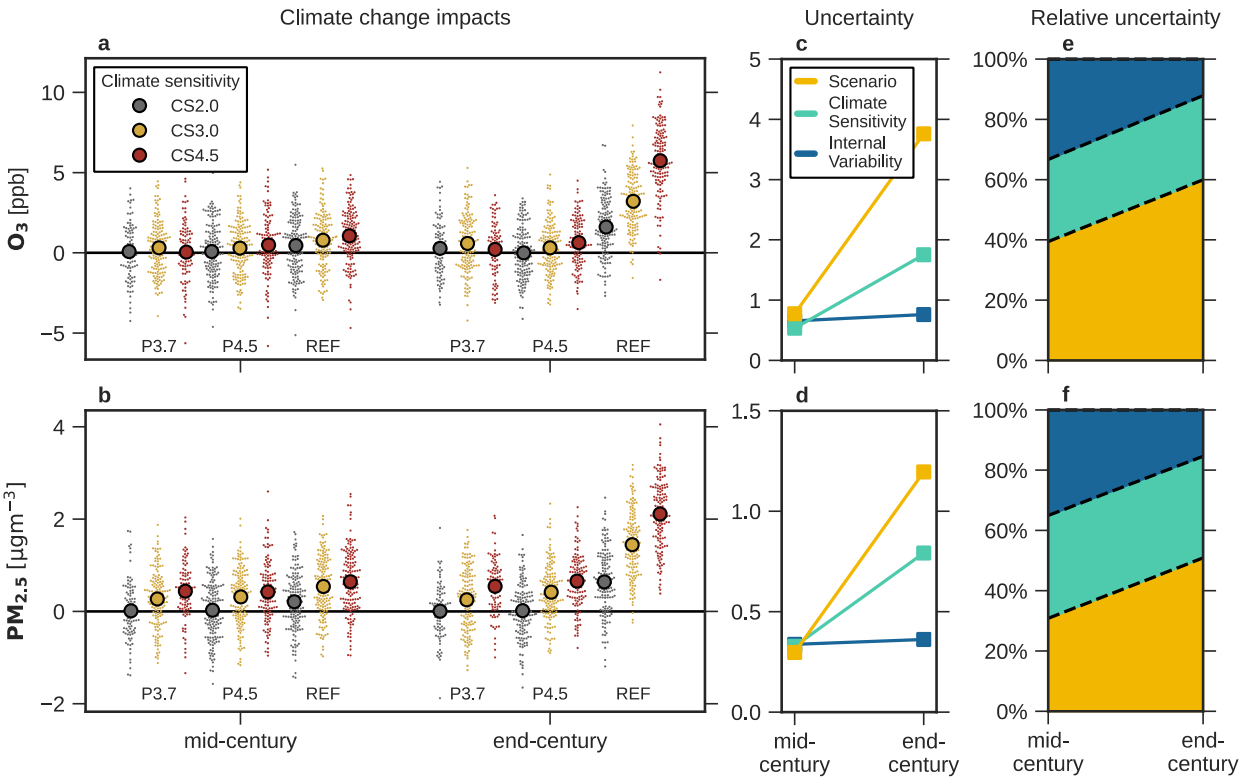

Figure 4.1: Climate change impacts on O_3 and $PM_{2.5}$ pollution in the U.S. and their associated uncertainties. Projected anomalies in annual average population-weighted O_3 (**a**) and $PM_{2.5}$ (**b**) ground-level annual concentrations, relative to historical simulation. Ensemble simulations include 5 initializations and 30-year periods covering 1981-2010, 2036-2065, and 2086-2115 to characterize the beginning, middle, and end of the 21st century. Large dots are ensemble means for each period and each small dot is a single-year average in the ensemble, grouped by climate sensitivity and GHG emissions scenario. Uncertainty in projected climate impacts on O_3 (**c**) and $PM_{2.5}$ (**d**) at middle and end of the century associated with GHG emissions scenario, climate sensitivity, and internal variability. Weights of uncertainties in projected climate impacts on O_3 (**e**) and $PM_{2.5}$ (**f**), relative to the sum of uncertainties from the three sources considered.

Uncertainties in projected climate change impacts on U.S. air quality arising from GHG emissions scenario, climate sensitivity, and internal variability are compared in Figure 4.1c-f. Here, the uncertainty associated with each source is defined as the average range across 30-year mean climate change impacts projected under the different combinations of emissions scenario, climate sensitivity, and internal variability included in the ensemble (see Fig. C.1 and Methods for a detailed description). At mid-century, the uncertainties in projected climate impacts associated with each source are similar for annual population-weighted concentrations of both O_3 (0.5-0.8 ppb) and $PM_{2.5}$ (0.3-0.4 μg m^{-3}). These uncertainties are significant relative to mid-century ensemble-mean

projections of climate penalties on U.S. air pollution. By the end of the century, the uncertainty associated with GHG emissions scenario is clearly largest, 30% higher than climate sensitivity uncertainty in $PM_{2.5}$ impacts and over twice as high as climate sensitivity uncertainty in O_3 impacts, while uncertainty due to internal variability is the smallest. However, even at the end of the century all sources of climate-related uncertainty remain significant. Internal variability remains consistent throughout the simulations. It is comparable with the uncertainties associated with GHG emissions scenario and climate sensitivity at mid-century and is approximately 20 to 40% of that imposed by emissions scenario at century end. Climate sensitivity uncertainty grows by the end of the century, becoming distinctly larger than internal variability. The relative weights of these climate-related uncertainties reflect those reported for projections of surface temperature and precipitation (Hawkins and Sutton, 2009; Lehner et al., 2020; Monier et al., 2015), with internal variability as the major source of uncertainty early in the 21st century and GHG emissions scenario becoming the largest towards the end of the century. These estimates also highlight the challenges imposed by internal variability and uncertain climate sensitivity on projecting a climate response in air quality to variations in greenhouse gas emissions for time horizons extending over few decades.

4.3.2 Uncertainty in projections of regional air quality impacts

Uncertainties in climate change impacts on air pollution across U.S. National Climate Assessment regions (U.S. Global Change Research Program, 2018) are compared in Figure 4.2. The largest uncertainties in projections of regional climate-induced changes to O_3-season and annual $PM_{2.5}$ concentrations are in the Northeast, Midwest, and Southeast, the regions anticipated to experience the largest climate penalties on air quality. Uncertainties associated with GHG emissions scenario and climate sensitivity are 2 to 8 times larger in the Northeast than the Western U.S. Internal variability is largest in the Midwest and Northeast, and lowest in the Southwest and Northwest. Although the relative weights of each uncertainty vary regionally, and internal variability has a larger influence on regional penalties, they are consistent with uncertainties for national population-weighted concentrations. At mid-century, differences between uncertainties associated with each source are small and, for most regions, internal variability is the largest uncertainty in projections of climate change impacts on O_3 and $PM_{2.5}$. Larger uncertainties related to GHG emissions scenario and climate sensitivity emerge by the

end of the century. In all regions, uncertainty arising from GHG emissions is the largest in 2100. Across the simulations, the influence of climate sensitivity grows in response to GHG emissions, while the level of internal variability remains relatively consistent, rendering it the smallest uncertainty source at century end.

Figure 4.3 shows the progression of climate-related uncertainty in projections of U.S. air quality from middle to end of the 21st century. While the uncertainties in climate change impacts on O_3 and $PM_{2.5}$ associated with different sources are comparable at mid-century, by end-century the spatial distributions of emissions scenario and climate sensitivity uncertainty resemble the patterns of climate-induced changes in pollutant concentrations. Although the projected climate impacts on $PM_{2.5}$ are largely driven by enhanced SO_2 oxidation and nitrate partitioning to the gas phase under higher temperatures, as discussed in Pienkosz et al., (2019) and Garcia-Menendez et al. (2015), the spatial patterns of $PM_{2.5}$ impacts and their uncertainty differ from those of climate impacts on surface temperature. Uncertainty in $PM_{2.5}$ impacts is lowest in the Western U.S., where uncertainty in the temperature projections, estimated in Monier et al. (2015), is greatest. High uncertainty in the Eastern U.S. is more consistent with reported uncertainty in projections of precipitation under climate change (Monier et al. 2015). The simulated climate impacts on O_3, examined in Garcia-Menendez et al. (2017, 2015), are driven by competing positive associations with temperature and biogenic emissions and negative associations with humidity and wind speed. The largest projected impacts and uncertainties in climate-induced changes in U.S. O_3 concentrations are in the East and Midwest, while temperature is projected to increase most and be most uncertain in the West and Great Plains regions (Monier et al., 2015). While uncertainty in future temperature and other weather variables propagates to projections of air quality, the complex relationships between pollutant concentrations, meteorological drivers, and emissions of pollutants and precursors make it difficult to infer the impacts and uncertainty in climate change impacts on air pollution based on projections of meteorology alone.

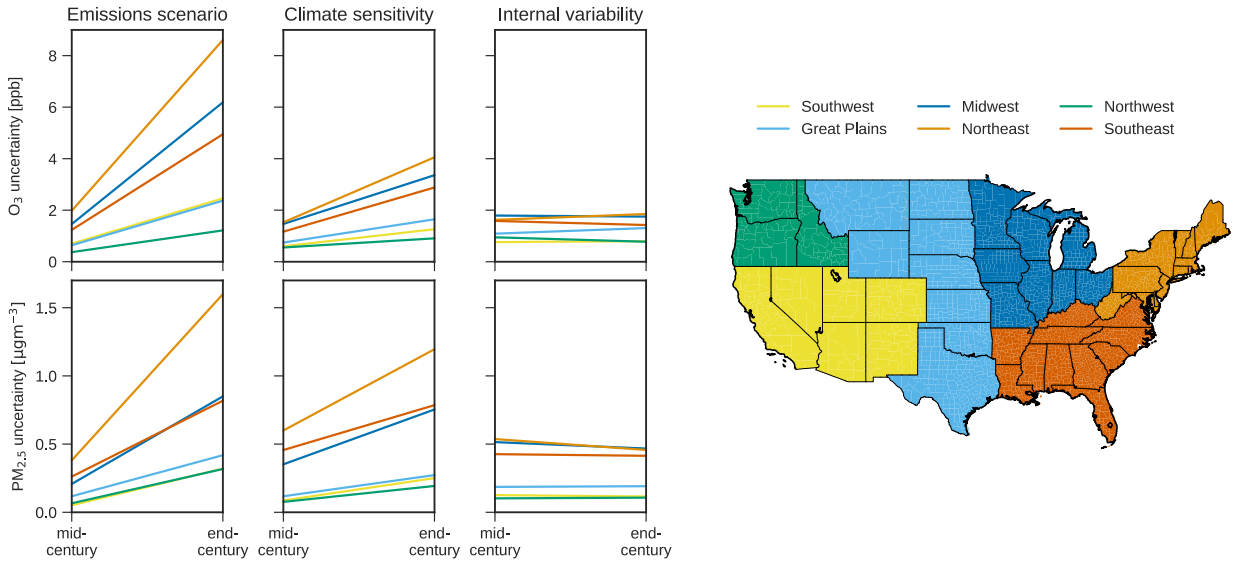

Figure 4.2: Uncertainty in climate change impacts on regional air pollution. Uncertainties in U.S. regional projections of climate-induced impacts on regional May-Sept. (O_3-season) O_3 and annual $PM_{2.5}$ ground-level concentrations associated with GHG emissions scenario, climate sensitivity, and internal variability at mid-century and end-century.

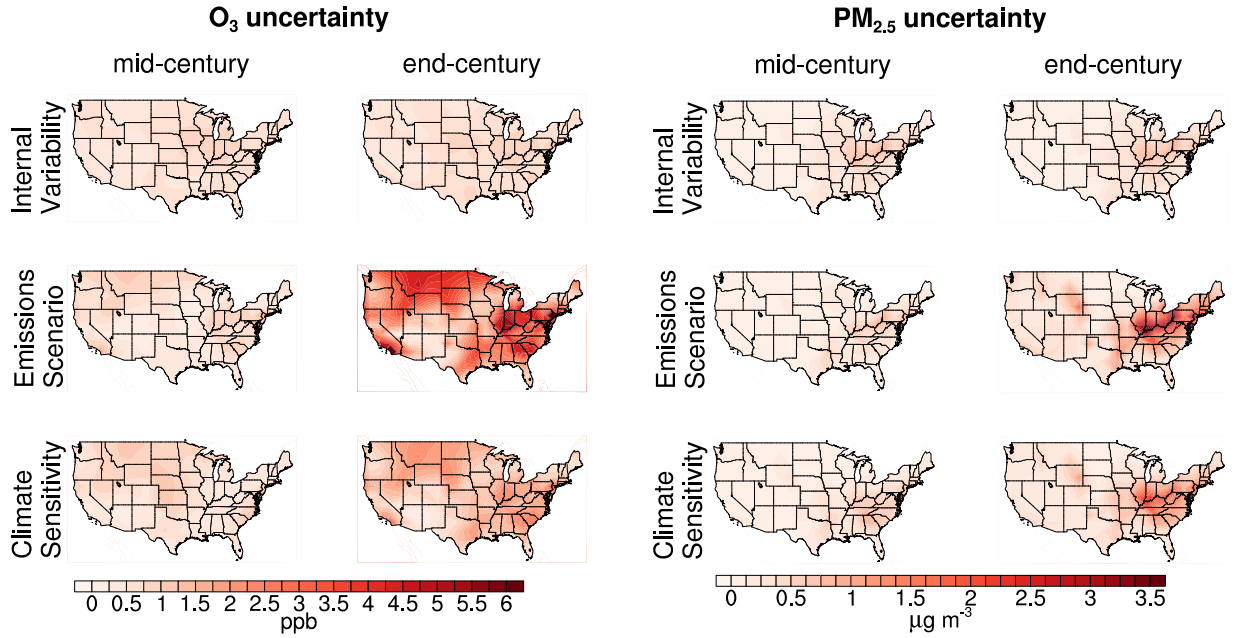

Figure 4.3: Varying uncertainty in climate change impacts on U.S. O_3 and $PM_{2.5}$ pollution across the 21st century. Uncertainties in projections of climate-induced impacts on annual O_3 and $PM_{2.5}$ ground-level concentrations over the U.S. associated with GHG emissions scenario, climate sensitivity, and internal variability at mid-century and end-century.

4.3.3 Influence of internal variability on projected air quality impacts

Internal variability, an irreducible uncertainty, propagates from climate simulations to projections of air quality. Most studies investigating the impact of climate change on air quality have relied on simulating time slices of 10 or fewer years to account for internal variability when comparing present-day and future air quality (Garcia-Menendez et al., 2017). However, sub-decadal simulations have proven insufficient to filter out the noise of internal variability in projections of climate impacts on O_3 and $PM_{2.5}$ and the associated health impacts (Garcia-Menendez et al., 2017; Pienkosz et al., 2019; Saari et al., 2019; Barnes et al., 2016). Although large ensembles offer an effective approach to capture internal variability in climate projections (Deser et al., 2020), few studies have used initial condition ensembles to simulate climate change impacts on air quality (Barnes et al., 2016; Westervelt et al., 2016; Xu and Lamarque, 2018; Fiore et al., 2022). Here, we rely on multiple model initializations and 30-year periods to represent a range of internal variability and weigh its influence on projected climate change impacts on pollutant concentrations. Fig. 4.4 shows that uncertainty in projected impacts on mean O_3 and $PM_{2.5}$ concentrations for shorter time slices is significantly higher. The uncertainty associated with internal variability is approximately 2.5 times larger when projecting the end-century 15-year mean impact of climate change on population-weighted O_3 and $PM_{2.5}$ concentrations, relative to 30-year mean impacts. Projecting 5-year mean or single-year impacts leads to internal variability exceeding the uncertainties associated with emissions scenario or climate sensitivity throughout the entire 21st century. In projections of climate impacts on air quality, sufficient timescales or initial condition ensembles are needed to avoid internal variability exceeding other climate-related uncertainties.

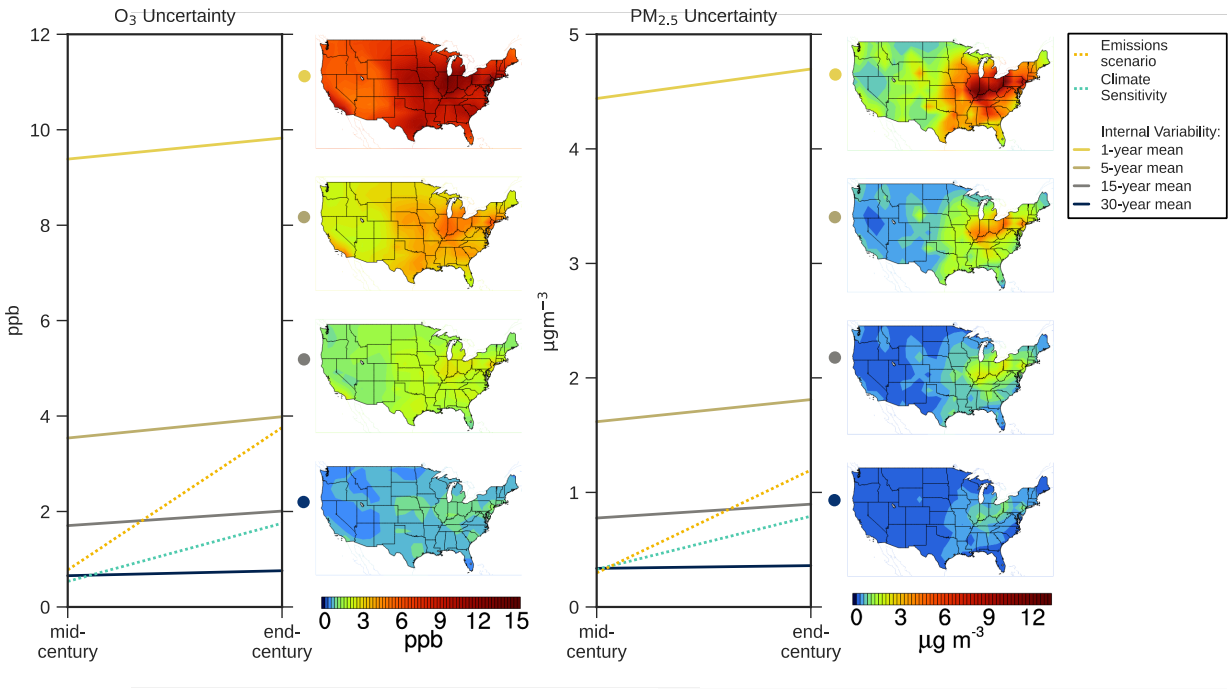

Figure 4.4: Internal variability in climate change impacts on air pollution at varying timescales. Uncertainty associated with internal variability in projected mean climate impacts on annual O_3 and $PM_{2.5}$ ground-level concentrations for periods ranging from 1 to 30 years. Solid colored lines indicate uncertainty in impacts on U.S. population-weighted concentrations at mid-century and end-century. Dashed lines indicate GHG emissions scenario and climate sensitivity uncertainties. Maps show uncertainty associated with internal variability at end of 21st century with the corresponding averaging period indicated by adjacent circles.

4.3.4 Influence of climate sensitivity on projected air quality impacts

In this simulation ensemble, the uncertainty in projected climate change impacts on U.S. air quality associated with climate sensitivity is comparable to that stemming from emissions scenario and internal variability at mid-century and remains substantial throughout the 21st century. Figure 4.5 shows projected impacts at century end under the REF scenario and based on a climate sensitivity of 3.0 °C, the IPCC's best estimate (Arias et al., 2021), and relative differences at lower and higher sensitivities. Climate sensitivity near the high end of the IPCC's "very likely" range, 4.5 °C, intensifies projected climate change impacts on annual O_3 and $PM_{2.5}$ concentrations by as much as 6 ppb and 2 μg/m^3, respectively, while climate sensitivity of 2.0 °C, at the low end of the "very likely" range, diminishes them by more than 4 ppb and 2 μg/m^3, relative to the best-estimate sensitivity. Relative to a 3.0 °C climate sensitivity, projected REF-scenario population-weighted

climate penalties on annual O_3 and $PM_{2.5}$ concentrations at the end of the century increase by 2.5 ppb (78%) and 0.7 μg/m³ (46%) under a 4.5 °C climate sensitivity, and decrease by 1.6 ppb (49%) and 0.8 μg/m³ (55%) under a 2.0 °C climate sensitivity. Under the climate stabilization scenarios considered, differences in projected impacts across the range of climate sensitivities are significantly lower. Ensemble-average climate change impacts under all GHG emissions scenarios are included in Table C.1.

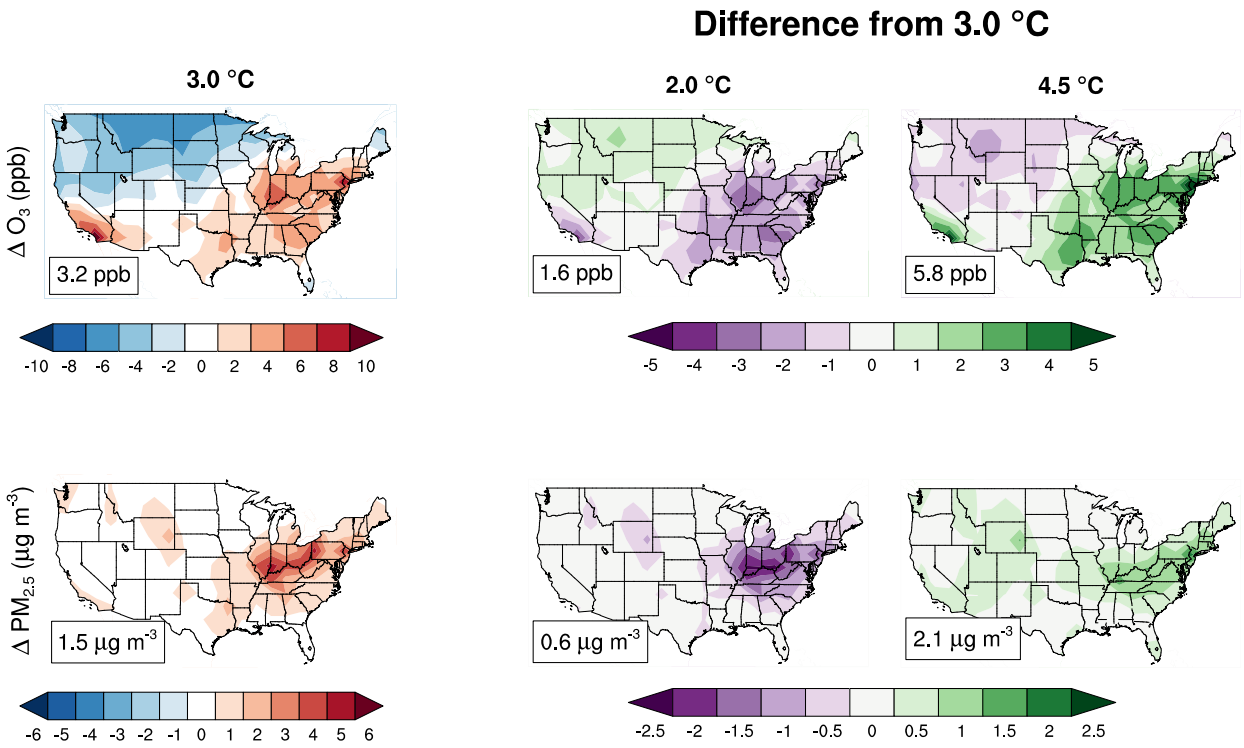

Figure 4.5: Climate change impacts on O_3 and $PM_{2.5}$ pollution under varying climate sensitivity. Climate change impacts on annual O_3 and $PM_{2.5}$ ground-level concentrations projected under climate sensitivities of 2.0, 3.0, and 4.5 °C at end-century. Left maps show ensemble-mean impacts under the REF scenario and a 3.0 °C climate sensitivity. Maps on the right side of the plot show differences in projected impacts, relative to 3.0 °C climate sensitivity, under climate sensitivities of 2.0 and 4.5 °C. Boxes indicate population-weighted mean climate change impacts under each sensitivity.

Uncertainties in climate impacts on air quality associated with emissions scenario, internal variability, and climate sensitivity are linked. Climate driven changes to meteorology that impact O_3 and $PM_{2.5}$ pollution depend on GHG emission levels, while the intensity of these impacts is determined in part by the earth system's climate sensitivity. Internal variability adds noise to these forced signals in air quality. In this simulation ensemble, for example, we find that stronger climate sensitivity leads not only

to a higher mean O_3 penalty over the U.S. by the end of the century, but also larger interannual spread in projected concentrations (Figure 4.6). The increased level of internal variability at higher climate sensitivity suggests greater risks of extreme air pollution at the high ends of the distribution. For example, while the mean REF-scenario climate penalty on annual population-weighted O_3 concentration projected at century end under the highest climate sensitivity (4.5 °C) is approximately 4 ppb larger than that projected under the lowest climate sensitivity (2.0 °C), the 95th percentile year simulated among all initial conditions and 30-year periods is 5 ppb larger between the highest and lowest climate sensitivities. In contrast, internal variability in climate-induced changes to $PM_{2.5}$ remains consistent across the climate sensitivities explored. While several studies have reported a climate penalty on the upper tails of projected future air pollution distributions (Fiore et al., 2022; Rieder et al., 2015; Hong et al., 2019; Nolte et al., 2018b, 2021), further work investigating high pollution events can shed additional light on the interconnections between different sources of climate uncertainty in simulations of air quality impacts.

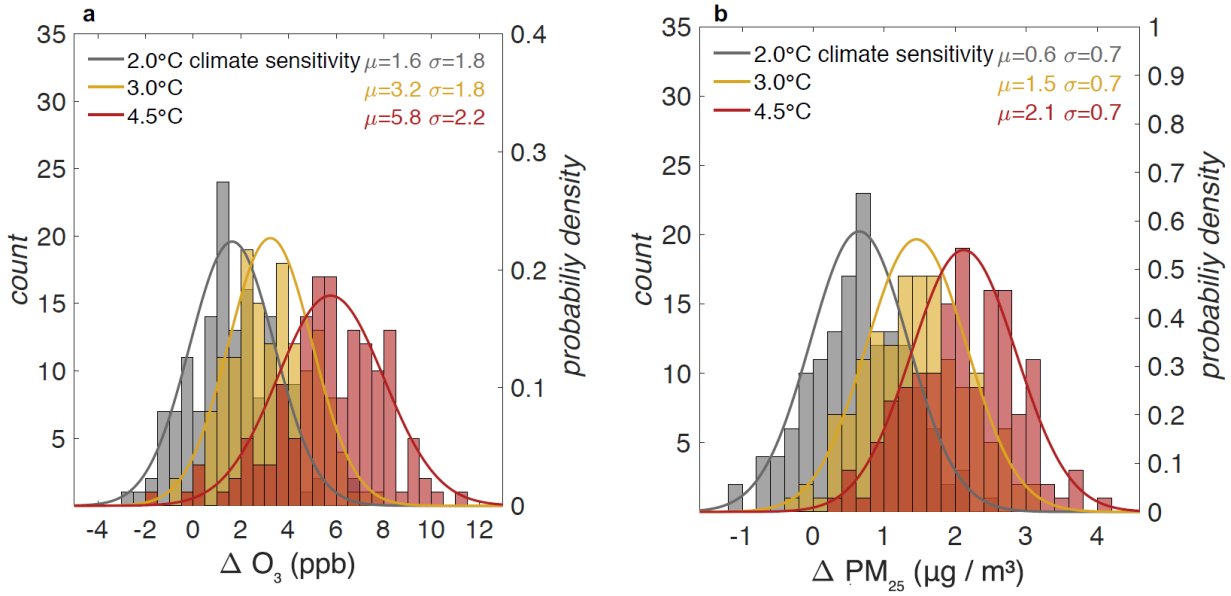

Figure 4.6: Interannual variability in climate change impacts on U.S. O_3 and $PM_{2.5}$ pollution at higher climate sensitivity. Distribution of single-year projected climate impacts on annual population-weighted O_3 (**a**) and $PM_{2.5}$ (**b**) ground-level concentrations at end-century, relative to the historical simulation, under the REF scenario and climate sensitivities of 2.0, 3.0, and 4.5 °C. Fitted normal distributions, means (μ), and standard deviations (σ), are shown for each climate sensitivity considered.

4.4 Discussion

4.4.1 Outlook for impacts assessment in the face of uncertainty

Extensive research has projected significant impacts of climate change on O_3 and particulate matter pollution (Fiore et al., 2015). Here, we show that important uncertainties in these projections arise from recognized sources of uncertainty in climate simulations. For example, in our simulation ensemble we find that internal variability has the potential to far exceed greenhouse gas emissions scenario and climate sensitivity uncertainties throughout the 21st century if adequate timescales are not considered. We also observe that when considering uncertainty in climate sensitivity, a previously unexplored risk of climate-induced extreme O_3 pollution is revealed. Further propagation of these uncertainties to projections of pollution-related health damages and costs has implications for climate policy assessments (Saari et al., 2019). More complete characterizations of uncertainty in climate-induced impacts on air pollution can aid decision-making.

Due to the high computational cost of atmospheric chemistry simulations, only 3 emissions scenarios and 5 model initializations, constrained by limited resolution and representations of atmospheric processes, are included in this analysis. Recent work has suggested that more initial conditions may be needed to fully capture internal climate variability (Deser et al., 2020). Our finding that emissions scenario uncertainty is the largest climate-related uncertainty at the end of the century depends on the scenarios considered. While the analysis is based on an ensemble covering a wide range of climate policy and GHG emissions futures, including additional scenarios or accounting for scenario likelihood could reduce uncertainty in estimates of climate impacts on air pollution. By including a high-emissions REF scenario, projections with a large climate penalty on air quality in the ensemble strongly contribute to this uncertainty. The ensemble explored here is based on a single modeling framework and a cloud radiative adjustment method is used to project impacts under different climate sensitivities (see Methods). However, structural uncertainty can make the framework's response to GHG concentrations differ from that in other models.

Models with more detailed chemical mechanisms may better capture atmospheric composition, but their computational costs currently render them challenging for the large ensembles required to assess uncertainty in climate change impacts. When coupled

climate-chemistry simulations are not feasible, climate model ensembles offer opportunities to characterize variability in meteorological conditions conducive to air pollution, such as stagnation (Garrido-Perez et al., 2022; Callahan and Mankin, 2020; Horton et al., 2014; Hou and Wu, 2016). Additionally, the influence of air pollutant emissions, and thus atmospheric composition, on climate-induced impacts on air quality and their uncertainties remains largely unexplored. We rely on emissions at the beginning of the century to isolate climate change impacts and the associated uncertainty estimates are based on these emissions levels. However, pollutant and precursor emissions will change continuously during the 21st century (Rao et al., 2017), and ensembles like the one used here do not model the impacts of variations in air pollutant emissions. Analyses that consider changing air pollutant emissions (e.g., Barnes et al., 2016; Rieder et al., 2015) can capture the effects of both climate and emissions, although separating these effects requires simulations in which radiative forcing or air pollutant emissions are fixed, and inconsistencies arising from the dual role of some species as both conventional air pollutants and climate forcers are addressed. Despite these limitations, the ensemble examined here, encompassing over 2,800 annual simulations of atmospheric chemistry, constitutes one the of the largest modeling efforts to date investigating the impacts of climate on air pollution and allows for an unprecedented view into how three primary sources of uncertainty in climate modeling propagate to projections of air quality under climate change.

Considering its adverse effects on public health, including disease and premature mortality, a climate-induced deterioration of air quality would have major economic consequences. Larger uncertainty will likely increase the costs of strategies to mitigate or adapt to these impacts (Hawkins and Sutton, 2009). The potential to reduce uncertainty in projections of air quality under climate change differs among uncertainty sources. Reducing uncertainty arising from emissions of climate forcers requires a more certain climate policy outlook. Uncertainty associated with climate sensitivity can be diminished as representations of the earth system in models are improved. Internal variability uncertainty is not expected to decrease, but its effects on projections can be captured by large initial-condition simulation ensembles. Additionally, future air quality depends on uncertain emissions of air pollutants and precursors. In light of this complexity, large ensembles offer an approach to elucidate the impacts of climate change and anthropogenic emissions on air quality. Multi-model and large initial-condition

ensembles, such as the Coupled Model Intercomparison Project and single model initial-condition large ensemble (SMILE) initiatives, have provided important insights into climate projections (Deser et al., 2020). Climate impacts on air quality have not yet been explored with ensembles at this scale. Although the analysis presented here is based on a smaller-scale simulation ensemble, it shows the opportunities large ensembles generated with fully coupled climate-chemistry models would offer to investigate uncertainty in the impacts of both climate change and anthropogenic precursor emissions on atmospheric composition. Such an advance would improve co-benefits assessments and inform climate policy design in light of the uncertainties presented here.

4.5 Acknowledgments

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Chapter 5: Changes in frequency and magnitude of future extreme ozone events under natural variability and uncertain climate sensitivity

Abstract

Climate change is projected to adversely affect ground-level ozone over many populated regions, with potentially larger impacts on extreme concentrations. Intensified extreme ozone pollution represents a risk to human health, but projecting these changes is complicated by internal variability in climate-chemistry simulations. In addition, model projections suggests that dispersion in future ozone concentration distributions may further increase under higher climate sensitivity. Here, we leverage a climate-chemistry modeling ensemble with multiple initial condition and climate sensitivity members to explore the connections among internal variability, climate sensitivity, and climate-induced changes to ozone pollution over the Unites States. To assess pollution extremes, we first bias correct historical simulations based on observed ozone concentrations. We then apply the bias correction to projections of climate impacts on ozone under various greenhouse gas emissions pathways and multiple climate sensitivities at mid- and end-century. Our results show an increased risk for extreme ozone in addition to a mean impact under future climate change. We further show that internal variability can lead to over- or under-prediction of extremes if not accounted for, and that climate sensitivity exerts an important control on projections of extreme pollution. To robustly project ozone extremes, adequately considering these sources of climate uncertainty in simulations of air quality under climate change is needed.

5.1 Introduction

Climate change is projected to impact future ozone (O_3) pollution through changes in meteorology and chemistry (Fiore et al., 2015; Jacob and Winner, 2009), with negative impacts on human health, the environment, and the economy (Fann et al., 2015, 2022; Garcia-Menendez et al., 2015; Nolte et al., 2018a). Climate change impacts are expected to lead to increases in the magnitude and frequency of extreme ozone events (Brown-

Steiner et al., 2015; Rieder et al., 2015; Schnell et al., 2016), as well as their co-occurrence with heat waves and extreme particulate matter pollution episodes (Fiore et al., 2022; Meehl et al., 2018; Shen et al., 2016). However, many studies investigating climate impacts on ozone primarily consider annual or seasonal means, potentially hiding changes to high O_3 pollution. By increasing the magnitude or frequency of the highest concentration O_3 days, climate change's impact could diminish gains achieved by emissions reductions in recent decades. Understanding this effect is a critical component of efforts to mitigate the impacts of O_3 on human health and the environment.

Assessing climate change impacts is complicated by uncertainties in projections due to internal variability, greenhouse gas (GHG) emissions scenario, and climate sensitivity (Hawkins and Sutton, 2009). Although previous studies indicate that anthropogenic climate change can increase pollution, a climate penalty on O_3, large uncertainties in the distribution and magnitude of these impacts remain (Fiore et al., 2012, 2015). Despite these uncertainties, climate policy has been shown to be a strong determinant of future impacts (Garcia-Menendez et al., 2015). Previous research characterized uncertainties in air quality impacts, noting that higher climate sensitivity may lead to a greater risk for future extreme air pollution (East et al., 2022a). However, few studies have projected ozone extremes while accounting for these uncertainties. Although some studies projecting climate driven changes to O_3 extremes have used multiple initializations to address internal variability (Fiore et al., 2022; Rieder et al., 2015), most rely on a single GHG emissions pathway (Meehl et al., 2018; Rieder et al., 2015), and the role of climate sensitivity uncertainty in projecting ozone extremes has not been quantified in isolation.

A challenge to examining modeled O_3 extremes is that climate-chemistry models exhibit a consistent high bias when compared to historical ozone observations (Brown-Steiner et al., 2015; Lamarque et al., 2012; Liu et al., 2022; Turnock et al., 2020). While analyses examining changes between historical and projected climate often account for the bias by only considering relative changes, concentration thresholds are important to assess air pollution extremes (Phalitnonkiat et al., 2016; Rieder et al., 2013). Correction methods have been applied to address model bias, but may only correct the bias in mean O_3 concentrations (Brown-Steiner et al., 2015), an approach which does not account for changes in O_3 distributions, or be applied over a limited geographic region (Rieder et al., 2015). To examine the impacts of climate change on extreme air pollution, approaches are

needed to bias correct model fields covering large geographic areas with heterogeneous statistical characteristics.

Here we analyze the impact of climate change on incidences of high ground-level O_3 concentration in the U.S. using an ensemble of simulations generated with a coupled model framework designed to capture main sources of uncertainty in climate projections. We test two bias correction approaches against U.S. ground-based observations and apply extreme value theory to calculate historical and projected probabilistic frequency and magnitude of extreme ozone pollution events. Leveraging the ensemble of over 2,800 annual climate and air quality simulations, we are able to assess the influence of uncertainty in climate sensitivity, greenhouse gas (GHG) emissions pathway, and internal variability within the projections. Our results show that the magnitude of O_3 pollution extremes in the U.S. is projected to increase more than mean ozone under high climate sensitivity and unconstrained GHG emissions, while the relative increases in extremes are approximately equal to or less than mean increases at lower climate sensitivities.

5.2 Methods

5.2.1 Modeling framework and simulation ensemble

To project the impacts of climate change on high ozone concentrations in the U.S., we rely on an ensemble of simulations generated with a climate and air quality modeling framework. In this framework, the Massachusetts Institute of Technology Integrated Global System Model (MIT IGSM) (Monier et al., 2013) is coupled to the National Center for Atmospheric Research Community Atmosphere Model (CAM) to simulate historical and future climate scenarios. The MIT IGSM couples a human activity model, the Economic Projection and Policy Analysis (EPPA) model (Chen et al., 2016; Paltsev et al., 2005), with an earth system model of intermediate complexity, the MIT Earth System Model (MESM) (Sokolov et al., 2018). EPPA is a computable general equilibrium model that simulates economic activity and GHG emissions according to policy constraints (Paltsev et al., 2005). GHG emissions and land use change from EPPA are used as inputs to the Earth system model, MESM, which simulates climate with a two-dimensional zonally averaged atmosphere. To generate 3-dimensional climate fields, MESM provides sea-surface temperatures, sea ice cover, GHG concentrations, and land use change to the

CAM (version 3), which simulates climate using coupled land and atmosphere components. Additionally, a feature of the MIT-IGSM CAM is the ability to adjust the model's climate sensitivity through a cloud radiative adjustment method (Sokolov and Monier, 2012). The MIT-IGSM CAM has been shown to realistically simulate historical climate system observations (Monier et al., 2013; Sokolov et al., 2018; Waldhoff et al., 2015) and U.S. temperatures in line with those reported by the National Climate Assessment (NCA) (Vose et al., 2017). The framework is described in detail in Monier et al. (2013) and Monier et al. (2018).

To project climate change impacts on air quality, meteorological fields from the MIT IGSM-CAM are used to simulate atmospheric chemistry with the CAM-Chem model (version 4) (Lamarque et al., 2012) using one-way coupling. CAM-Chem simulates over 100 gas and aerosol species at $1.9° \times 2.5°$ horizontal resolution with 26 vertical layers extending to the lower stratosphere. In the CAM-Chem simulations, anthropogenic emissions are from the Precursors of Ozone and their Effects in the Troposphere (POET) inventory (Lamarque et al., 2012) and are held constant at year-2000 levels to isolate the impact of climate on O_3. Biogenic emissions of isoprene and monoterpenes, simulated with the Model of Emissions of Gases and Aerosols from Nature (MEGAN) version 2 (Guenther et al., 2012) within CAM-Chem, include online feedbacks to changes in temperature. Lightning, soil, wetland, and wildfire emissions in the CAM-Chem simulations do not respond to variations in climate. The dry deposition scheme in CAM-Chem is optimized following Val Martin et al. (2014). CAM-Chem has been previously evaluated for simulations of U.S. air quality (Brown-Steiner et al., 2015; Lamarque et al., 2012; Tilmes et al., 2015) and used to project the potential impacts of climate on air quality and health (Brown-Steiner et al., 2018; East et al., 2022a; Garcia-Menendez et al., 2015, 2017; Pienkosz et al., 2019; Saari et al., 2019).

Future climate and air quality are simulated with CAM-Chem in 30-year time-slices centered on the beginning (1980-2010), middle (2036-2065), and end (2085-2115) of the 21st century. The simulation at the beginning of the century is referred to throughout the chapter as the historical simulation. Three future climate policy scenarios are modeled: emissions control policies P4.5 and P3.7, in which end-of-century radiative forcings reach 4.5 and 3.7 W/m^2, respectively, and a reference scenario (REF) in which no GHG emissions policy is implemented, representing an upper bound for potential climate change impacts. Climate policies take effect after the historical simulation. The

economic and emissions projections for each climate scenario are further described in (Monier et al., 2015; Paltsev et al., 2015). Under each emissions policy considered, air quality is modeled using climate sensitivities of 2.0, 3.0, and 4.5 °C global mean temperature increase per doubling of average atmospheric CO_2 concentration, representing the range of uncertainty in Earth's climate sensitivity according to IPCC AR6 (IPCC, 2021). Internal variability is represented by five unique climate model initializations under each emissions policy scenario and climate sensitivity. To represent internal variability with these initial condition ensemble members, different initial conditions in the atmosphere and land components of the MIT IGSM-CAM are drawn from a random year in a pre-industrial control simulation (Monier et al., 2013). In total, over 2,800 years of climate and air quality are simulated.

5.2.2 Bias correction

CAM-Chem simulations, like those from other global climate-chemistry models, exhibits a high bias in ground-level O_3 concentrations (Brown-Steiner et al., 2015; Lamarque et al., 2012; Liu et al., 2022; Turnock et al., 2020). We address CAM-Chem's high bias relative to historical daily maximum 8-hour (MDA8) O_3 observations by applying a quantile mapping (QM) bias correction against MDA8 O_3 observations from the U.S. Environmental Protection Agency's (EPA) Clean Air Status and Trends Network (CASTNET) (U.S. EPA, 2020). CASTNET sites are located in rural locations that better capture regional O_3 concentrations. This makes them more suitable to evaluate coarse resolution model fields compared to urban and near-road sites. For bias correction, we use CASTNET observations from all available sites overlapping the historical simulation with a valid measurement quality flag, totaling 73 sites across the U.S. First, the modeled and observed MDA8 O_3 concentrations are separated into quantile bins with width 0.001 and evenly spaced between 0 and 1. Next, we apply QM to adjust the modeled MDA8 O_3 so that each quantile of the modeled distribution matches the corresponding quantile of the observation distribution (Maraun, 2013). As a result, the modeled and observed MDA8 O_3 distributions' shapes exactly match, a critical feature to assess extremes over a threshold that is not achieved by correcting the mean bias only. QM is then applied to O_3 projections at mid- and end-century. To bias correct future ozone distributions, first, the ratio between the uncorrected future and the uncorrected historical simulations is calculated at each quantile. Then, MDA8 O_3 values from the corrected historical

simulation within each quantile are multiplied by the corresponding future-to-historical quantile ratio. The resulting future O_3 concentrations are bias-corrected with respect to CASTNET values but reflect climate impacts on future O_3. The QM approach applied here is motivated by that presented in Rieder et al. (2015).

As O_3 concentrations and variability observed differ across CASTNET sites, we organize the sites and the model grid cells into clusters and apply QM within each clusterd. We test two approaches to cluster sites and model grid cells, and evaluate each approach's ability to improve agreement between simulated MDA8 O_3 concentrations and CASTNET observations. First, we group CASTNET sites and model grid cells according to NCA climate regions (U.S. GCRP, 2018). Because NCA regions were partially selected for their climatological resemblance, we expect CASTNET within each region to exhibit some similarities. Next, we group CASTNET sites and grid cells statistically based on location and O_3 concentrations and variability. In this approach, we first use k-means clustering to group the CASTNET sites by site features including latitude, longitude, O_3 mean, O_3 standard deviation, and 95th percentile O_3. Then, we assign model grid cells to the CASTNET clusters determined using the same features. We use a k-means unsupervised machine learning algorithm and a nearest neighbors unsupervised learning function to achieve the clustering (Pedregosa et al., 2011). In k-means clustering, the features on which the data is clustered and the number of clusters must be prescribed—we test combinations of number of clusters and grouping features to identify which leads to most effective bias correction. We tested using 2 to 20 clusters, and for each number of clusters, tested all combinations of grouping by O_3 mean, O_3 standard deviation, and 95th percentile O_3 at each site or grid cell. Latitude and longitude were always included as clustering features. As k-means is sensitive to the data order, this was repeated 100 times with data in randomized order each time to produce robust estimates. We select the combination of clustering features and number of clusters that minimizes normalized mean bias (NMB) and normalized mean error (NME) in modeled values after bias-correction. In the final analysis using k-means, observation data are ordered by site ID in alphabetical order and then by observation time. We perform 10-fold cross validation, in which a random 10% of the data is removed and the bias correction is performed. This is repeated 10 times with a different 10% of the data removed, producing 10 sets of bias-corrected model MDA8 O_3. The final bias corrected

MDA O_3 point values are the mean correction over the 10 iterations in the cross validation for both statistical and NCA grouping.

A decreasing trend in ground-level O_3 was observed in the U.S. after NO_x (NO_x = NO + NO_2) State Implement Plans (SIP) were implemented starting in 1998 (Simon et al., 2015). However, because anthropogenic precursor emissions are held constant in the CAM-Chem simulations used, such a trend is not reflected in the model's emissions or O_3 concentrations. Here, we focus on the model's ability to capture variability across the historical period, rather than its ability to capture trends within the period. Thus, we remove the mean trend centered on the year 2000 in CASTNET data before applying bias correction. A similar approach was applied in the O_3 analysis presented by Brown-Steiner et al. (2018).

5.2.3 Extreme value theory

Extreme value theory has been previously used to estimate return values, the maximum value expected to be observed during a specified time period, of observed and modeled ozone (Brown-Steiner et al., 2015; Phalitnonkiat et al., 2016; Rieder et al., 2013, 2015; Shen et al., 2016). We apply the general approach taken in these studies following Coles (2001) and briefly describe it here. To identify modeled and observed extremes, we define extremes according to the points-over-threshold (POT) method as any MDA8 O_3 concentration exceeding the current National Ambient Air Quality Standard (NAAQS) of 70 ppb. Following the Pickands–Balkema–De Haan theorem, extremes identified using POT asymptotically tend to a generalized Pareto (GP) distribution (Coles, 2001). Fitting a GP distribution to observed extremes allows calculation of probabilistic return levels and intervals. We fit a GP distribution to the extremes by using maximum likelihood estimation (MLE) to quantify the shape and size parameters, with the location parameter set to 70 ppb. Then, we calculate probabilistic return levels and return intervals for ozone as described in Coles (2001).

We evaluate the model's ability to reproduce observed O_3 extremes by comparing 1-year return levels at CASTNET sites and the corresponding model return levels for uncorrected modeled MDA8 O_3, modeled MDA8 O_3 bias corrected using NCA regions, and modeled MDA8 O_3 bias corrected using statistical clustering. We then compute 1-year return levels and 70-ppb return intervals in historical and future modeled ozone for each 30-year member in the CAM-Chem simulation ensemble. For each model grid cell,

we use the Kolmogorov-Smirnov (KS) test to evaluate whether the extremes in each grid cell follow the GP distribution at the 95% confidence level.

5.3 Results and discussion

5.3.1 Bias correction of simulated O_3 concentrations

O_3 extremes modeled in the historical simulated exhibits a high bias compared to CASTNET. The bias correction methods evaluated can reduce differences between observed and simulated extremes. In Figure 5.1, we compare 1-year MDA8 O_3 return levels ("1-year O_3") calculated from CASTNET to 1-year O_3 calculated with uncorrected CAM-Chem (M_orig), CAM-Chem bias-corrected according to NCA regions (M_NCA), and CAM-Chem bias-corrected according to statistical groupings (M_kmeans). M_orig, M_NCA, and M_kmeans are all drawn from the historical simulation at CS3.0 and averaged across all initial condition members. In the figure, CASTNET and corresponding CAM-Chem modeled MDA8 O_3 from 1996-2005 are used to calculate 1-year O_3. Only CASTNET sites which have at least 6 years of data within this time frame are included. Uncorrected CAM-Chem O_3 leads to high biases in 1-year O_3 estimates exceeding 60 ppb at individual sites and reaching 28 ppb on average across all sites. Applying either bias correction significantly improves the model's representation of historical extremes. M_NCA results in a bias of less than 3 ppb across all sites, although biases exceed ±15 ppb at several sites. M_kmeans bias-correction results in 1-year O_3 biased low by 7 ppb compared to CASTNET 1-year O_3. Many sites in the west, which are more rural and exhibit low mean observed O_3, contain insufficient number of extremes to calculate return levels. In the eastern U.S., almost all sites have a sufficiently high number of exceedances between 1995-2006 to compute the return level. Although there is spatial heterogeneity in the sign of the bias in M_NCA, sites further to the northeast and southwest where CASTNET 1-year O_3 is larger tend to be biased lower, while in M_kmeans the low bias is ubiquitous.

In addition to the comparison described above, we also test M_orig, M_NCA, and M_kmeans 1-year O_3 against pre-SIP observations from CASTNET in order to compare against an observation period without a mean trend. In this comparison, shown in Figure D.1, all CASTNET data prior to 2001 is considered, no detrending is applied, and sites with a statistically significant mean trend different than zero are removed. While the

total number of sites and observations is reduced, the observation that M_NCA better represents historical extremes remains.

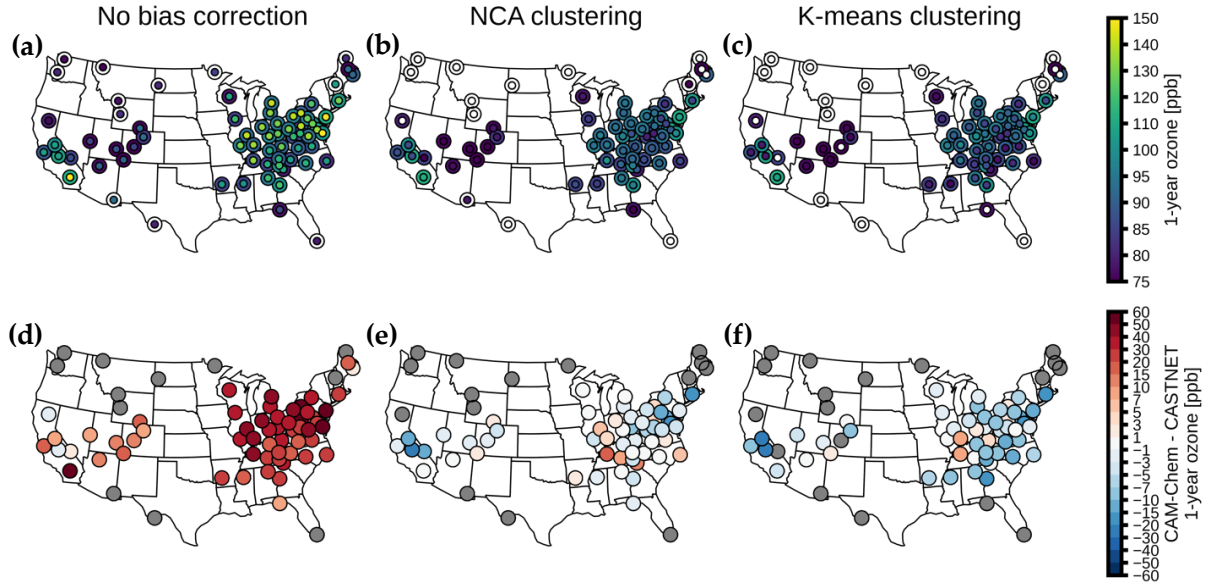

Figure 5.1: Modeled and observed historical ozone summertime return levels. Maps **(a), (b),** and **(c)** show 1-year ozone return levels calculated at CASTNET sites (outer circles) and with **(a)** M_orig (no bias correction), **(b)** M_NCA (bias correction using NCA clustering), and **(c)** M_kmeans (bias correction using k-means clustering) modeled ozone (inner circles). White-filled inner and outer circles indicate an insufficient number of extreme values to calculate a return level. Maps **(d)**, **(e)**, and **(f)** show the difference between the modeled return levels and the CASTNET return levels. Positive values indicate modeled return levels are high biased compared to CASTNET. Grey circles indicate insufficient number of extreme values to calculate a return level in either the model or CASTNET. CASTNET and corresponding model values are from 1995-2004 and only sites which have at least 6 total seasons of data within this time frame are included.

To produce the bias correction applied to M_kmeans, the best performance in reducing MDA8 O_3 NMB was achieved with 15 clusters, grouped by latitude, longitude, O_3 mean, and O_3 standard deviation. Figure D.2 shows the error and bias for the 1, 2, 5, 10, 25, and 50-year ozone return levels for M_NCA and M_kmeans compared to CASTNET return levels. M_NCA has reduced bias compared to M_kmeans at all return intervals, although both methods result in a low-bias. M_NCA's better performance suggests that, in the absence of observations available at each grid cell, location-based clustering encompassing multiple sites and grid cells may be more effective for bias correction than statistically based groupings.

5.3.2 Impact of internal variability on modeling extreme ozone

Several recent studies have identified the importance of considering internal variability in modeling the impacts of climate on ozone (Barnes et al., 2016; Brown-Steiner et al., 2018; Garcia-Menendez et al., 2017). Figure 5.2 shows the potential for using multiple initial conditions to improve representation of ozone extremes and decrease the potential error caused by underrepresenting internal variability. Here, potential error is maximum error across all initial conditions in M_NCA 1-year O_3 at a given sites. Figure 5.2 shows a histogram of the potential error across all sites and the mean initial condition error across all sites, along with their gaussian kernel densities. Our results show that internal variability can lead to errors in return levels compared to observations when using a single initial condition ensemble member. We find that using multiple initial conditions can decreases bias in modeled 1-year O_3 by 1.5 ppb on average, compared to the potential error using a single initial condition, although the potential error can reach an additional 4 ppb beyond the average initial condition error at individual sites. While this modeling framework is not designed to capture specific exceedance events, applying extreme value theory across a long time series, in this case 10-years, and using multiple initial condition members allows for a larger sample size of exceedances and a better fit of the GP distribution, improving the probabilistic estimates of extreme events. The mean of all five initial condition members is used in all subsequent results.

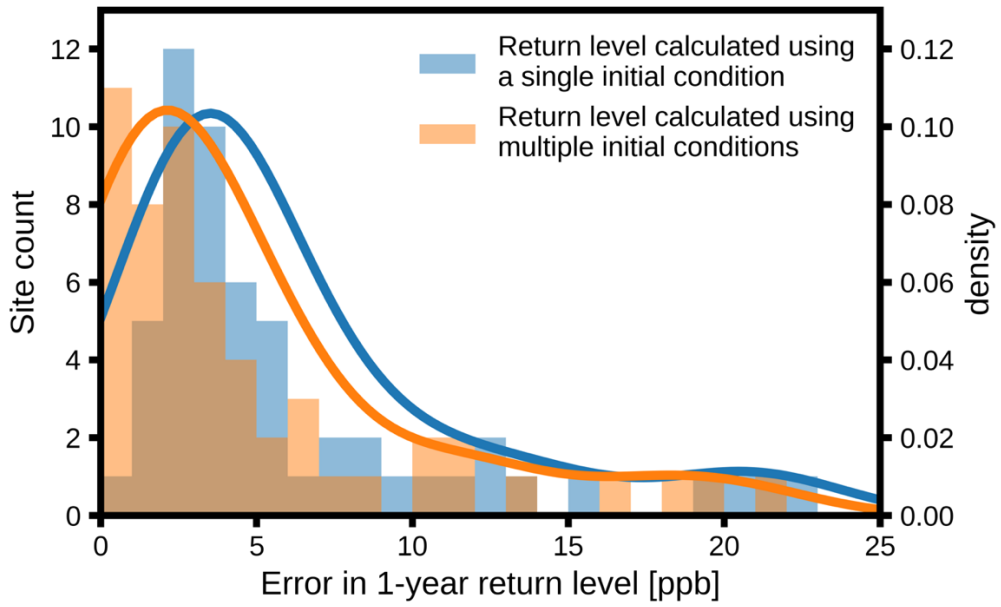

Figure 5.2: Histogram of error in 1-year return level at all CASTNET sites overlaid with a gaussian kernel density empirical distribution of the data. The x-axis is the error in the modeled 1-year O_3 compared to the corresponding CASTNET 1-year O_3, calculated as the difference between the modeled and CASTNET 1-year return level at each site. Orange and blue histograms show the number of CASTNET sites in each error bin. Blue (darker) bars are based on model extremes calculated using a single initial condition ensemble member, and orange (lighter) bars are the mean of extremes calculated from five initial conditions.

5.3.3 Extremes projected to future climate scenarios

Climate change under a moderate policy scenario (P4.5 scenario) is projected to increase the frequency and magnitude of high concentration O_3 incidences over much of the U.S. Figure 5.3 shows the magnitude (1-year ozone return level) and frequency (return interval for an MDA8 O_3 of 70 ppb) of extreme O_3 events in the historical simulation, and the change in each under climate policy P4.5 and climate sensitivity CS3.0 at mid and end century. The 1-year O_3 return level is 82 ppb on average across the U.S. in the historical simulation but can exceed 108 ppb over specific urban areas, such as Los Angeles and New York City. Grey areas on the map reflect grid cells where an insufficient number of exceedances was simulated, so extremes could not be calculated. Hatching represents grid cells in which the GP distribution does not pass the KS test at 95% confidence for all initial conditions. In a few cells where a return level can be computed in the climate projections but not the historical simulation, those levels do not exceed 72 ppb. By mid-century, climate change is projected to increase the magnitude and frequency of high O_3 concentrations across much of the U.S. On average, the 1-year ozone return value

increases by 1.9 ppb at mid-century and 1.8 ppb at end century relative to start-of-century conditions, with the smaller mean increase at end century due to smaller return values in the U.S. Southeast and Gulf coast. Historical values and projected climate-induced changes to 1-year O_3 in the Northeast U.S. at mid and end-century are very similar to those reported in Rieder et al. (2015), who also reported larger increases at mid-century than at end-century under a policy scenario similar to scenario P4.5. Examining the return interval shows that 70 ppb days will become more common across much of the U.S. The largest increases in frequency, up to 90 days and represented in red on the map, occur in areas where the 1-year O_3 level is barely above the 70 ppb threshold, and the historical return interval is close to 1 year. Still, these increases represent a potential for climate to increase extreme pollution frequency in areas where it was previously rarely experienced. Exceedance frequency is projected to decrease at over Southern California, Louisiana, Texas, Arkansas, and Missouri. In areas where the return interval decreases and the return level increases, both the magnitude and the frequency of high O_3 incidents is projected to increase due to climate change.

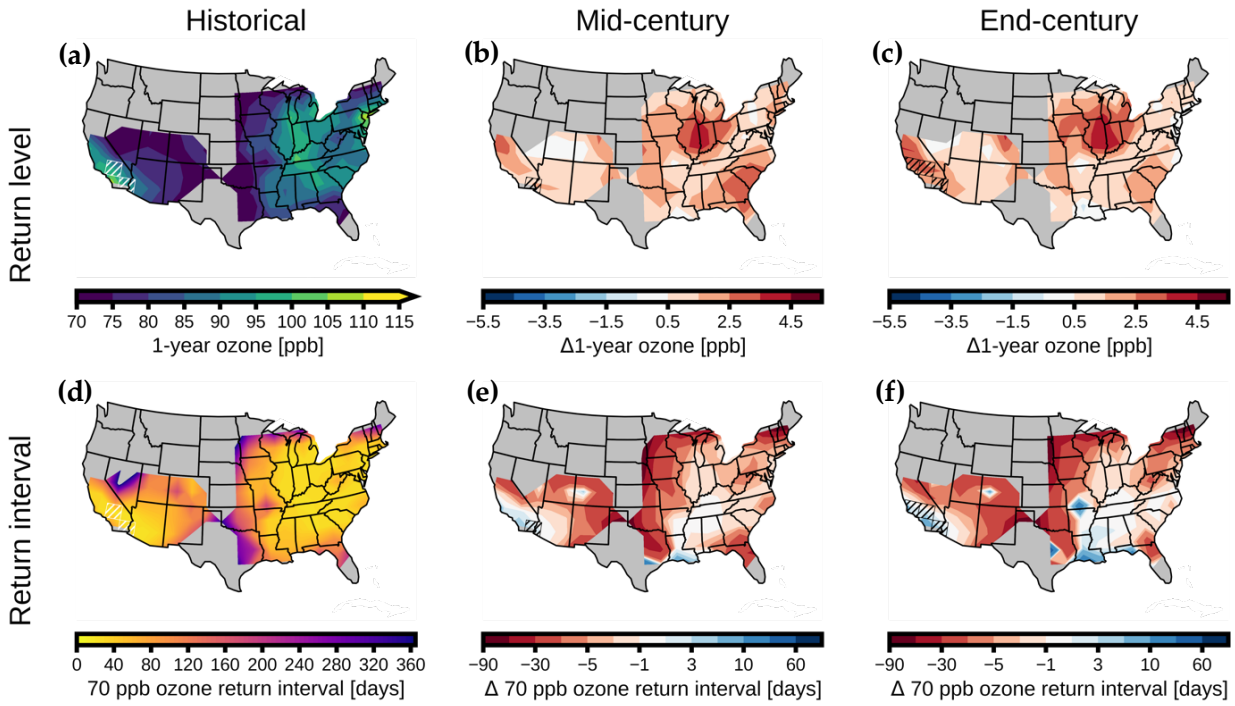

Figure 5.3: Historical intensity and frequency of extreme O_3 and projected changes under policy scenario P4.5 and climate sensitivity CS3.0. 1-year O_3 return level for the historical simulation **(a)**, and changes at **(b)** mid- and **(c)** end-century. 70 ppb O_3 return interval for the historical simulation **(d)**, and changes at **(e)** mid- and **(f)** end-century. Cells are hatched where modeled extremes do not match the fitted GP distribution at the 95% confidence interval according to the K-S test. Grey areas represent cells with an insufficient number of threshold exceedances to calculate a return level and return interval in the historical simulation.

5.3.4 Climate sensitivity and policy scenarios

Alternate future GHG emissions scenarios and climate sensitivities are projected to either magnify or diminish future extremes relative to projections based on the P4.5 GHG emissions scenario and climate sensitivity 3.0 °C, depending on the direction of the climate forcing. Figure 5.4 depicts mid-century changes to 1-year O_3 for each of the GHG emissions scenarios and climate sensitivities included in the modeling ensemble, relative to start-of-century. The corresponding changes at end-century are shown in Figure D.3 and reflect similar patterns to those seen at mid-century but with larger increases, exceeding 30 ppb relative to the historical simulation, under the highest GHG emissions and climate sensitivity scenario combination, the REF policy scenario and CS4.5. At mid-century, projected climate-induced increases to 1-year O_3 are more evident at higher climate sensitivity (CS4.5) compared to lower climate sensitivity (CS2.0) across all policy scenarios. However, while 1-year O_3 generally increases on average with decreasing GHG

emissions policy constraints, policy scenarios are difficult to distinguish under a low climate sensitivity (CS2.0). There would be few discernable differences in climate impacts on ozone under a low-climate sensitivity, no-policy future compared to an alternate high-climate sensitivity future with strict greenhouse gas controls. On the other hand, the differences among different policies are clear at the highest climate sensitivity, CS4.5, with U.S. average 1-year ozone increases of up to 6.6 ppb under the REF scenario compared to <1 ppb under the P3.7 scenario. The similarities between impacts under high climate sensitivity/low GHG emissions versus low climate sensitivity/high GHG emissions is most clear when comparing among the combinations P3.7 and CS4.5, P4.5 and CS3.0, and REF and CS2.0. In these divergent scenarios, mean differences between each are less than 1 ppb. This sheds new light on the role of climate sensitivity on modeled climate impacts on air quality, a previously unexplored aspect of climate and air quality interactions, and highlights the importance of considering uncertainty due to climate sensitivity.

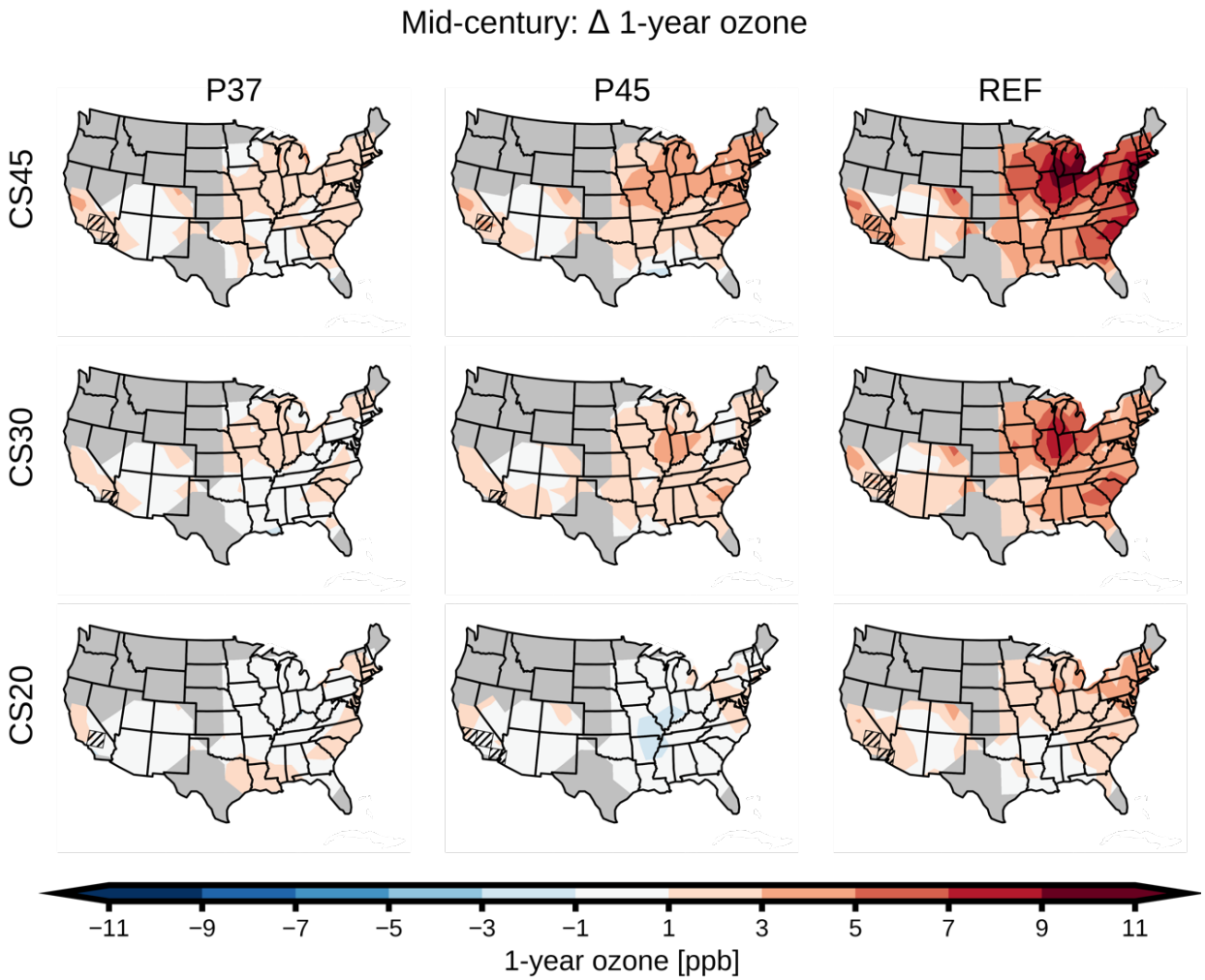

Figure 5.4: Change in 1-year ozone return level at mid-century across multiple policies and multiple climate sensitivities. Change are relative to the historical simulation with the corresponding climate sensitivity. Cells are hatched where modeled extremes do not match the fitted GP distribution at the 95% confidence interval according to the K-S test. Grey areas represent cells with an insufficient number of threshold exceedances to calculate a return level and return interval in the historical simulation.

5.3.5 Changes to future extremes compared to the future mean

Several previous studies have noted larger increases in the upper quantiles of the projected O_3 distribution (e.g. Nolte et al., 2018; Pfister et al., 2014; Schnell et al., 2016). Here, we show that, while this is generally true, climate sensitivity plays an important role in modulating the extent of this impact. Figure 5.5 shows U.S. average 1-year and mean O_3 at start-of-century, mid, and end-century for all emissions scenarios and climate sensitivities, and their U.S. average changes relative to the historical simulation. In the figure, grid cells which do not have 1-year O_3 return levels in the historical simulation are

not included in the mean ozone change calculation. Both the mean and 1-year O_3 increase at mid and end-century across all cases, with largest concentrations projected under REF across all climate sensitivities. In all most cases, increases in the extremes are larger than increases in the mean. However, when considering higher compared to lower climate sensitivities, extremes increase more than the mean under P4.5 and REF. Figure D.4 shows this clearly by showing the relative change in means and extremes. Changes in the extremes at higher climate sensitivities further highlight the importance of considering climate sensitivity uncertainty in future climate impacts on air quality.

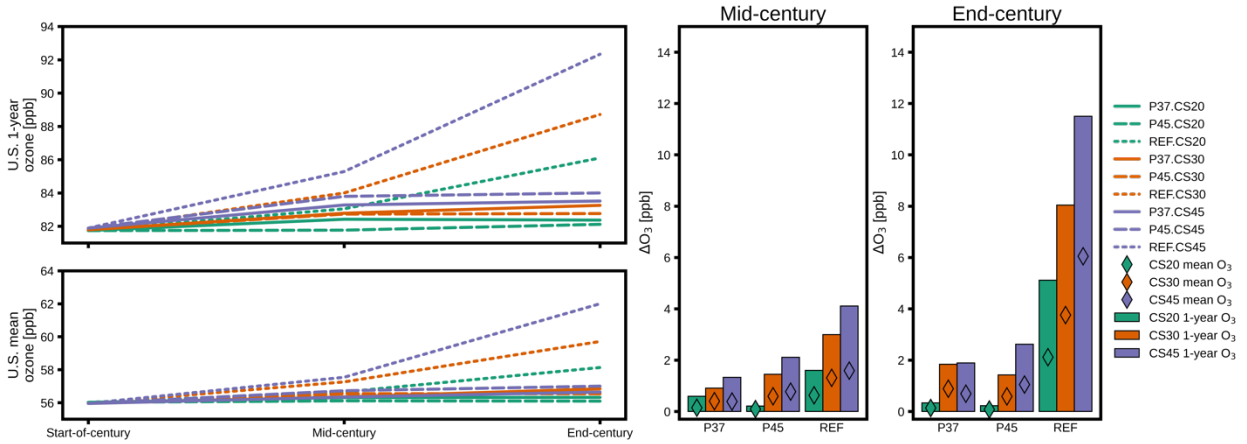

Figure 5.5: U.S. average 1-year O_3 and mean MDA O_3 concentration at beginning, middle, and end-century under each climate sensitivity and each policy scenario. Top left: U.S. average 1-year O_3 using bias-corrected simulated O_3. Bottom left: Same, but showing U.S. average mean MDA8 O_3 concentration. Middle: Change in U.S. average 1-year O_3 and mean MDA8 O_3 concentration relative to start-of-century conditions. Right: Same, but for end-century.

5.4 Conclusions

Climate change is projected to negatively impact U.S. air quality by increasing ozone pollution, both in terms of the mean concentrations as well as frequency and magnitude of high ozone events. In this study, we examined this impact using a multiple-initial-condition simulation ensemble of climate-chemistry simulations that consider multiple GHG emissions scenario and multiple climate sensitivities. We show that the modeling can capture historical extremes after applying a location-based quantile mapping bias correction framework, and that the historical extremes are better represented when multiple initial conditions are considered to capture internal variability. In projections of a future climate, extremes are heavily dependent on both the GHG emissions scenarios and climate sensitivity, which can play a potential role in magnifying or diminishing climate impacts on O_3. Under unconstrained GHG emissions

at mid-century and a high climate sensitivity, 1-year ozone may increase by up to 4.5 ppb on average over the U.S.; however, a lower climate sensitivity would dampen this impact to less than 2 ppb, highlighting the importance of considering uncertainty due to climate sensitivity in climate impacts on air quality. GHG emissions-limiting policies can also limit concentration impacts to less than 2 ppb. Our results provide further evidence that variability in O_3 concentrations will likely increase in addition to future absolute changes. The increase in the upper tail of the concentration distribution is larger than the mean at high climate sensitivities, while at low and medium climate sensitivities, the mean increase is approximately the same or larger than the extremes increase.

The potential for climate change to impact high ozone values in distributions of concentrations is of high relevance to air quality management in the U.S., particularly in regions in or near non-attainment of the NAAQS, where small changes in frequency and magnitude of extreme O_3 days can impact attainment. Our results provide policy relevant information about projected climate impacts on the magnitude and frequency extreme ozone events, independent of human intervention through precursor emissions reductions. Insights revealed here provide further evidence that policies limiting GHG emissions can support efforts to improve air quality by limiting future climate increases to extreme O_3 pollution as well as increased risk stemming from uncertainty in climate system response and internal variability.

These results carry implications for modeling future climate impacts on air quality. Recent attention has been given to using large ensembles in modeling climate and its impacts (e.g. Deser et al., 2020; Fiore et al., 2022; Lehner et al., 2020; Mankin et al., 2020). This study provides additional evidence supporting use of multiple initializations in climate and air quality simulations, by showing that using a single model initialization can lead to errors in modeled extremes, while using multiple initial conditions can reduce error in modeled historical extremes and increase confidence in projections. Further, these results show that large ensembles which cover a range of climate sensitivities or different climate models may better sample internal variability and therefore better capture uncertainty in future mean and extremal changes. The role of climate sensitivity in modulating climate impacts cannot be ignored while future climate forcing scenarios remain uncertain.

Chapter 6: Conclusions

In this chapter, key findings of each previous chapter are summarized into major concepts and broader implications of the research, both for the field of air quality modeling and policies relating to air pollution management, are discussed. Then, conclusions synthesizing the key findings of each chapter, including research goals towards which this work may be viewed as a first step. Finally, I identify specific potential research extensions of this work.

6.1 Findings and implications

6.1.1 Chapter 2 summary

Deteriorated air quality in megacities in LMICs is a pressing global environmental issue with severe human health impacts. Our work (Chapter 2; East et al., 2021) investigated a framework that applies chemical transport models in a megacity which has sparse air quality data and limited prior applications of comprehensive photochemical modeling systems. This framework was used to identify the largest local contributors to $PM_{2.5}$ air pollution and analyze potential air pollution control policies.

6.1.1.1 Key findings and advancements

1. Based on our source scaling results, we identified resuspended particulate matter (RPM) as largest source of PM in Bogota, followed by on-road and industrial emissions sources. While RPM was suspected to be a large emissions source in the urban area, our results quantified the amount and location of RPM's contribution to total $PM_{2.5}$.
2. In our projections of potential future air quality control scenarios, we found that road-paving initiatives can lead to $PM_{2.5}$ pollution decreases, but that air pollution in much of the city will increase without additional actions due to projected growth in industry and traffic related emissions.
3. Regulatory actions in Bogotá targeting the traffic and industrial sectors can decrease air pollution levels in the future. Improved emissions standards applied to Bogotá's light duty vehicle fleet can result in pollution decreases across much

of the city, while implementing additional emissions controls for the industrial sector can reduce air pollution hotspots in industry-heavy areas.

6.1.1.2 Implications for the air quality modeling research

Air quality modeling is increasingly being applied in LMICs and data sparse regions for air quality management. While the research application of Chapter 2 was specific to a particular city, the work carries implications for future air quality modeling research applied in similar settings. The following implication for air quality modeling is identified:

1. The application of air quality modeling to assess Bogotá's air quality and future emissions controls provides an approach that could be applied to other cities. Our work in Bogotá highlights the importance of local emissions inventories. Inventory development should be a priority for enabling air quality modeling in regions where it has not previously been applied due to data and resource constraints.

6.1.1.3 Implications for policy

Implications for air quality management policy can be drawn from this work:

1. Using air quality models, the combined use of emissions source-scaling sensitivities and scenario analysis offer an approach to first identify priorities for emissions reductions by quantifying the contribution of different sectors to the air pollution burden, and then assess the effectiveness of control measures targeting those sectors.
2. As many developing regions that are data sparse begin to address local air quality issues, our case study in Bogotá reveals the minimum elements necessary to effectively leverage chemical transport models for air quality management goals. Priorities for enabling effective modeling tools should include: (1) development of an observation network. In Bogotá, a small network of filter-based PM measurements laid the groundwork for identifying pollution sources and for evaluating air quality models. (2) Development of a local emissions inventory. While global inventories are available, they have been shown to be inaccurate in many LMICs and of limited use for urban scale modeling. (3) A baseline emissions projection scenario along with emissions control policies. These provide the basis

for assessing the effectiveness of control strategies. None of these elements are trivial to implement. However, they should be viewed as critical path items for effective application of air quality models in the development of control policies.

6.1.2 Chapter 3 summary

As the contribution of local sources of ozone declines in many areas of the U.S., background and non-controllable contributions to ground level ozone become increasingly important. Accurate global emissions are critical for representing this component in air quality modeling for regulatory and research purposes. Our research (Chapter 3; East et al., 2022b) develops and applies a framework to bridge the gap between uncertain global emissions inventories and true emission levels through satellite chemical data assimilation. We implemented satellite NO_2 chemical data assimilation in the CMAQ model for the first time and applied a finite-difference mass-balance inversion to estimate NO_x emissions in the Northern Hemisphere using NO_2 retrievals from the OMI and TROPOMI satellites, separately. Key findings and implications are outlined below.

6.1.2.1 Key findings and advancements

1. A new inversion framework was developed implementing chemical data assimilation in the CMAQ model, enabling assimilation of OMI and TROPOMI NO_2 retrievals and iterative updates to LNO_x and ANO_x emissions. The system is flexible to be adapted to new species.
2. The assimilation and inversion found large differences between NO_2 assimilated from OMI and TROPOMI (retrieval versions 1.2 and 1.3), as well as NO_x emissions inferred from either of the two satellites, partially attributed to a previously identified low bias in TROPOMI version 1.2 and 1.3 NO_2 retrievals.
3. A new approach was developed to reduce emissions misallocation errors in the finite-difference mass-balance inversion. In the new approach, made possible by the implementation of chemical data assimilation in the framework, the model vertical columns densities of NO_2 used in the assimilation are split into upper and lower portions. The lower portion is used to adjust near-surface anthropogenic emissions, eliminating interference from NO_2 in the mid- and upper-troposphere.

4. Updates to LNO_x emissions resulted in larger impacts on tropospheric ozone than updates to near-surface ANO_x emissions and led to the largest impact on trans-Pacific ozone impacting U.S. air quality.

6.1.2.2 Implications for air quality modeling research

1. The CMAQ model is widely used in the regional air quality and regulatory communities. By implementing chemical data assimilation in the platform, emissions and boundary conditions for U.S. regional simulations produced by the framework will be available to the wider research communities.
2. While emissions inversions using satellite retrievals are useful for providing observational constraints on global emissions, this research points to the effects that existing uncertainties and biases in satellite products can have in inverse estimated emissions. The uncertainty in the satellite retrieval product must be considered in emissions inversions.
3. Our research highlights the potential for biases in modeled LNO_x emissions to introduce bias to data assimilation and emissions inversions due to the satellite instrument's higher sensitivity to NO_2 in the upper troposphere and the longer lifetime of NO_2 in the upper troposphere.

6.1.2.3 Implications for policy

1. Our results showed that anthropogenic NOx (ANOx) emissions inferred using the OMI satellite instrument could lead to a +5 ppb increase in estimates of ozone transported across the Pacific Ocean from anthropogenic sources in Asia to the U.S., compared to ozone simulated with the *a priori* inventory. This change, the difference between a global emissions inventory nominally representative of 2017 and observational constraints from 2019, shows that outdated global emissions inventories may lead to inaccurate representations of background ozone in models. Our framework and its application highlight the value of combining inverse approaches generally for addressing this problem, and demonstrate the effectiveness of this approach, specifically. Policies designed for ozone management should consider the uncertainties in global emissions inventories used to simulate long range transported ozone.

6.1.3 Chapter 4 summary

Climate change is projected to negatively impact air quality during the 21st century. Ozone and $PM_{2.5}$ concentrations are projected to increase across much of the U.S., posing an increased risk to human health. Coupled climate chemistry models, which are used to project this climate penalty, are subject to uncertainties inherent to climate projections: uncertainty due to GHG emissions scenario, due to climate model response, and due to internal variability. These uncertainties have not been previously characterized for impacts on air quality. In our work (Chapter 4; East et al., 2022a), we leverage an ensemble of over 2,800 years of climate impacts on air quality to quantify these climate uncertainties in air quality projections.

6.1.3.1 Key findings and advancements

1. We systematically quantified major climate uncertainties in projections of the climate impacts on air quality for the first time. We found that, while policy uncertainty dominates by end century, internal variability and climate sensitivity uncertainty, closely related to climate model response uncertainty, are approximately equal to policy uncertainty at mid-century and remain important at end-century.
2. Using the climate and air quality ensemble to simulate a range of climate sensitivities that encompass the uncertainty range reported by the IPCC, we characterized climate impacts on air quality under different climate sensitivities for the first time. We found that climate sensitivity amplifies or dampens the climate impact at high or low climate sensitivities, respectively. This result introduces a new dimension into analysis of air quality in a future climate.
3. We analyzed the time scales of internal variability uncertainty and find that internal variability uncertainty will dominate at mid-century on time scales shorter than multidecadal. In our analysis, an averaging period of 30 years resulted in internal variability uncertainty less than of approximately equal to climate sensitivity and policy uncertainty, while an averaging time of 15 years caused internal variability to surpass all uncertainties except that of policy at end-century.

6.1.3.2 Implications for air quality modeling research

4. To reduce internal variability uncertainty, the average change in air quality over multidecadal simulations should be considered. Otherwise, the climate signal may be misattributed or may not be discernable over natural variability. Our results indicated that a 30-year averaging time was sufficient, while a 15-year may lead to internal variability uncertainty that surpasses all others at mid-century and is larger than climate sensitivity uncertainty at end-century.
1. Large initial-condition ensembles are critical for adequately representing internal variability and should be used in climate-chemistry projections.

6.1.3.3 Implications for policy

1. Although internal variability is irreducible and climate sensitivity is limited by the scientific community's current understanding of the earth system, there is potential to reduce uncertainty, and therefore more confidently project future climate impacts, through increasing certainty and commitment to a GHG emissions policy future.
2. Future air quality is a result of changing meteorology and changing primary and precursor emissions. While our analysis isolates the effects of climate by simulating constant anthropogenic primary and precursor emissions, large ensembles with changing precursor emissions have the potential to provide an additional dimension of co-benefits assessments of climate policies.

6.1.4 Chapter 5 summary

In addition to impacts on future average air quality, simulates of atmospheric composition project an intensification of extreme air quality episodes in a future climate that goes beyond the climate penalty to average air quality. However, the potential for climate to exacerbate extreme air quality has not been studied along the uncertainty dimension of climate sensitivity. Further, climate-chemistry models generally have been shown to exhibit a high bias for ozone compared to historical observations, and bias-correction methods are needed to analyze threshold-based extremes. Using the same climate and air quality modeling ensemble introduced in Chapter 4, our work (Chapter 5) evaluated two different grouping methods in which quantile mapping bias correction

is applied. Then, we identified extremes and applied extreme value theory to project historical and future intensity and frequency of high surface level ozone concentrations in the U.S. under multiple climate sensitivities and multiple climate policies.

6.1.4.1 Key findings and advancements

1. When evaluating a historical model simulation against MDA8 O_3 observations, we found that using a single model initialization can lead to larger errors in modeled extremes, while using multiple initial conditions can reduce error in modeled historical extremes and increase confidence in projections.
2. Our results show that, on average, climate change is projected to increase both the magnitude and frequency of extreme ozone pollution events in the U.S. across a range of policy scenarios and climate sensitivities. This was the first time that the effect of different climate sensitivities on future air quality extremes was analyzed.
3. Climate sensitivity is linked to variability. We found that the magnitude of ozone extremes is projected to increase more than mean ozone in the U.S. under high climate sensitivity and unconstrained GHG emissions. However, when relative as opposed to absolute changes from the historical simulation are considered, increases in extremes are approximately equal to or less than mean increases at lower climate sensitivities, revealing a previously unexplored role of climate sensitivity.

6.1.4.2 Implications for air quality modeling research

1. This research provides further evidence of the importance of using multiple initializations in modeling climate impacts. Using a single model initialization can lead to larger errors in modeled extremes. Multiple initial conditions can be used to reduce error in modeled historical extremes and increase confidence in projections.
2. The differences in projected extremes under different climate sensitivities revealed in our research suggests that large ensembles which include multiple climate sensitivities or multiple climate models better sample internal variability and therefore better capture the range of future mean and extremal changes.

6.1.4.3 Implications for policy

1. Increases in future extreme ozone events are highly dependent on emissions scenario. Emissions control scenarios show potential for mitigating increases to extreme size and frequency, while the largest increases are projected under a no-policy future. Implementing GHG emissions control policies can lead to co-benefits in the form of reduced impacts to air quality extremes.
2. Climate impact on air quality projected to pose challenges to meeting ozone NAAQS. In our research, we identified extremes over the threshold of the current ozone NAAQS, 70ppb, and showed the potential for future climate to increase the frequency of exceedances. While our study isolates the impact of climate by not considering changes to anthropogenic ozone precursor emissions, air quality managers should consider the potential for climate changes to counter efforts to meet NAAQS requirements when designing emissions control strategies.

6.2 Overarching conclusions

The research presented here describes contributions to understanding air quality impacts of emissions changes across a range of spatial and temporal scales. Many of the key findings and implications revealed are self-contained within each chapter. However, synthesizing the findings and impacts of each chapter leads to several broad conclusions that can enhance air quality research and policy analyses using chemical transport models more generally.

Taken together, the findings show that informative policy relevant analyses require knowledge of emissions changes and impacts across varying scales. Focusing only on local or near-term emissions changes and air quality impacts can be short sited, failing to account for broad scale impacts such as those from climate change and those from long range transported pollutants. On the other hand, focusing only on background impacts, long-range transport, and large-scale sources may ignore important subgrid-scale processes as well as highly localized air quality impacts. Knowledge of impacts at all scales gives stakeholders and policymakers more complete information for deciding which sources and processes to target in the design of near- and long-term air quality management strategies. An integrated system accounting for observational constraints on emissions and atmospheric composition that couples global scale

simulations to urban scale case studies is ideal for achieving this goal. In practical terms, resource limitations often render implementation of such a system unfeasible. Nevertheless, understanding how emissions changes and impacts on multiple scales interact can guide decisions about where stakeholders and policymakers should focus available resources.

This research reveals the potential of various incremental steps towards such a fully integrated system to advance air quality analyses. In our emissions inversion framework, emissions constraints and impacts are done at relatively coarse horizontal scales. At these scales, the assumptions of the finite-difference mass-balance inversion employed are valid. However, at finer regional to urban scales, these assumptions break down, limiting the applicability of our inversion system, as well as other global scale inversion systems, to provide fine-scale constraints. Linking coarse scale emissions constraints and impacts to fine-scale air quality simulations through dynamic boundary conditions and emissions downscaling that accounts for subgrid-scale variations can connect coarse scales constraints to fine scale impacts. Several such downscaling approaches have been proposed and evaluated in Wang et al., (2020) and Li et al. (2021), but further development is needed.

Similarly, our projections of climate impacts on air quality are simulated at coarse global scales of 1.9°× 2.5°. While such scales make the ensemble simulations possible, they are unable to represent fine scale processes and urban scale changes. Simulating climate impacts using dynamically downscaled meteorology in regional weather and chemical transport models can resolve finer scale changes and typically simulate air quality using more complex chemical mechanisms. In addition, as computational resources grow, climate can be simulated at finer scales and directly used to run more fine-scale and complex chemical transport models than the one used here. Our research reveals new information about the interactions of climate and air quality and motivates such additional development.

6.3 Research extensions

The research in each chapter can be extended in new directions. Our research applying a chemical transport model in Bogotá revealed the challenges presented in assessing air quality and evaluation modeling efforts in regions with sparse observation

availability. Satellite retrievals of trace gases including NO_2, CO, and NH_3, as well as retrievals of aerosol products such as aerosol optical depth can add an additional layer of observations to air quality analyses in Bogotá and similarly situated urban centers. The TROPOMI satellite instrument provides retrievals at scales relevant for urban air quality analysis and its products are continually improved through ongoing validation and retrieval algorithm development efforts. Applying satellite retrievals for model evaluation is a realistic extension of the research presented here, with potential to reveal more information about air quality in Bogotá and similar cities and to help identify priorities for improving model performance.

Our emissions inversion framework was evaluated and applied for NO_2 retrievals and NO_x emissions. However, it is capable of being updated to incorporate observations of different trace gases for inversion of different emitted species. Enabling assimilation of ammonia (NH_3) from the Cross-track Infrared Sounder (CrIS) and implementing NH_3 emissions estimation in the inversion framework presented here would extend our framework and facilitate new insights into emissions of a species important to air quality and climate.

Finally, our analysis of changes to extreme air quality events in various future climate scenarios was limited to ozone. As a pollutant with negative impacts on human and ecological health whose future changes have implications for air quality management policies, understanding future changes to ozone extremes was critical. However, $PM_{2.5}$ pollution is responsible for severe health impacts worldwide, and future climate change is expected to exacerbate $PM_{2.5}$ pollution. A study of the climate impact on $PM_{2.5}$ extremes would extend the extremes analysis started in this dissertation. Because $PM_{2.5}$ is a complex mixture of different components, any future analyses would require careful consideration of biases in each component and an extremes analysis method appropriate for handling such biases. If a threshold based approach as described here for ozone is not appropriate, other methods based on empirical orthogonal functions have been applied for evaluating future $PM_{2.5}$ and may be effective (Fiore et al., 2022).

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APPENDICES

Appendix A: Supplementary material for Chapter 2

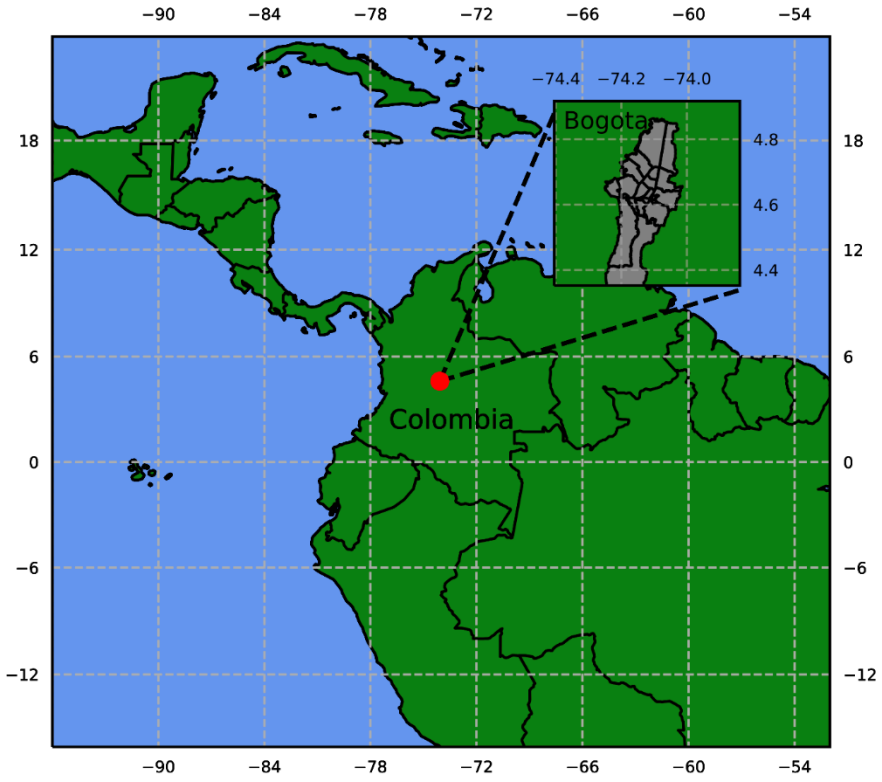


Figure A.1: Northern South America and the modeling domain used in Chapter 3.

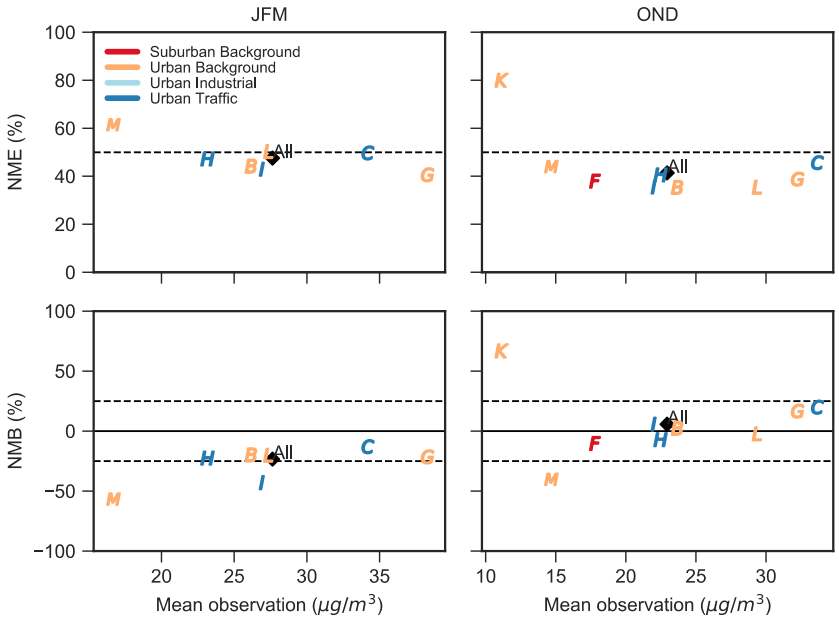


Figure A.2: Model performance for $PM_{2.5}$ simulated in Bogotá. Each letter represents a monitoring site in the city and labels match those in Figure 2.1a. The mean across all sites, weighted by number of available observations, is indicated by the black marker. NME and NMB are shown for the JFM and OND seasons. Recommended performance criteria for each statistic (Emery et al., 2017) are shown as dashed lines.

Table A.1: Sector contributions to $PM_{2.5}$ at all monitoring sites during JFM season. Modeled $PM_{2.5}$ is the CMAQ-modeled $PM_{2.5}$ before source apportionment. Rescaling error is sum of $PM_{2.5}$ contributions from individual sectors relative to CMAQ-modeled $PM_{2.5}$. Letters next to site names match the site labels in Figure 2.1a.

Site	Site category	Modeled $PM_{2.5}$ ($\mu g/m^3$)	On road	Industrial	Construction RPM	Quarrying RPM	Main road RPM	Secondary road RPM	Unpaved industrial road RPM	Unpaved public road RPM	All others	Boundary	Rescaling error
Bolivia (A)	Suburban background	10.0	17.9%	4.4%	4.2%	1.1%	1.6%	3.7%	10.0%	3.3%	2.9%	53.5%	-2.5%
CDAR (B)	Urban background	15.1	23.6%	10.0%	3.2%	0.2%	3.8%	4.9%	9.3%	14.9%	8.7%	21.6%	-0.3%
Carvajal (C)	Urban traffic	22.8	19.4%	11.0%	0.6%	2.6%	1.4%	3.5%	17.9%	30.5%	2.2%	12.8%	-1.9%
Convenio 176_10 (D)	Urban traffic	14.5	21.8%	9.9%	4.4%	0.3%	3.1%	4.5%	11.2%	20.8%	6.0%	20.1%	-2.2%
Fontibon (E)	Urban Industrial	21.3	26.8%	11.5%	4.3%	1.0%	4.5%	10.9%	9.9%	5.3%	6.1%	21.1%	-1.5%
Guaymaral (F)	Suburban background	8.9	25.6%	7.6%	1.7%	1.6%	3.2%	1.8%	7.3%	3.9%	1.8%	48.8%	-3.4%
Kennedy (G)	Urban background	22.0	23.4%	11.9%	1.0%	2.0%	2.6%	5.9%	13.1%	21.6%	3.3%	17.5%	-2.3%
Las Ferias (H)	Urban traffic	13.2	24.2%	12.0%	4.8%	0.6%	2.9%	5.0%	8.1%	12.3%	4.4%	27.7%	-2.1%
Min Ambiente (I)	Urban traffic	14.5	23.0%	13.2%	2.4%	0.2%	3.4%	5.0%	12.6%	14.8%	5.4%	20.7%	-0.7%
Puente Aranda (J)	Urban industrial	23.6	24.1%	24.3%	1.4%	0.8%	3.2%	4.4%	11.7%	9.7%	5.7%	15.9%	-1.3%
San Cristobal (K)	Urban background	11.0	19.5%	15.8%	1.0%	1.0%	1.9%	3.6%	13.1%	15.2%	5.8%	26.3%	-3.2%
Tunal (L)	Urban background	16.1	23.9%	10.2%	0.4%	4.8%	2.0%	3.8%	17.7%	22.2%	2.1%	15.8%	-3.1%
Usaquen (M)	Urban background	5.2	11.7%	4.7%	5.8%	0.9%	1.1%	1.5%	4.0%	7.3%	3.0%	66.5%	-6.6%

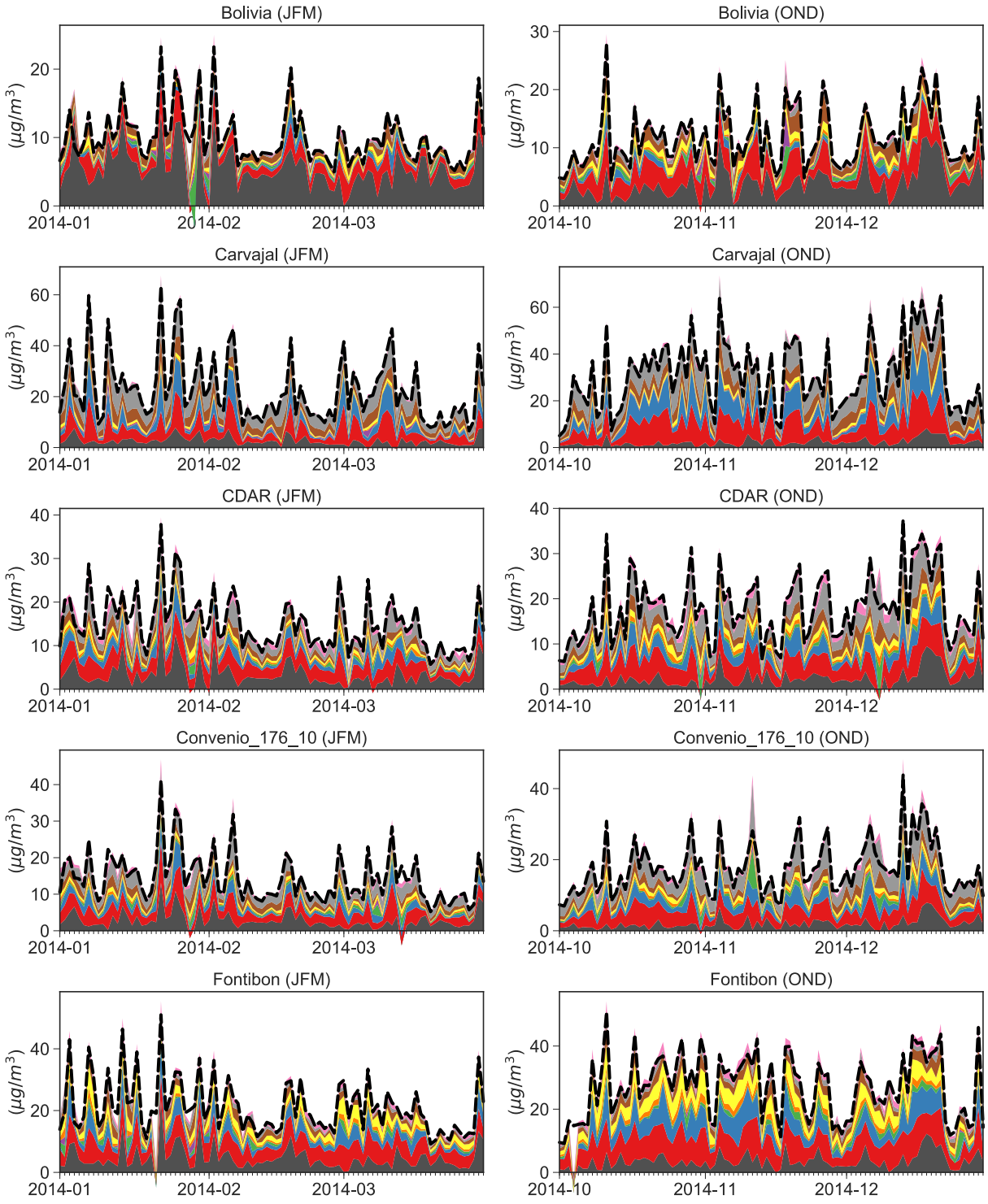
Table A.2: Sector contributions to $PM_{2.5}$ at all monitoring sites during OND season. Modeled $PM_{2.5}$ is the CMAQ-modeled $PM_{2.5}$ before source apportionment. Rescaling error is sum of $PM_{2.5}$ contributions from individual sectors relative to CMAQ-modeled $PM_{2.5}$. Letters next to site names match the site labels in Figure 2.1a.

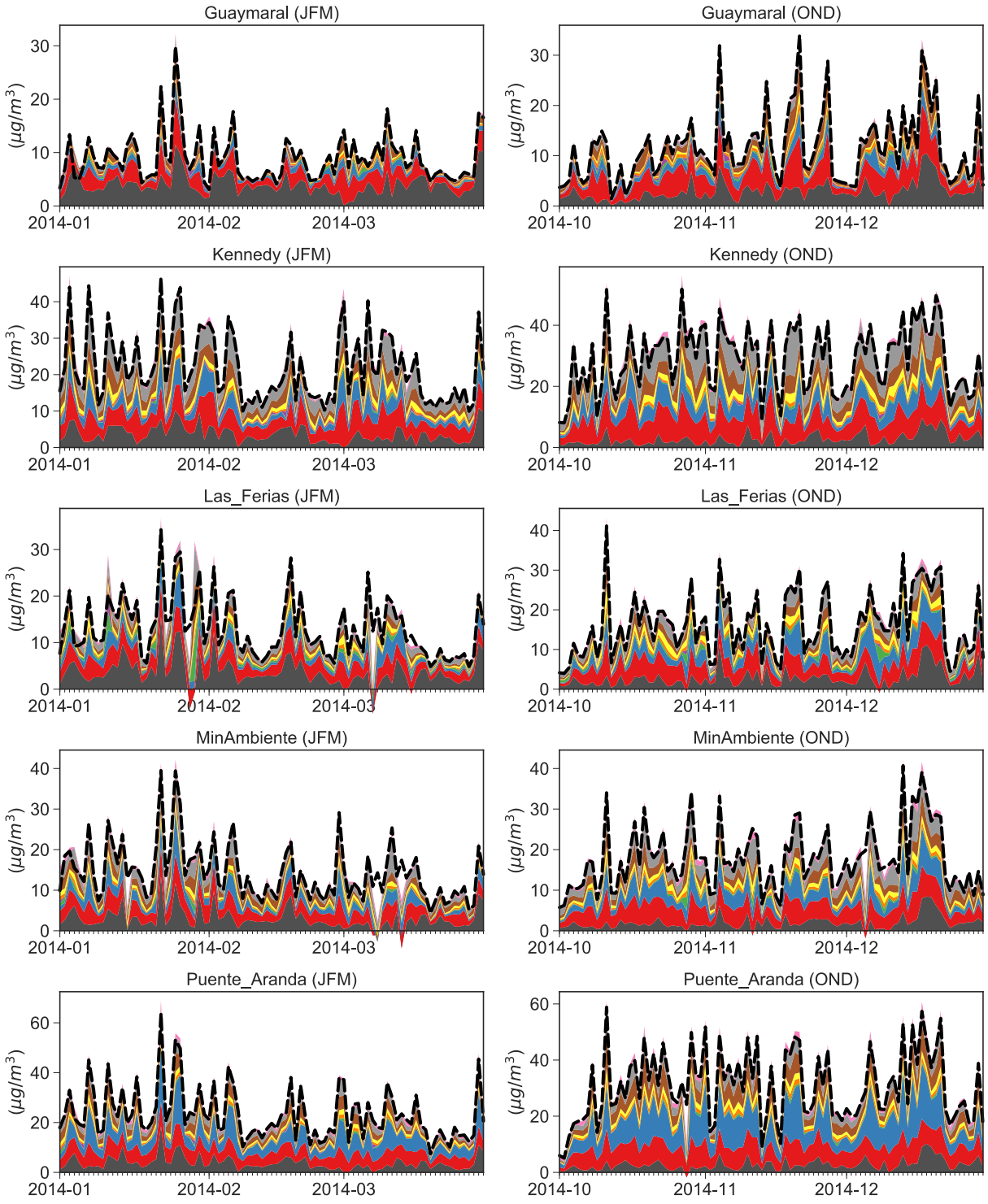
Site	Site category	Modeled $PM_{2.5}$ ($\mu g/m^3$)	On road	Industrial	Construction RPM	Quarrying RPM	Main road RPM	Secondary road RPM	Unpaved industrial road RPM	Unpaved public road RPM	All others	Boundary	Rescaling error
Bolivia (A)	Suburban background	11.9	27.1%	4.6%	4.9%	1.4%	2.5%	6.4%	13.1%	5.1%	4.0%	32.0%	-1.1%
CDAR (B)	Urban background	18.1	24.2%	11.9%	3.7%	0.2%	4.3%	5.9%	10.3%	19.2%	8.7%	12.4%	-0.7%
Carvajal (C)	Urban traffic	31.1	24.0%	17.2%	0.5%	1.2%	1.5%	3.8%	16.0%	28.6%	2.6%	7.6%	-3.1%
Convenio 176_10 (D)	Urban traffic	17.8	24.0%	12.4%	4.8%	0.2%	3.5%	5.2%	10.1%	25.3%	6.1%	10.8%	-2.4%
Fontibon (E)	Urban Industrial	27.2	27.6%	15.3%	4.6%	0.3%	5.4%	13.8%	9.7%	5.2%	7.1%	12.6%	-1.5%
Guaymaral (F)	Suburban background	11.6	32.4%	9.7%	2.0%	2.3%	4.4%	2.5%	12.1%	5.8%	2.2%	27.6%	-1.0%
Kennedy (G)	Urban background	29.2	26.1%	14.9%	1.1%	0.7%	2.7%	6.4%	14.3%	24.2%	3.6%	9.6%	-3.6%
Las Ferias (H)	Urban traffic	16.3	25.6%	15.6%	4.8%	0.5%	3.6%	6.5%	9.4%	14.3%	4.7%	17.0%	-1.9%
Min Ambiente (I)	Urban traffic	17.9	25.0%	16.1%	2.5%	0.1%	3.9%	5.8%	12.2%	18.4%	5.9%	11.1%	-1.0%
Puente Aranda (J)	Urban industrial	30.8	23.7%	28.3%	1.4%	0.1%	3.7%	5.0%	13.8%	10.8%	5.8%	8.8%	-1.5%
San Cristobal (K)	Urban background	14.2	22.2%	21.5%	1.0%	0.3%	2.2%	4.2%	13.8%	17.2%	6.6%	12.8%	-1.9%
Tunal (L)	Urban background	22.2	26.7%	14.4%	0.5%	2.4%	2.4%	4.7%	15.7%	24.5%	2.6%	8.8%	-2.7%
Usaquen (M)	Urban background	6.7	18.3%	6.6%	9.8%	1.8%	1.9%	2.5%	7.9%	14.4%	3.8%	34.9%	-1.9%

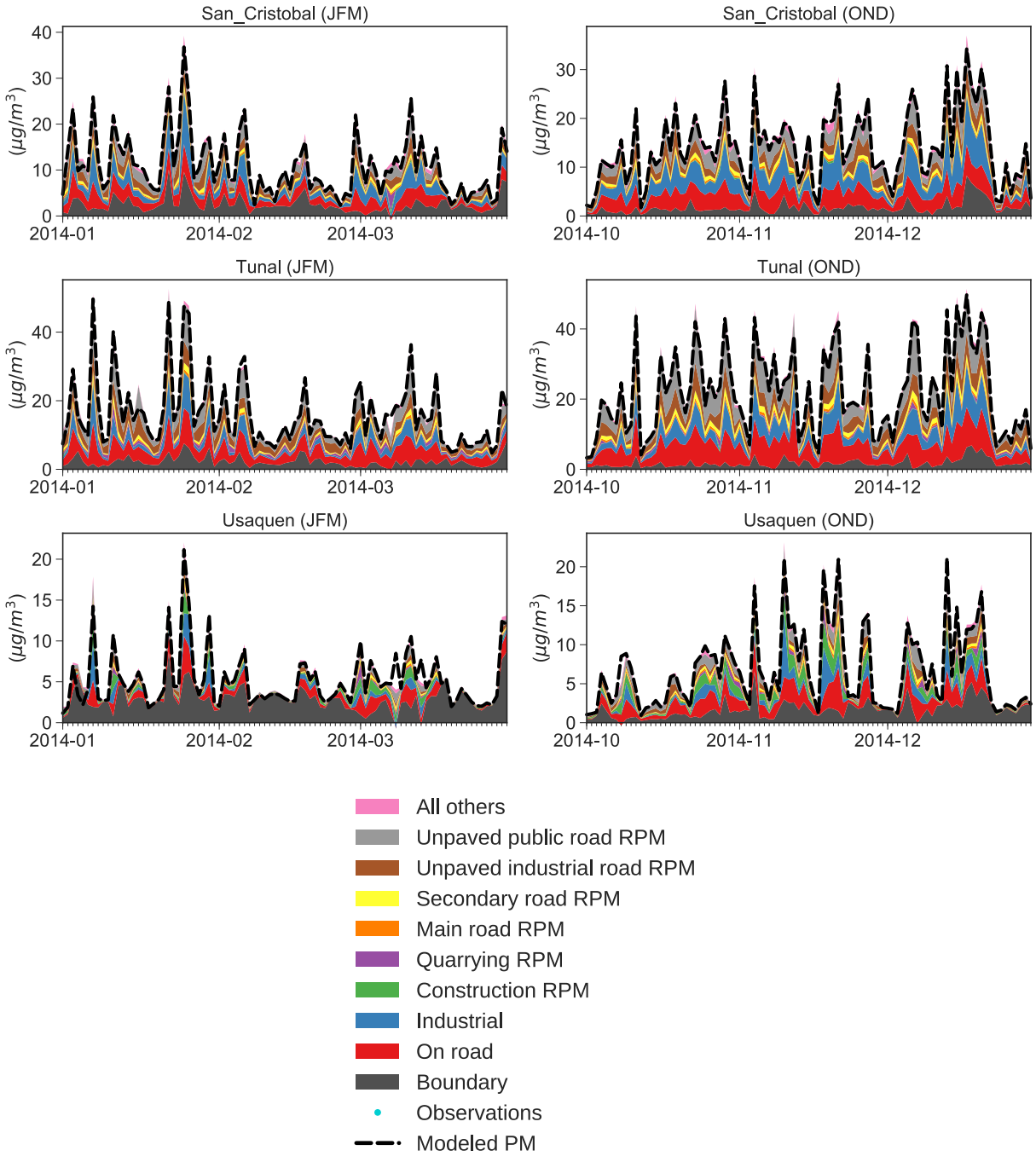
Table A.3: Reductions in modeled $PM_{2.5}$ at urban/suburban and traffic/industrial monitoring site locations for JFM and OND seasons.

Scenario	Urban/suburban sites		Traffic/industrial sites	
	JFM	OND	JFM	OND
SC1	-4.3%	-5.9%	-5.8%	-7.2%
SC2	-2.7%	-3.6%	-4.7%	-5.5%
SC3	-6.4%	-7.0%	-7.5%	-7.4%
SC4	-7.9%	-10.2%	-7.3%	-8.6%

Figure A.3: 24-hr $PM_{2.5}$ sector contributions during JFM and OND seasons at each monitoring site. Black dashed line is CMAQ-modeled $PM_{2.5}$.


Bolivia (JFM)
Bolivia (OND)
Carvajal (JFM)
Carvajal (OND)
CDAR (JFM)
CDAR (OND)
Convenio_176_10 (JFM)
Convenio_176_10 (OND)
Fontibon (JFM)
Fontibon (OND)
(µg/m3)
2014-01
2014-02
2014-03
2014-10
2014-11
2014-12


Guaymaral (JFM)
Guaymaral (OND)
Kennedy (JFM)
Kennedy (OND)
Las_Ferias (JFM)
Las_Ferias (OND)
MinAmbiente (JFM)
MinAmbiente (OND)
Puente_Aranda (JFM)
Puente_Aranda (OND)
(µg/m3)
2014-01
2014-02
2014-03
2014-10
2014-11
2014-12


San_Cristobal (JFM)
San_Cristobal (OND)
Tunal (JFM)
Tunal (OND)
Usaquen (JFM)
Usaquen (OND)
(µg/m³)
2014-01
2014-02
2014-03
2014-10
2014-11
2014-12
All others
Unpaved public road RPM
Unpaved industrial road RPM
Secondary road RPM
Main road RPM
Quarrying RPM
Construction RPM
Industrial
On road
Boundary
Observations
Modeled PM

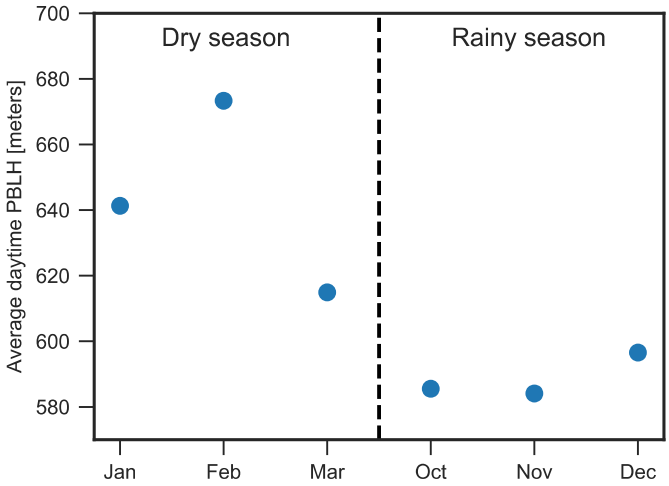


Figure A.4: Average daytime modeled planetary boundary layer height (PBLH) for each simulated month.

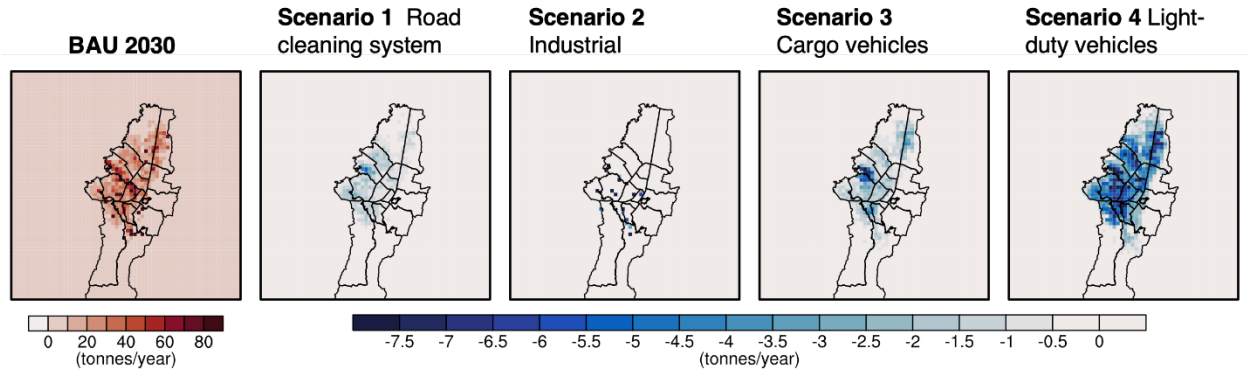


Figure A.5: Mean $PM_{2.5}$ emissions rates for 2030 BAU scenario and relative change under each mitigation strategy.

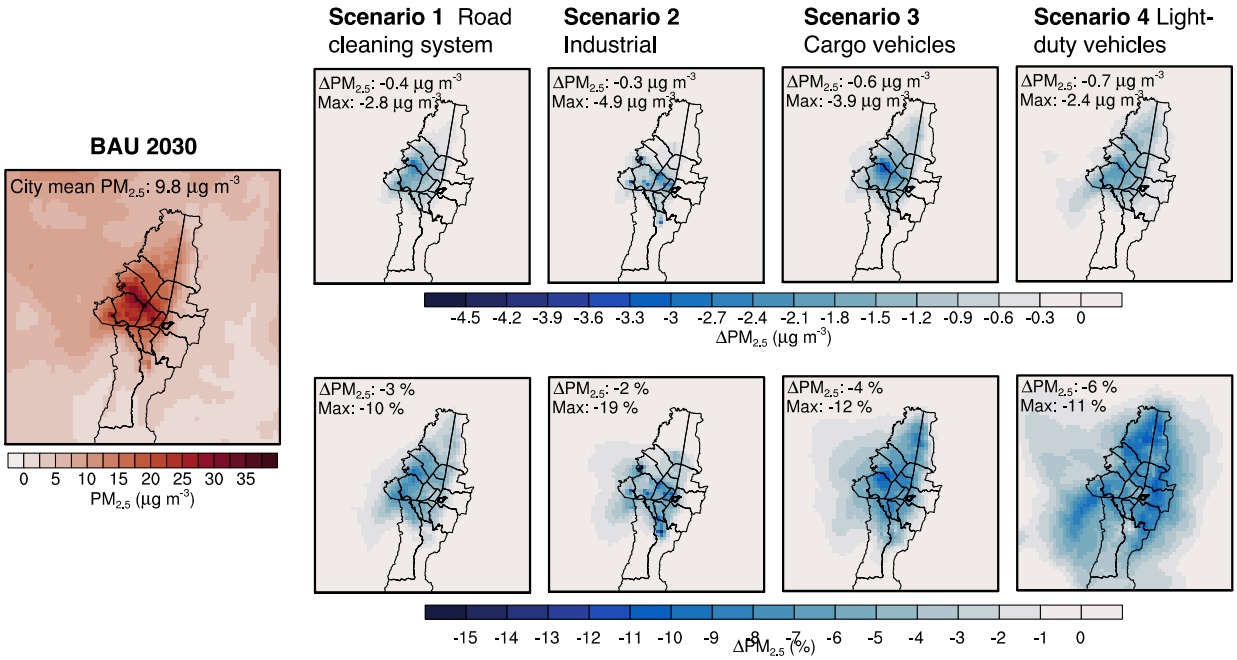


Figure A.6: 2030 BAU average $PM_{2.5}$ concentrations during JFM season (left), with absolute (top) and relative (bottom) changes for each emissions control scenario. City-average ($\Delta PM_{2.5}$) and maximum changes are listed for each mitigation scenario. Analogous results for the OND season are shown in Figure 2.8.

Appendix B: Supplementary material for Chapter 3

This appendix contains 11 figures and 3 tables. Further information is provided on monthly average inferred lightning NO_x (LNOx) emissions (Figs. B.1 and B.2), analysis increment in the lower and upper troposphere (Fig. B.3), inversion framework testing results (Fig. B.4), seasonal mean Jacobian (β) values (Fig. B.5), impact of NO_x emissions updates on modeled NO_2 VCDs during each season (Figs. B.6-B.9), emissions increments for TROPOMI v1.2.2 and TROPOMI v2.3.1 (Fig. B.10), prior and posterior emissions totals for January 2019 including TROPOMI v2.3.1 posterior (Fig. B.11), all CMAQ simulations performed for this study (Table B.1), and CMAQ model performance evaluated against ground O_3 observations (Table B.2) and NO_2 observations (Table B.3).

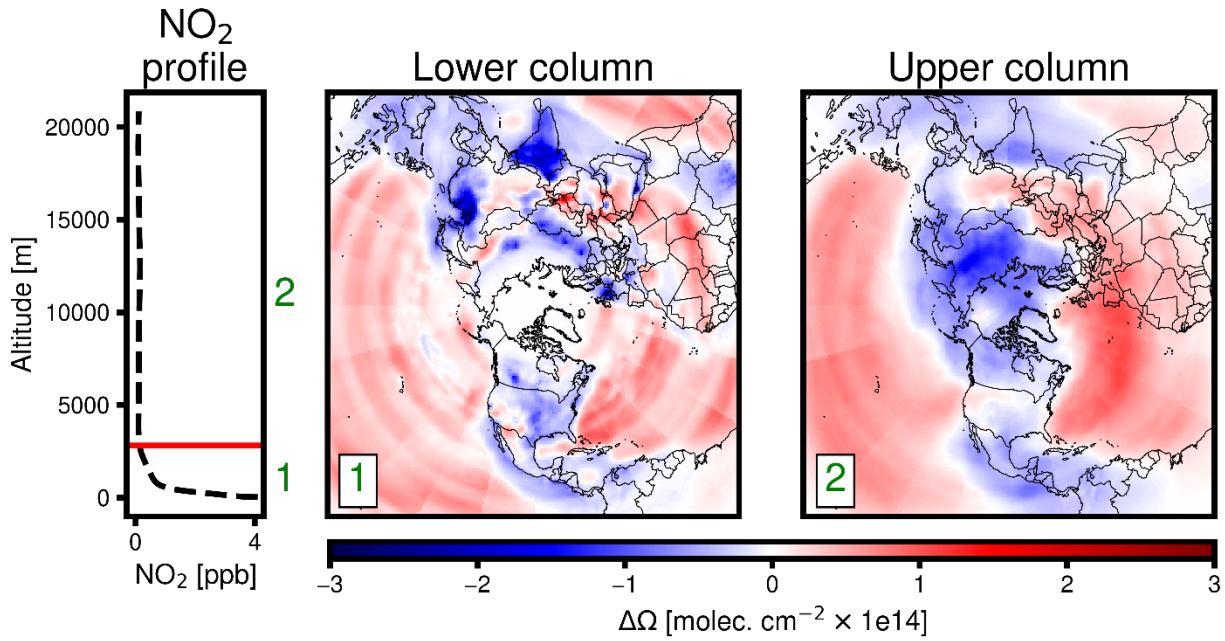


Figure B.1: Left panel: Vertical concentration profile of monthly-averaged July NO_2 for the model grid cell containing New York City. Red line indicates the cut-off used for lower troposphere in the emissions inversion. Center and right panels: Monthly average difference in modeled NO_2 columns between simulations with and without assimilated NO_2 satellite data ($\Delta\Omega$), for lower (1) and upper (2) column regions defined by the emissions inversion cutoff. $\Delta\Omega$ is shown for one iteration of assimilating July 2019 TROPOMI NO_2 with background errors for the boundary layer and LNO_x emissions updates applied.

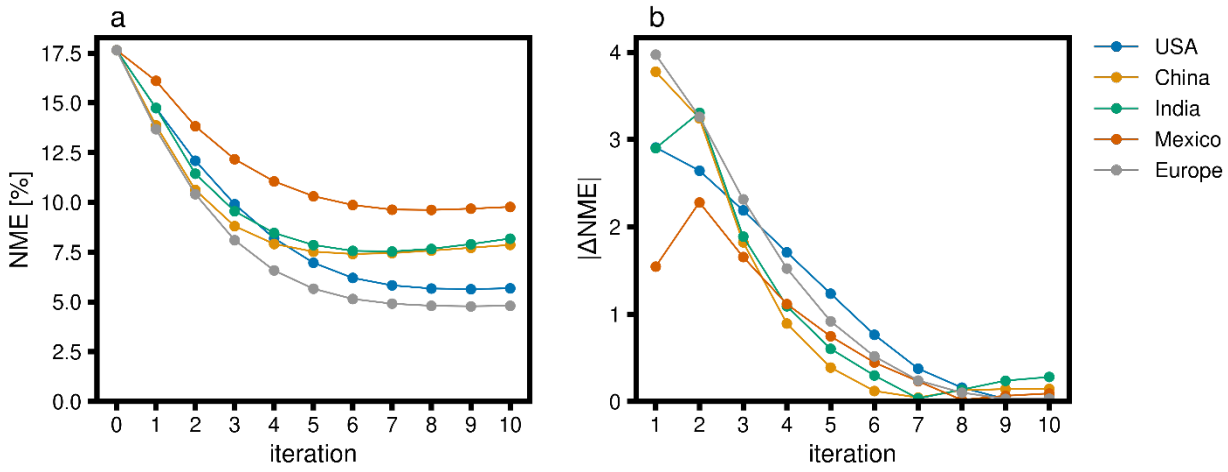


Figure B.2: Synthetic inversion test results showing (a) NME (%) by region for each iteration of FDMB, and (b) change in NME (%) by region for each iteration of FDMB. Synthetic observations are created by a simulation initialized with a uniform 15% reduction in anthropogenic NO_x emissions. Errors are computed using the VCDs within five regions of the Northern Hemisphere including only grid cells that are included in the inversion according to the filtering criteria described in Sect. 3.2.5.

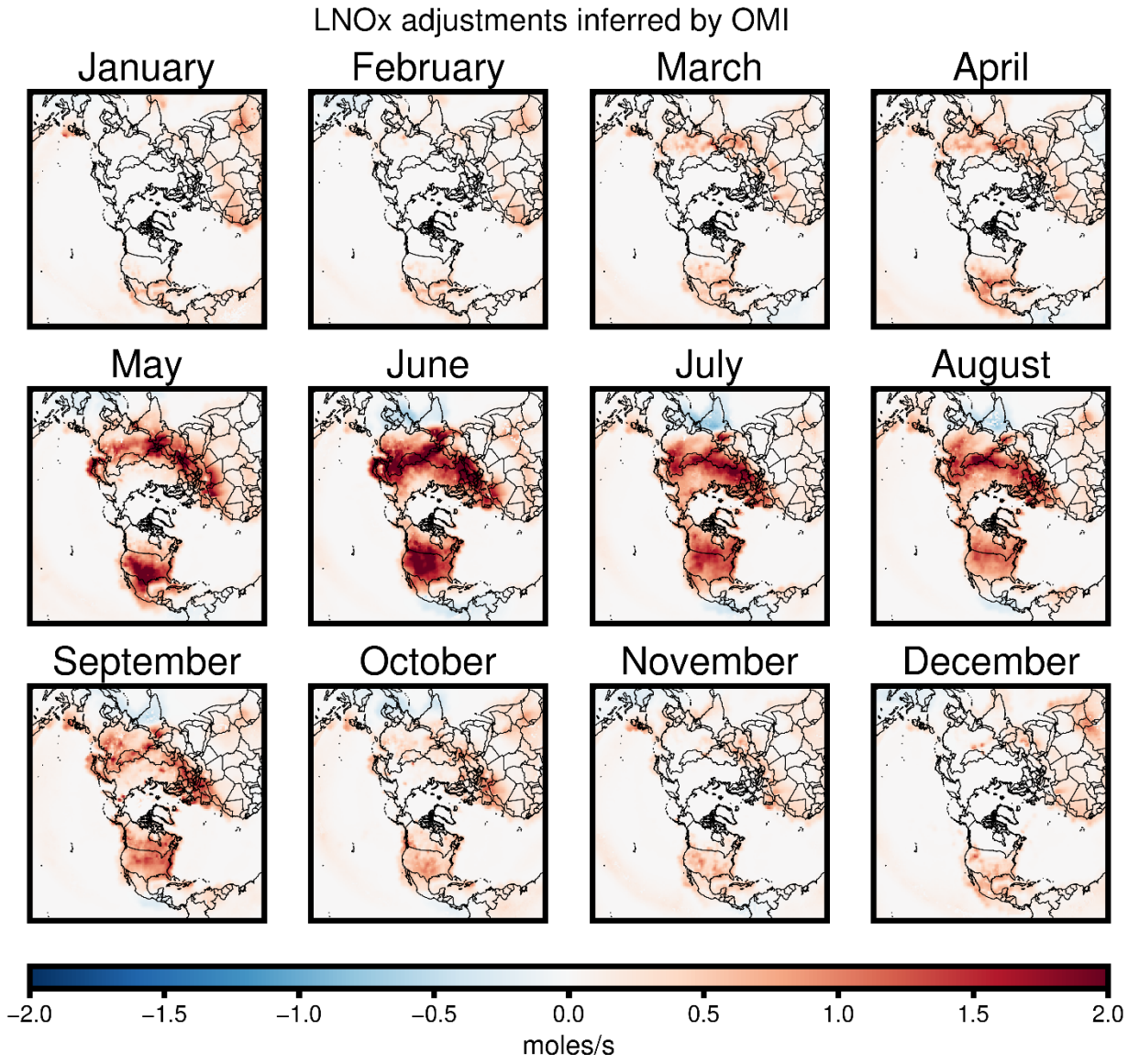


Figure B.3: Monthly-average 2019 LNOx emissions changes inferred from OMI observations.

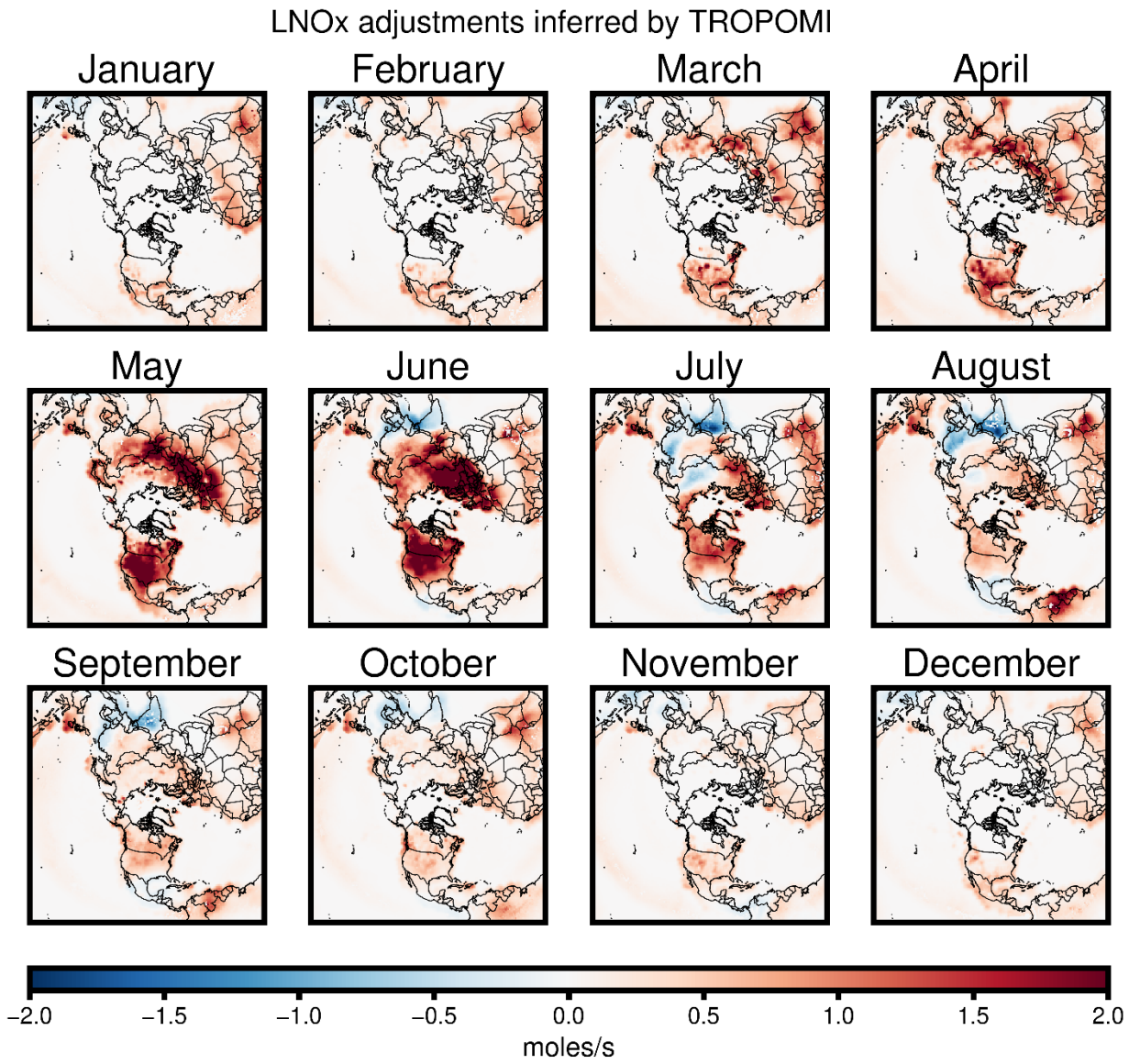


Figure B.4: Monthly-average 2019 LNOx emissions changes inferred from TROPOMI observations.

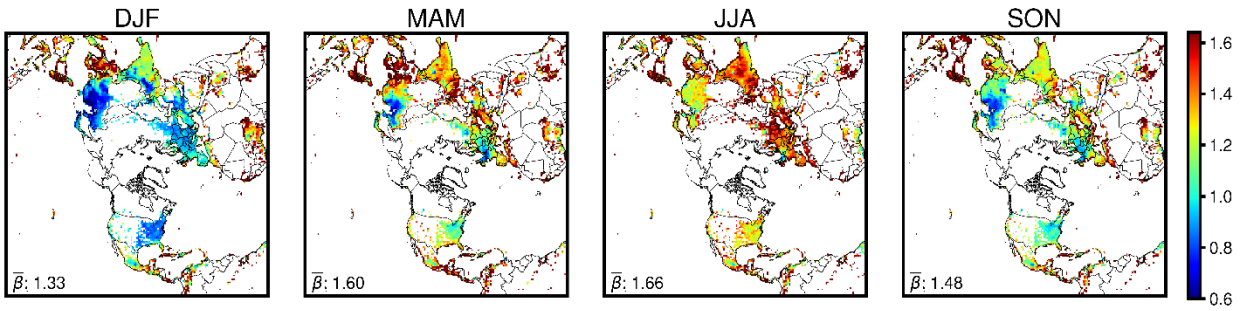


Figure B.5: Seasonal mean Jacobian ($\boldsymbol{\beta}$) relating changes in NO_x emissions to changes in the lower tropospheric NO_2 vertical column. $\boldsymbol{\beta}$ shown for winter (DJF), spring (MAM), summer (JJA), and fall (SON). Domain-average values across ($\bar{\boldsymbol{\beta}}$) are indicated on each map.

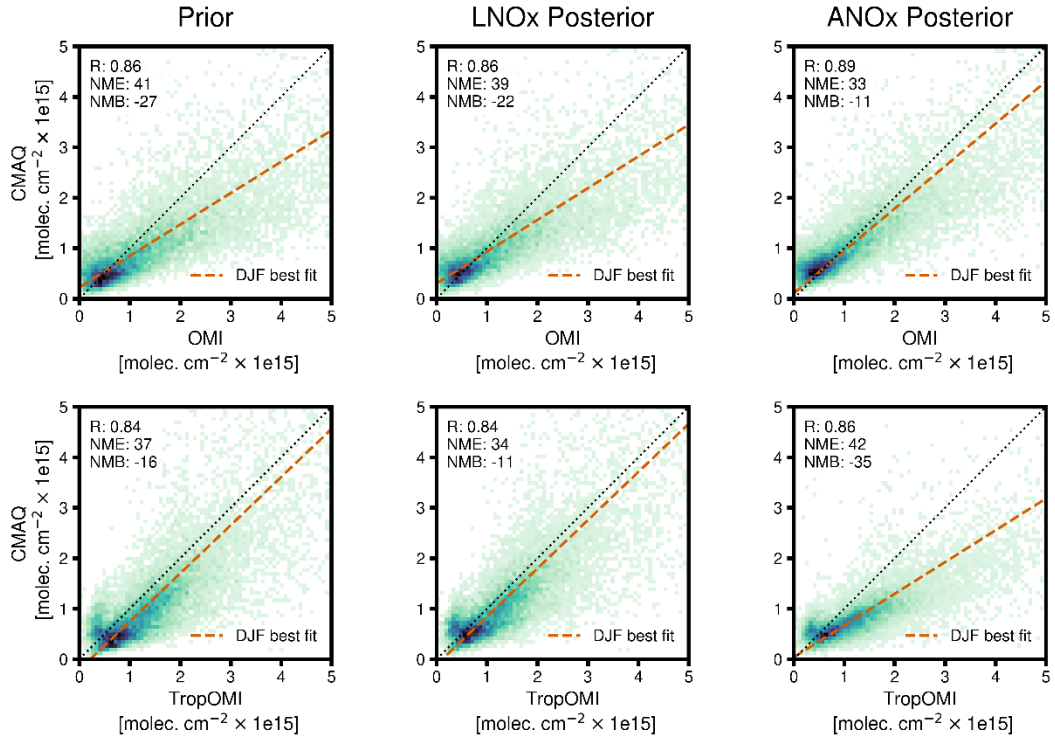


Figure B.6: Impact of NO_x emissions updates on modeled NO_2 VCD during winter (December, January, and February). Plots compare 2019 monthly-average CMAQ-modeled NO_2 VCD at each domain grid cell in which NO_x emissions were updated by the inverse modeling system against OMI and TROPOMI tropospheric NO_2 VCD retrievals averaged in each cell. Modeled NO_2 VCD using prior emissions (Prior), inferred LNOx emissions (LNOx posterior), and inferred lightning and anthropogenic NOx emissions (ANOx posterior) are each compared with NO_2 VCD retrievals. Top row plots compare retrievals and modeled VCD based on OMI observations, while bottom row plots compare retrievals and modeled VCD based on TROPOMI observations. Linear regression line, coefficient of determination (R), normalized mean error (NME), and normalized mean bias (NMB), relative to tropospheric NO_2 VCD retrievals, are shown for each CMAQ simulation.

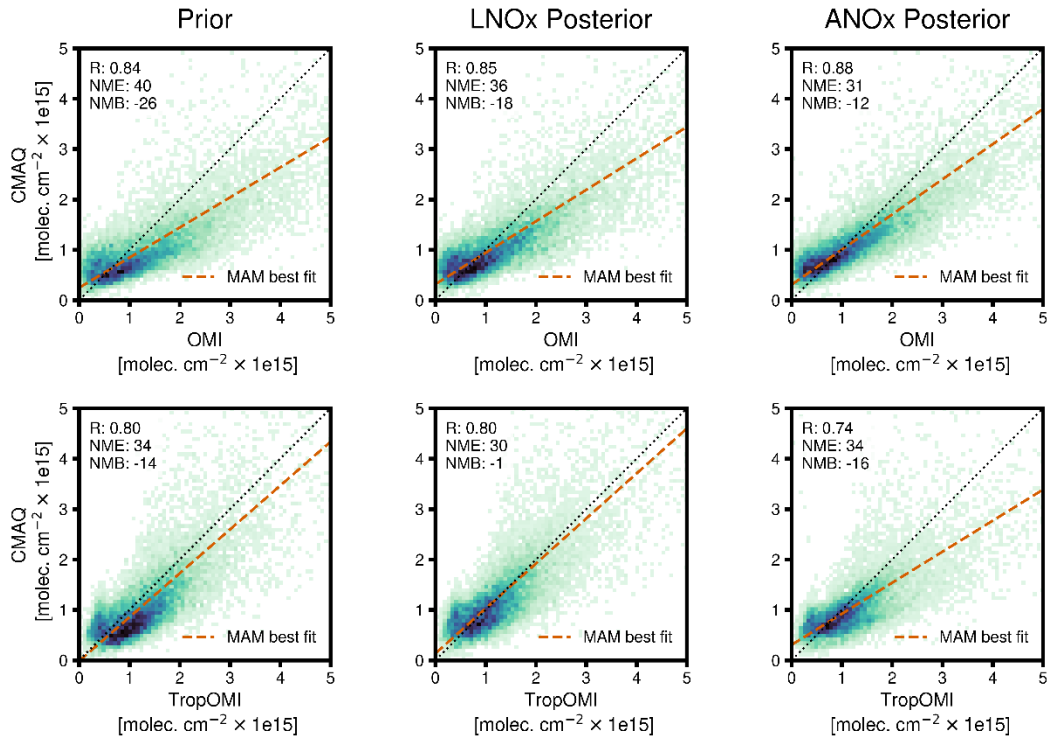


Figure B.7: Impact of NO_X emissions updates on modeled NO_2 VCD during spring (March, April, and May). Plots compare 2019 monthly-average CMAQ-modeled NO_2 VCD at each domain grid cell in which NO_X emissions were updated by the inverse modeling system against OMI and TROPOMI tropospheric NO_2 VCD retrievals averaged in each cell. Modeled NO_2 VCD using prior emissions (Prior), inferred LNOx emissions (LNOx posterior), and inferred lightning and anthropogenic NOx emissions (ANOx posterior) are each compared with NO_2 VCD retrievals. Top row plots compare retrievals and modeled VCD based on OMI observations, while bottom row plots compare retrievals and modeled VCD based on TROPOMI observations. Linear regression line, coefficient of determination (R), normalized mean error (NME), and normalized mean bias (NMB), relative to tropospheric NO_2 VCD retrievals, are shown for each CMAQ simulation.

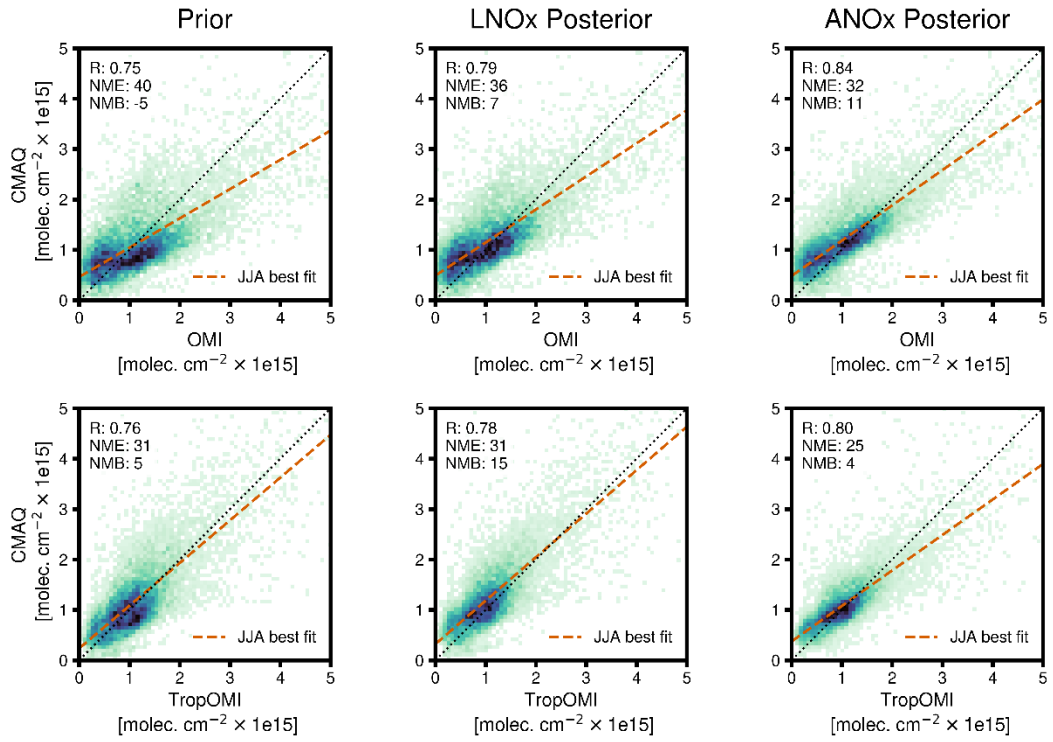


Figure B.8: Impact of NO_X emissions updates on modeled NO_2 VCD during summer (June, July, and August). Plots compare 2019 monthly-average CMAQ-modeled NO_2 VCD at each domain grid cell in which NO_X emissions were updated by the inverse modeling system against OMI and TROPOMI tropospheric NO_2 VCD retrievals averaged in each cell. Modeled NO_2 VCD using prior emissions (Prior), inferred LNOx emissions (LNOx posterior), and inferred lightning and anthropogenic NOx emissions (ANOx posterior) are each compared with NO_2 VCD retrievals. Top row plots compare retrievals and modeled VCD based on OMI observations, while bottom row plots compare retrievals and modeled VCD based on TROPOMI observations. Linear regression line, coefficient of determination (R), normalized mean error (NME), and normalized mean bias (NMB), relative to tropospheric NO_2 VCD retrievals, are shown for each CMAQ simulation.

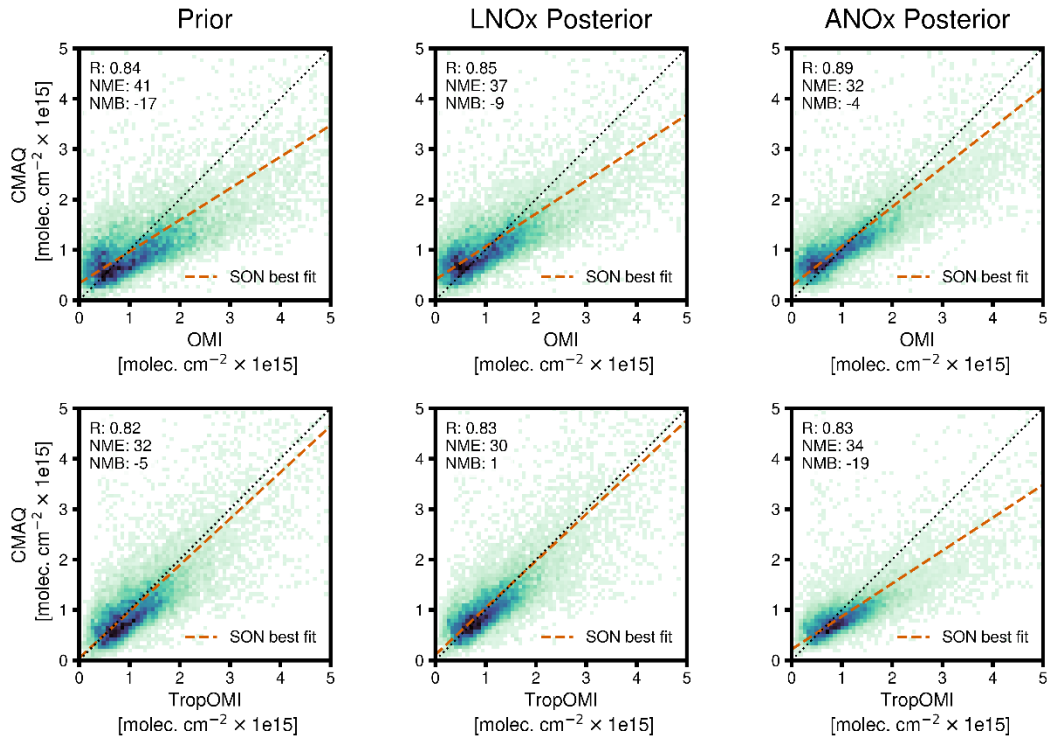


Figure B.9: Impact of NO_X emissions updates on modeled NO_2 VCD during autumn (September, October, and November). Plots compare 2019 monthly-average CMAQ-modeled NO_2 VCD at each domain grid cell in which NO_X emissions were updated by the inverse modeling system against OMI and TROPOMI tropospheric NO_2 VCD retrievals averaged in each cell. Modeled NO_2 VCD using prior emissions (Prior), inferred LNOx emissions (LNOx posterior), and inferred lightning and anthropogenic NOx emissions (ANOx posterior) are each compared with NO_2 VCD retrievals. Top row plots compare retrievals and modeled VCD based on OMI observations, while bottom row plots compare retrievals and modeled VCD based on TROPOMI observations. Linear regression line, coefficient of determination (R), normalized mean error (NME), and normalized mean bias (NMB), relative to tropospheric NO_2 VCD retrievals, are shown for each CMAQ simulation.

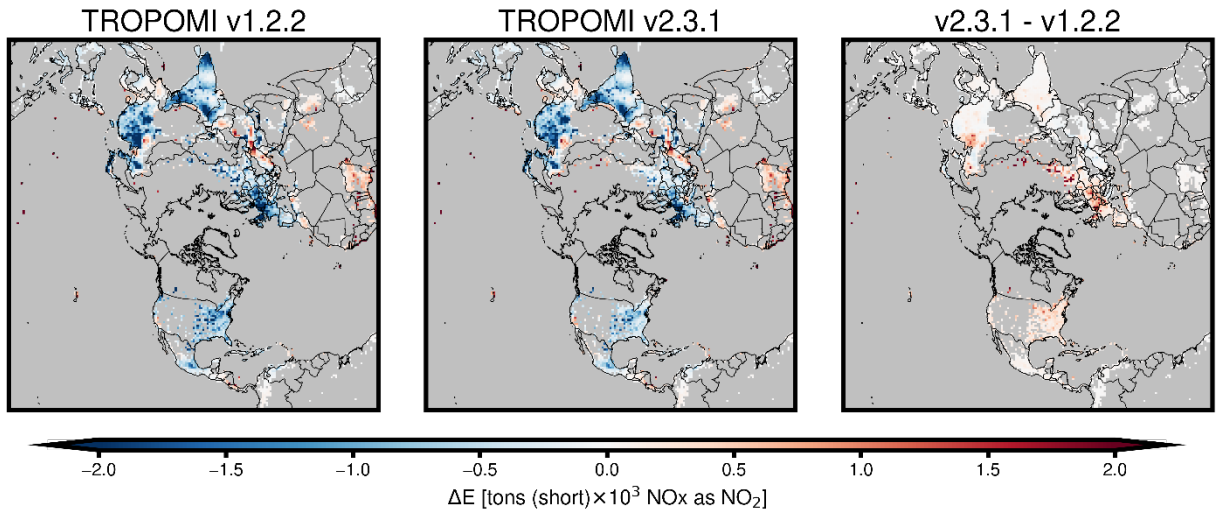


Figure B.10: Anthropogenic NO_x emissions increment for January 2019 inferred from TROPOMI version 1.2.2 and version 2.3.1 NO_2 observations, and difference in the analysis increment based on the two datasets.

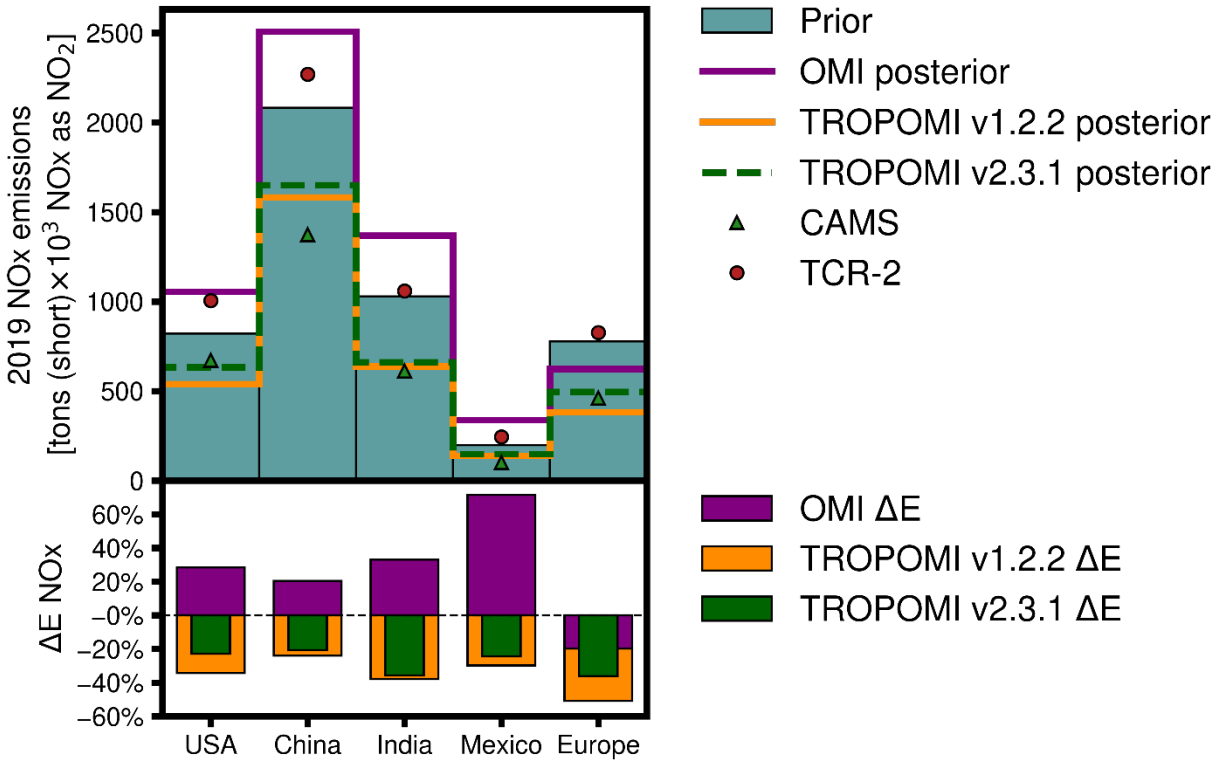


Figure B.11: Prior and satellite-inferred January 2019 anthropogenic NOx emissions in select global regions. Top plot shows total emissions (as NO_2) from prior emissions estimates, inference with TROPOMI version 1.2.2 and 2.3.1 observations (TROPOMI v1.2.2 and v2.3.1 posterior), inference with OMI (OMI posterior), and CAMS or TCR-2 inventories in the U.S., China, India,

Mexico, and Europe. The bottom plot shows the percent change ($\Delta E\ NO_x$) inferred with OMI or TROPOMI data, relative to prior emission estimates, for each region.

Table B.1: Hemispheric CMAQ simulations conducted as part of this study

Simulation	Satellite data assimilated	Anthropogenic Emissions (initial)	LNO_x emissions	Emissions perturbation	Iteration	Used for emissions adjustment
Base	-	Prior	Prior	-	No	-
No LNO_x	-	Prior	Prior	0% LNOx	No	-
OMI LNO_x update	OMI NO_2	Prior	Prior	-	No	LNO_x
TROPOMI LNO_x update	TROPOMI NO_2	Prior	Prior	-	No	LNO_x
OMI Base LNO_x	-	Prior	Posterior (OMI)	-	No	-
OMI ANO_x reduced	-	Prior	Posterior (OMI)	ANOx 15% reduced	No	-
OMI ANO_x update	OMI NO_2	Prior	Posterior (OMI)	-	Yes	ANO_x
OMI posterior	-	Posterior (OMI)	Posterior (OMI)	-	No	-
TROPOMI Base LNO_x	-	Prior	Posterior (TropOMI)	-	No	-
TROPOMI ANOx reduced	-	Prior	Posterior (TropOMI)	ANOx 15% reduced	No	-
TROPOMI ANOx update	TROPOMI NO_2	Prior	Posterior (TropOMI)	-	Yes	ANO_x
TROPOMI posterior	-	Posterior (TropOMI)	Posterior (TropOMI)	-	No	-

Table B.2: CMAQ model performance evaluated against daily maximum 8-hour O_3 concentrations (MDA8 O_3) observed in 2019 by 1,218 AQS monitoring sites in the U.S., during winter (Dec.-Feb.), spring (Mar.-May), summer (Jun.-Aug.), and fall (Sep.-Nov.) months. Near-road monitors are not considered. Statistics are shown for simulations using prior emissions (Prior), lightning and anthropogenic NOx emissions inferred with OMI data (OMI-inferred), and lightning and anthropogenic NOx emissions inferred with TROPOMI data (TROPOMI-inferred). Coefficient of determination (R), normalized mean error (NME), and normalized mean bias (NMB), relative to AQS observations, are estimated for each CMAQ simulation.

Season	NO_x emissions	R	NME	NMB
Winter	Prior	0.60	15.8%	-4.8%
	OMI-inferred	0.61	15.0%	-0.5%
	TROPOMI-inferred	0.56	16.0%	-3.0%
Spring	Prior	0.53	15.9%	-10.2%
	OMI-inferred	0.57	13.4%	-4.9%
	TROPOMI-inferred	0.55	16.6%	-11.6%
Summer	Prior	0.58	17.2%	6.8%
	OMI-inferred	0.65	17.8%	11.8%
	TROPOMI-inferred	0.67	14.4%	3.2%
Fall	Prior	0.73	14.5%	2.3%
	OMI-inferred	0.74	15.1%	6.6%
	TROPOMI-inferred	0.73	14.3%	-1.0%

Table B.3: CMAQ model performance evaluated against daily 24-h average NO_2 concentrations observed in 2019 by 1,218 AQS monitoring sites in the U.S., during winter (Dec.-Feb.), spring (Mar.-May), summer (Jun.-Aug.), and fall (Sep.-Nov.) months. Near-road monitors are not considered. Statistics are shown for simulations using prior emissions (Prior), lightning and anthropogenic NOx emissions inferred with OMI data (OMI-inferred), and lightning and anthropogenic NOx emissions inferred with TROPOMI data (TROPOMI-inferred). Coefficient of determination (R), normalized mean error (NME), and normalized mean bias (NMB), relative to AQS observations, are estimated for each CMAQ simulation.

Season	NO_x emissions	R	NME	NMB
Winter	Prior	0.42	64.8%	-60.5%
	OMI-inferred	0.47	57.4%	-49.3%
	TROPOMI-inferred	0.47	76.3%	-75.4%
Spring	Prior	0.45	64.2%	-59.7%
	OMI-inferred	0.48	57.5%	-48.7%
	TROPOMI-inferred	0.48	70.9%	-69.0%
Summer	Prior	0.37	57.2%	-49.0%
	OMI-inferred	0.39	55.4%	-46.0%
	TROPOMI-inferred	0.43	63.3%	-59.5%
Fall	Prior	0.41	60.8%	-55.5%
	OMI-inferred	0.53	58.5%	-53.1%
	TROPOMI-inferred	0.44	72.4%	-71.0%

Appendix C: Supplementary material for Chapter 4

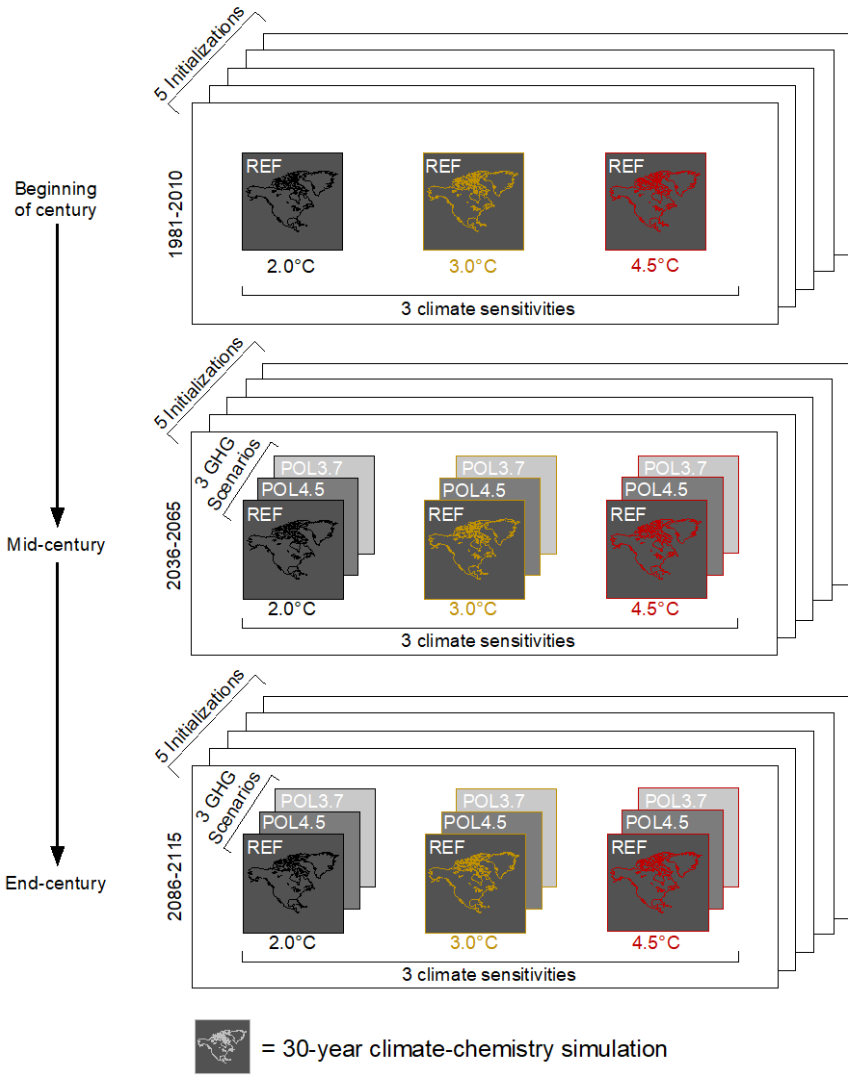


Figure C.1: Climate and atmospheric chemistry simulation ensemble design. The ensemble used for this study is designed to capture the three main climate-related uncertainties in projections air quality impacts. Internal variability is represented through five climate model initializations, model response uncertainty is represented through three climate sensitivities, and scenario uncertainty is represented through three GHG emissions scenarios. At the beginning of the century, a single GHG emissions scenario (REF) is considered. Additional climate-mitigation scenarios (POL4.5 and POL3.7) are introduced later in the century. Thirty-year periods of atmospheric chemistry are simulated at mid-century (2036-2065) and end-century (2086-2115) under each initialization, climate sensitivity, and GHG emission scenario considered. See Methods for additional details.

Table C.1: Mean climate-induced increases in U.S. population-weighted O_3 and $PM_{2.5}$ concentrations projected at the end of 21st century, relative to start of the century, for each climate sensitivity and policy scenario considered.

Policy	CS	May-Sept. O_3 (ppb)	Annual O_3 (ppb)	Annual $PM_{2.5}$ ($\mu g\ m^{-3}$)
P37	2.0	0.7 ± 0.7	0.3 ± 0.4	0.0 ± 0.1
	3.0	1.7 ± 0.5	0.6 ± 0.3	0.3 ± 0.1
	4.5	1.7 ± 0.6	0.2 ± 0.3	0.6 ± 0.1
P45	2.0	0.4 ± 0.5	0.0 ± 0.3	0.0 ± 0.1
	3.0	1.4 ± 0.5	0.3 ± 0.3	0.4 ± 0.1
	4.5	2.6 ± 0.5	0.7 ± 0.3	0.7 ± 0.1
REF	2.0	4.8 ± 0.5	1.6 ± 0.3	0.6 ± 0.1
	3.0	8.1 ± 0.5	3.2 ± 0.3	1.5 ± 0.1
	4.5	13.1 ± 0.6	5.8 ± 0.4	2.1 ± 0.1

Table C.2: Ensemble-mean U.S.-average temperature increase projected under each policy scenario and climate sensitivity considered in this study.

Policy	Climate Sensitivity (°C)	Mid-century (°C)	End-century (°C)
P37	2.0	0.6	0.7
	3.0	1.1	1.3
	4.5	1.4	1.8
P45	2.0	0.7	0.9
	3.0	1.3	1.6
	4.5	1.6	2.3
REF	2.0	1.5	3.7
	3.0	2.1	5.4
	4.5	2.6	7.3

Appendix D: Supplementary material for Chapter 5

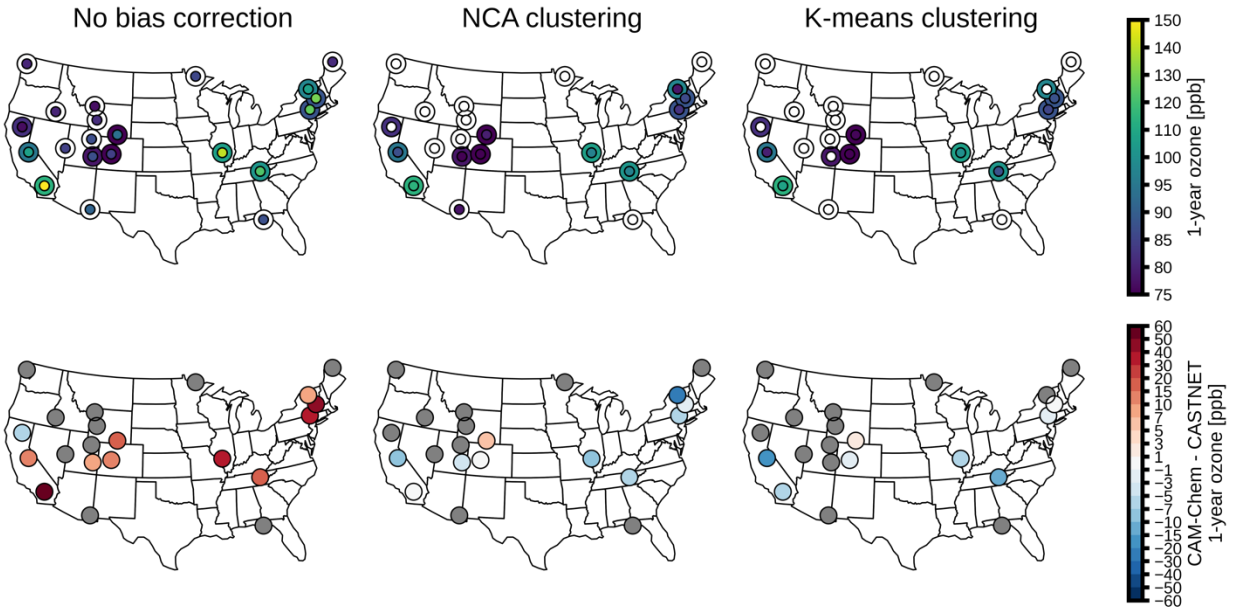


Figure D.1: Pre-SIP comparison in sites where no statistically significant trend was detected. Only CASTNET data prior to 2001 was considered. Sites which had a statistically significant mean trend were removed from the comparison. Detrending was not applied to CASTNET before calculating extremes. Top plot shows 1-year ozone return levels calculated at CASTNET sites (outer circles) and with M_NCA modeled ozone (grouped by NCA regions for the bias correction) (inner circles). White-filled inner and outer circles indicate an insufficient number of extreme values to calculate a return level. Bottom plot shows the difference between the M_NCA return level and the CASTNET return level. Positive values indicate M_NCA is high biased compared to CASTNET. Grey circles indicate insufficient number of extreme values to calculate a return level in either M_NCA or CASTNET.

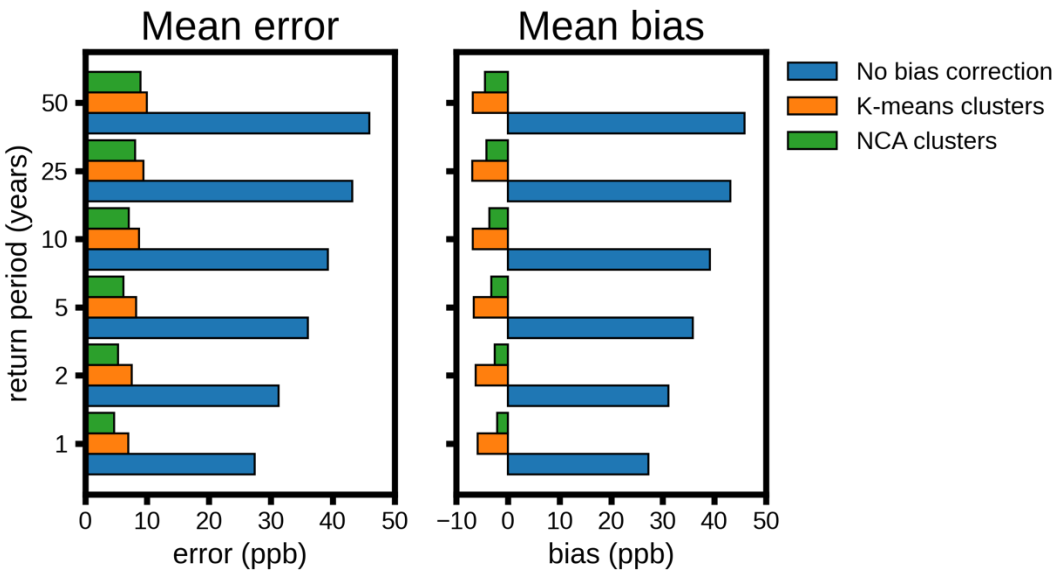


Figure D.2: Mean error and mean bias in ozone return levels computed with clusters by CASTNET site latitude, longitude, mean MDA8 O_3, MDA8 O_3 standard deviation; with the same and additionally clustered by the 95th percentile MDA8 O_3 at each CASTNET site; and by NCA geographic region.

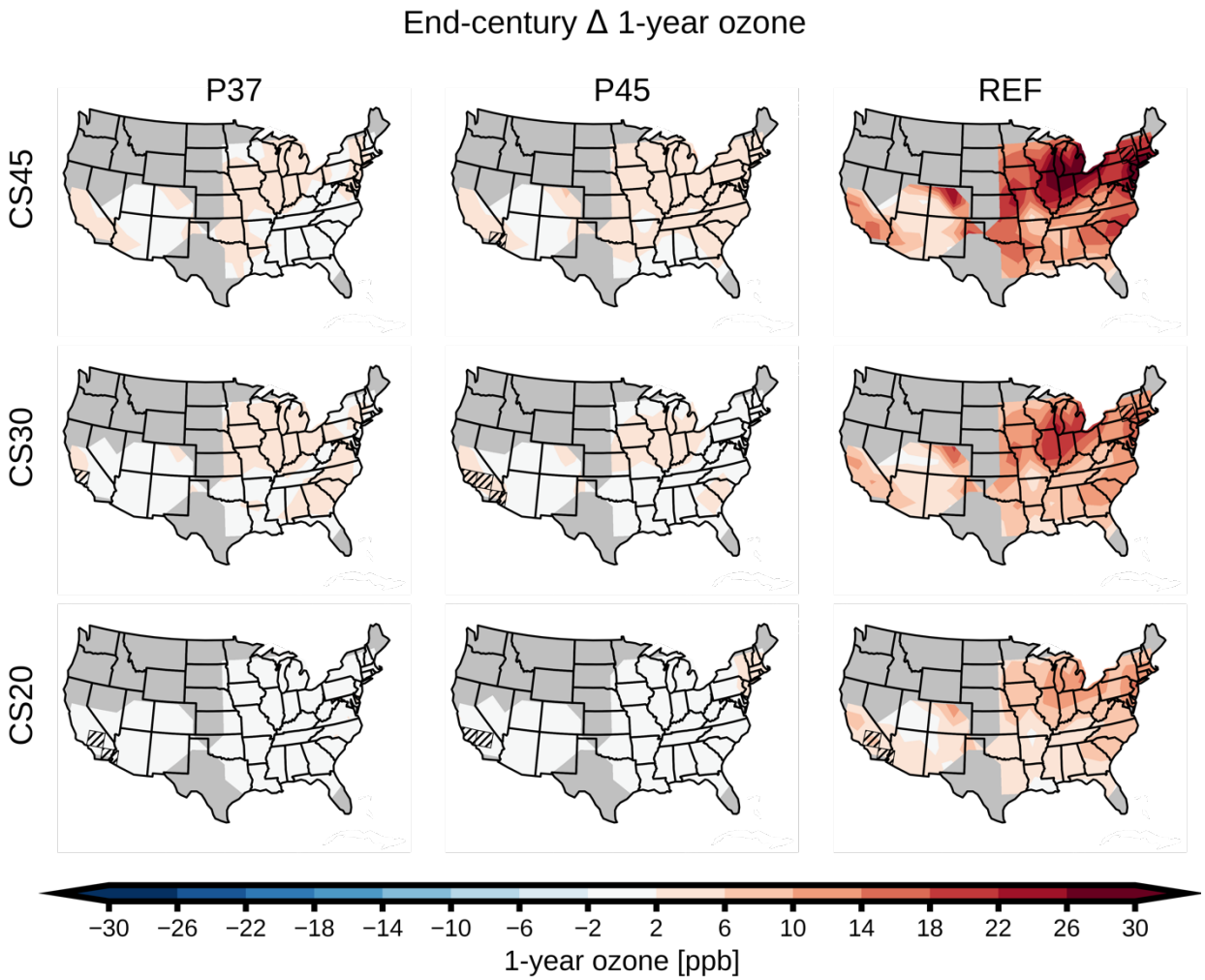


Figure D.3: Change in 1-year ozone return level at end-century across multiple policies and multiple climate sensitivities. Changes are relative to the historical simulation with the corresponding climate sensitivity. Cells are hatched where modeled extremes do not match the fitted GP distribution at the 95% confidence interval according to the K-S test. Grey areas represent cells with an insufficient number of threshold exceedances to calculate a return level and return interval in the historical simulation.

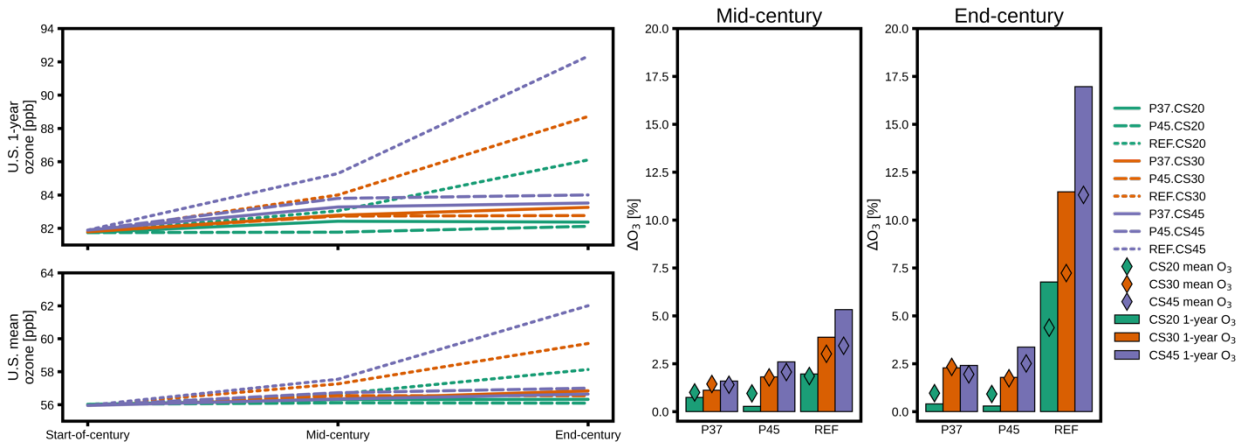


Figure D.4: U.S. average 1-year O_3 and mean O_3 concentration at beginning, middle, and end-century under each climate sensitivity and each policy scenario. Top left: U.S. average 1-year O_3 using bias-corrected simulated O_3. Bottom left: Same, but showing U.S. average mean O_3 concentration. Middle: Relative change in U.S. average 1-year O_3 and mean O_3 concentration relative to start-of-century conditions. Right: Same, but for end-century.